Advanced Mathematics for Engineering with Python

Solutions Manual

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Solutions to the end-of-chapter problems of Chapters 1–14

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Contents

Preface iii

1 A Brief Python Introduction for Symbolic Mathematics 1

2 First-Order Ordinary Differential Equations 7

3 Higher-Order Ordinary Differential Equations 13

4 Systems of Ordinary Differential Equations (Representative Problems) 19

5 Laplace Transforms 27

6 Matrices, Vectors, and Linear Systems 33

7 Eigenvalues and Eigenvectors in Engineering Applications 37

8 Vector Calculus 39

9 Fourier Series 41

10 Fourier Transform 45

11 Partial Differential Equations 51

12 Optimization 55

13 Neural Network Methods for Partial Differential Equations 61

14 Physics-Informed Neural Networks 81

References 95

Preface

This manual collects worked solutions to the end-of-chapter problems of *Advanced Mathematics for Engineering with Python – A Computational Approach to Engineering Mathematics*. Solutions are grouped by chapter and numbered as in the chapter problem sets (for Chapters 4, 8–12 and 14 the numbering follows the representative problems of this manual; see the notes at the beginning of those chapters). Figures labelled "Figure S $n.m$" were generated with the Python scripts of the companion repository (`solutions_figures/`); every script is named after the problem it illustrates and can be re-run with `python generate_all.py`.

Chapter 1

A Brief Python Introduction for Symbolic Mathematics

Problem 1.1: NumPy Linear Algebra: Matrix Operations

Solution. The coefficient matrix is $A = \begin{pmatrix} 2 & 3 & -1 \\ 1 & -2 & 4 \\ 3 & 1 & 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 5 \\ -3 \\ 8 \end{pmatrix}$. Using NumPy's linear algebra solver:

$$\begin{aligned} \mathbf{x} &= A^{-1}\mathbf{b} \\ &= \begin{pmatrix} 61/7 \\ -43/7 \\ -6 \end{pmatrix} \approx \begin{pmatrix} 8.714 \\ -6.143 \\ -6.000 \end{pmatrix} \end{aligned}$$

The determinant is $\det(A) = 7$, rank is 3 (full rank), and condition number $\kappa_2(A) \approx 18.3$, indicating a well-conditioned system.

Problem 1.2: NumPy Linear Algebra: Eigenvalue Decomposition

Solution. For matrix $A = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}$, the characteristic polynomial is $\det(A - \lambda I) = \lambda^2 - 6\lambda + 1 = 0$, giving eigenvalues:

$$\lambda_1 = 3 + 2\sqrt{2} \approx 5.83, \quad \lambda_2 = 3 - 2\sqrt{2} \approx 0.17$$

The corresponding normalized eigenvectors are $v_1 \approx \begin{pmatrix} 0.924 \\ 0.383 \end{pmatrix}$ and $v_2 \approx \begin{pmatrix} -0.383 \\ 0.924 \end{pmatrix}$. Verification: $Av_1 = \lambda_1 v_1$ holds exactly. The diagonalization is $A = PDP^{-1}$ where P has eigenvectors as columns and $D = \text{diag}(\lambda_1, \lambda_2)$.

Problem 1.3: NumPy Linear Algebra: Singular Value Decomposition

Solution. For $B = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$, the SVD is $B = U\Sigma V^T$ with singular values $\sigma_1 \approx 9.51$ and $\sigma_2 \approx 0.77$. The rank is 2. Reconstruction: $B = \sigma_1 u_1 v_1^T + \sigma_2 u_2 v_2^T$ recovers the original matrix exactly (up to numerical precision). The row space is spanned by the first two right singular vectors, the column space by the first two left singular vectors.

Problem 1.4: NumPy Linear Algebra: Matrix Norms

Solution. For any matrix A: (1) Frobenius norm: $\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2}$, (2) Spectral norm: $\|A\|_2 = \sigma_{\max}(A)$ (largest singular value), (3) Nuclear norm: $\|A\|_* = \sum_i \sigma_i$ (sum of singular values). Under scaling by factor c: all norms scale by $|c|$. Under orthogonal transformations (rotation): $\|A\|_2$ and $\|A\|_F$ are invariant, as are singular values.

Problem 1.5: SymPy Calculus: Limits

Solution. Using SymPy's limit function:

$$\lim_{x\to 0}\frac{\sin(3x)}{2x}=\frac{3}{2}\quad\text{(L'Hôpital or series)}$$
$$\lim_{x\to\infty}\frac{5x^3+2x^2}{3x^3+x}=\frac{5}{3}\quad\text{(divide by highest power)}$$
$$\lim_{x\to 0^+}x\ln(x)=0\quad\text{(using }\lim\frac{\ln(x)}{1/x}=\lim\frac{1/x}{-1/x^2}=0\text{)}$$
$$\lim_{n\to\infty}\left(1+\frac{2}{n}\right)^{3n}=e^6\quad\text{(standard exponential limit)}$$

Problem 1.6: SymPy Calculus: Derivatives and Optimization

Solution. For $f(x)=x^4-4x^3+6x^2-4x+1=(x-1)^4$:

$$f'(x)=4(x-1)^3,\quad f''(x)=12(x-1)^2$$

Critical point: $f'(x)=0\Rightarrow x=1$. At $x=1$: $f''(1)=0$ (second-derivative test inconclusive). Using higher derivatives: $f'''(1)=0$, $f^{(4)}(1)=24>0$. Since the first nonvanishing derivative is of even order and positive, $(1,0)$ is a local minimum; indeed $f(x)=(x-1)^4\geq 0$ makes it the global minimum.

Problem 1.7: SymPy Calculus: Integration

Solution. Computing integrals with SymPy:

$$\int_0^{\pi/2}x\sin(x)\,dx=1\quad\text{(integration by parts)}$$
$$\int\frac{x^2}{x^3+1}\,dx=\frac{1}{3}\ln|x^3+1|+C$$
$$\int_1^{\infty}\frac{1}{x^2(1+x)}\,dx=1-\ln(2)\quad\text{(partial fractions: }\tfrac{1}{x^2(1+x)}=-\tfrac{1}{x}+\tfrac{1}{x^2}+\tfrac{1}{1+x}\text{)}$$
$$\int e^{-x^2}\cos(x)\,dx\quad\text{(no elementary closed form; SymPy returns error functions of complex argument)}$$

Problem 1.8: SymPy: Taylor Series Approximation

Solution. For $f(x)=\cos(x)+e^x$:

$$T_5(x;0)=2+x+\frac{x^3}{6}+\frac{x^4}{12}+\frac{x^5}{120}+O(x^6)\quad\text{(the }x^2\text{ terms of }\cos x\text{ and }e^x\text{ cancel)}$$
$$T_4(x;\pi)=(e^\pi-1)+e^\pi(x-\pi)+\frac{(e^\pi+1)}{2}(x-\pi)^2+\ldots$$

At $x=0.5$: $T_5(0.5;0)\approx 2.5263$ vs. exact $f(0.5)=\cos(0.5)+e^{0.5}\approx 2.5263$ (agreement to four decimals). At $x=\pi+0.1$: the Taylor series around π provides better accuracy than the series around 0.

Problem 1.9: SymPy: Solving Equations

Solution. Solving with SymPy:

$$x^4-5x^2+4=0\Rightarrow(x^2-1)(x^2-4)=0\Rightarrow x\in\{-2,-1,1,2\}$$
$$\sin(x)=\frac{x}{2}\text{ has solutions approximately }x\approx 0,\pm 1.895$$
$$e^x=3x\text{ has solutions approximately }x\approx 0.619,1.512$$

$\ln(x) + x = 5$ has solution approximately $x \approx 3.693$

Problem 1.10: SymPy: Symbolic Matrix Operations

Solution. For symbolic matrices $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$:

$$AB = \begin{pmatrix} ae + bg & af + bh \\ ce + dg & cf + dh \end{pmatrix}$$

$$BA = \begin{pmatrix} ae + cf & be + df \\ ag + ch & bg + dh \end{pmatrix}$$

$$\det(A - \lambda I) = (a - \lambda)(d - \lambda) - bc = \lambda^2 - (a + d)\lambda + (ad - bc)$$

$$A^2 - 2A + I = \begin{pmatrix} a^2 + bc - 2a + 1 & b(a + d - 2) \\ c(a + d - 2) & bc + d^2 - 2d + 1 \end{pmatrix}$$

Problem 1.11: Matplotlib: Complex Function Visualization

Solution. For $f(z) = z^3 - 1$, the three cube roots of unity are at $e^{2\pi ik/3}$ for $k = 0, 1, 2$. (1) Magnitude plot: $|f(z)| = |z^3 - 1|$ is a 3D surface with minima at the roots. (2) Phase plot: $\arg(f(z))$ shows color variation representing argument, with discontinuities along branch cuts. (3) Roots marked at $(1, 0)$, $(-1/2, \sqrt{3}/2)$, $(-1/2, -\sqrt{3}/2)$ in complex plane. (4) Contours connect points of constant magnitude and phase.

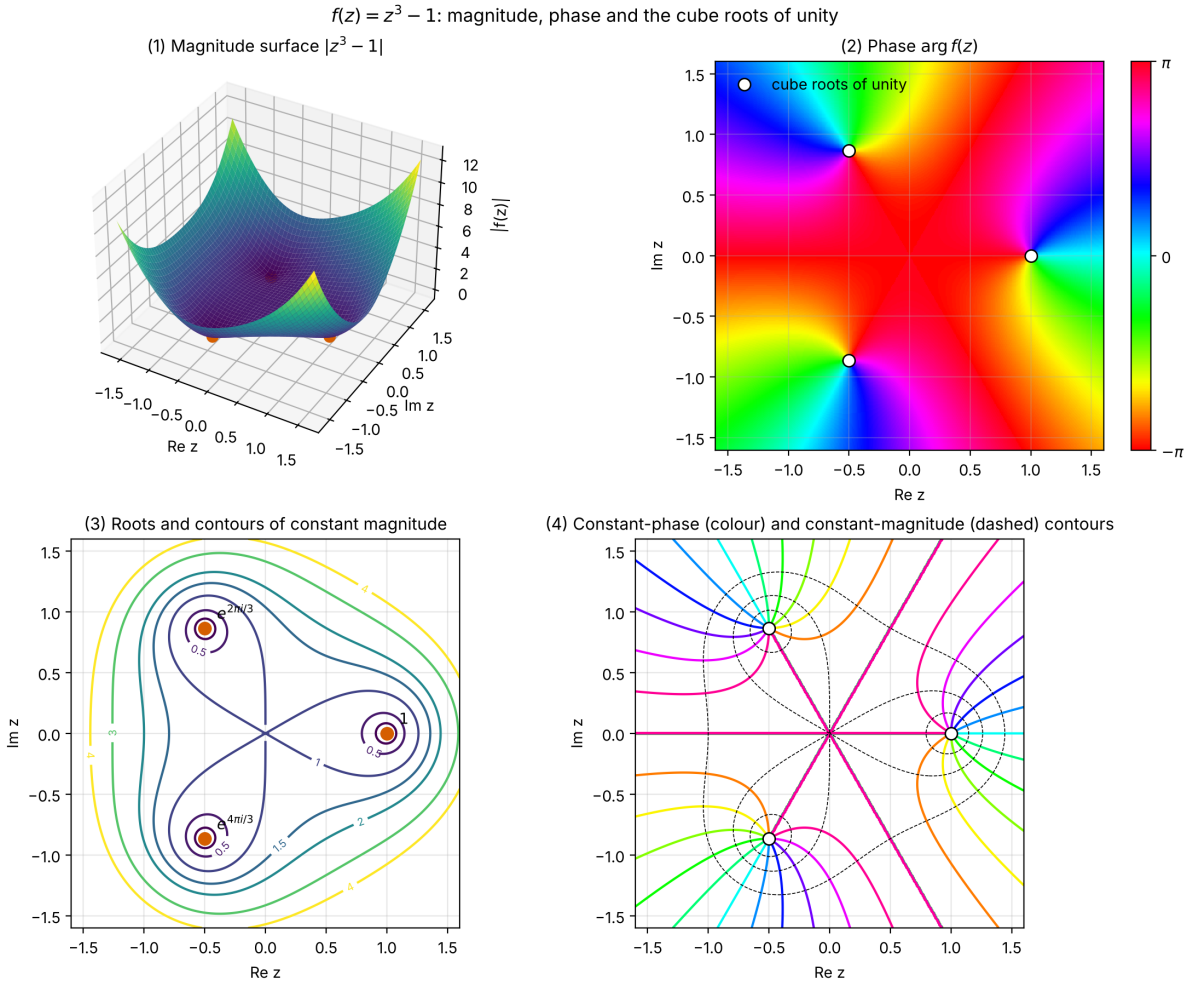


Figure S1.11. $f(z) = z^3 - 1$: magnitude surface, phase map, cube roots of unity and contours of constant magnitude and phase.

Problem 1.12: Matplotlib: 3D Plotting

Solution. The Rosenbrock function $f(x, y) = (1 - x)^2 + 100(y - x^2)^2$ has a global minimum at $(1, 1)$ with value 0. The surface shows a narrow valley oriented along the parabola $y = x^2$. (1) Surface plot from multiple angles reveals the valley structure. (2) Contour plot shows elliptical level curves concentrated in the valley. (3) Global minimum marked at $(1, 1)$. (4) Logarithmic scale highlights gradual terrain and the difficulty of optimization.

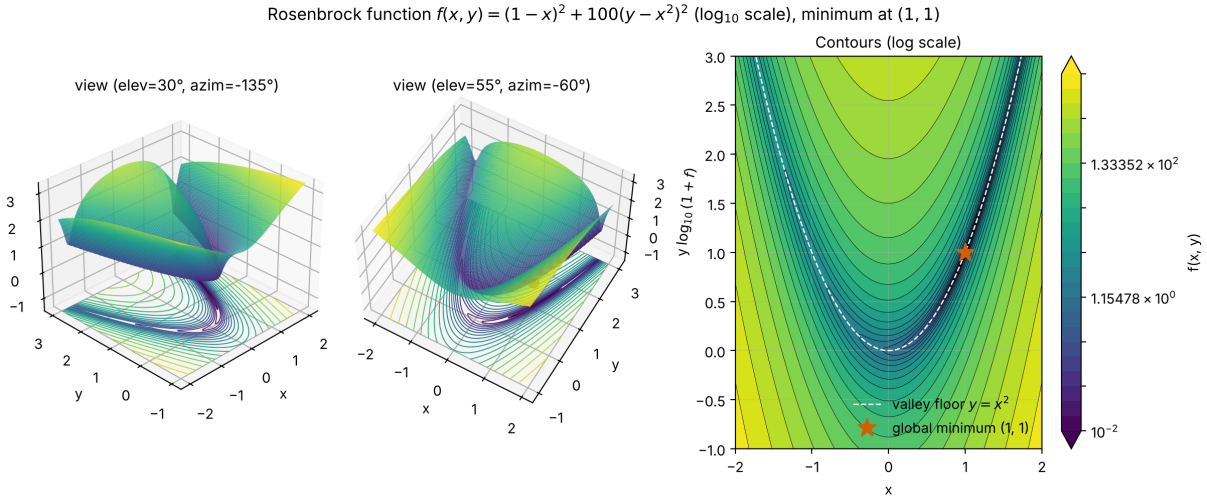


Figure S1.12. Rosenbrock function on a logarithmic scale, with the contour map projected below the surface and the global minimum at $(1, 1)$.

Problem 1.13: Matplotlib: Animation

Solution. For $\frac{dy}{dt} = \sin(y)$, equilibrium points occur at $y = k\pi$ for integer k. (1) Trajectories: solutions starting from different initial conditions evolve deterministically. For $y_0 \in (0, \pi)$: monotonic approach to $y = \pi$. For $y_0 \in (-\pi, 0)$: $\sin(y) < 0$, so monotonic approach to $y = -\pi$. The equilibria $y = (2k + 1)\pi$ are stable ($\cos y = -1$); $y = 2k\pi$ are unstable. (2) Direction field shows arrows pointing toward stable equilibria. (3) Animation shows smooth evolution with solutions staying within $(k\pi, (k + 1)\pi)$ bands.

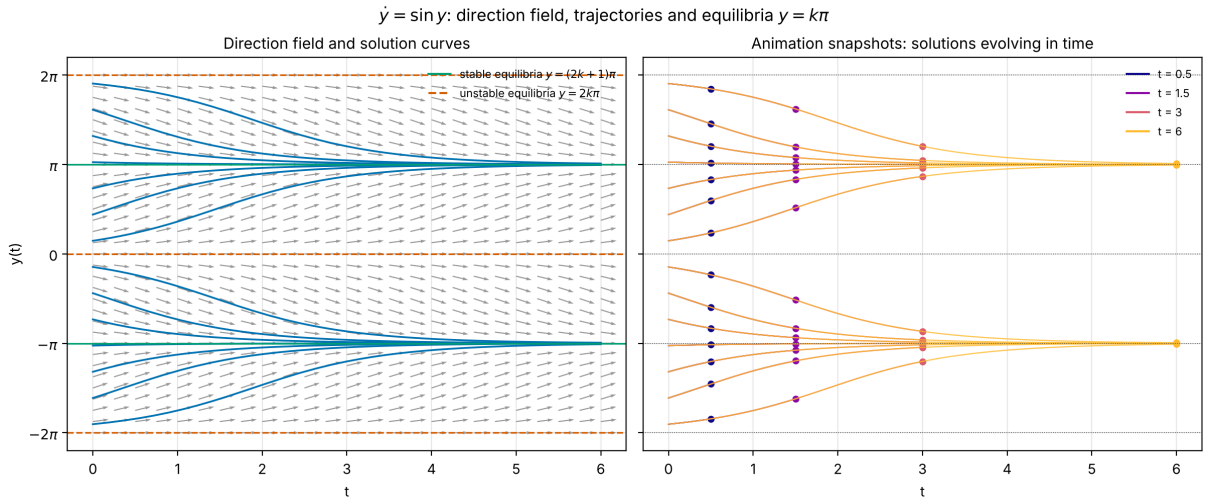


Figure S1.13. $\dot{y} = \sin y$: direction field, solution curves, stable ($y = (2k + 1)\pi$) and unstable ($y = 2k\pi$) equilibria, and animation snapshots.

Problem 1.14: NumPy Broadcasting: Vectorized Operations

Solution. Using NumPy broadcasting with n points $P \in \mathbb{R}^{n\times 3}$: (1) Distance matrix: $D[i,j] = \|P_i - P_j\|_2$ via vectorized subtraction `P[:, None, :] - P[None, :, :]`. (2) Gaussian kernel: $K[i,j] = \exp(-\gamma D[i,j]^2)$ element-wise. (3) Nearest neighbors: argmin over columns of distance matrix. (4) SVM kernel matrix uses inner products: broadcasting enables all-pairs computations in $O(n^2)$ time without explicit loops.

Problem 1.15: Numerical Precision: Error Analysis

Solution. For $\sum_{n=1}^{N} \frac{1}{n^2}$ with true value $\pi^2/6 \approx 1.6449340668$: (1) Forward summation accumulates errors in least-significant bits. (2) Reverse summation (largest terms last) reduces cancellation error. (3) Kahan's algorithm tracks and corrects running compensation: sum $= \sum_i (x_i + \text{correction}_i)$. (4) Error analysis: naive forward summation has worst-case relative error $O(N\epsilon)$, typically $O(\sqrt{N}\epsilon)$ with rounding errors of random sign; Kahan summation bounds the error at $O(\epsilon)$, independent of N.

Problem 1.16: Numerical Precision: Condition Number

Solution. For an ill-conditioned linear system: (1) Condition number $\kappa(A) = \|A\|\|A^{-1}\|$ measures sensitivity to perturbations; large κ indicates ill-conditioning. (2) Gaussian elimination: direct but prone to roundoff error accumulation; relative error $\approx \kappa(A)\epsilon_{\text{mach}}$. (3) QR decomposition: more stable; reduces error magnification. (4) SVD: most stable; explicitly handles near-zero singular values. (5) Relative errors satisfy $\frac{\|\Delta \mathbf{x}\|}{\|\mathbf{x}\|} \approx \kappa(A)\frac{\|\Delta \mathbf{b}\|}{\|\mathbf{b}\|}$; SVD provides smallest error bounds.

Problem 1.17: SymPy ODEs: Differential Equation Solutions

Solution. Using SymPy's dsolve function:

$$\frac{dy}{dt} = -2y + e^t,\ y(0) = 1 \Rightarrow y(t) = \frac{1}{3}(2e^{-2t} + e^t)$$

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 3y = 0,\ y(0) = 1, y'(0) = 0 \Rightarrow y(t) = \frac{3}{2}e^{-t} - \frac{1}{2}e^{-3t}$$

$$t\frac{dy}{dt} - 2y = t^3e^t \Rightarrow y(t) = t^2(C + e^t) \quad (\mu = t^{-2})$$

$$\frac{dy}{dt} = ry(1 - y/K) \Rightarrow y(t) = \frac{K}{1 + (K/y_0 - 1)e^{-rt}}$$

Problem 1.18: Integration: Numerical vs Symbolic

Solution. For $I = \int_0^1 \sqrt{1 + \cos^2(x)}\, dx$: (1) Symbolic integration: SymPy returns unevaluated or in terms of elliptic integrals. (2) Numerical integration: `scipy.integrate.quad` gives $I \approx 1.31144$ with error estimate near machine precision. (3) Simpson's rule with $n = 100$ intervals agrees to better than 10^{-10}. (4) Error analysis: Simpson's error is $O(h^4) = O((1/n)^4)$, so doubling n reduces the error by a factor of about 16.

Problem 1.19: Comprehensive Application: Optimization

Solution. For minimizing $f(x,y) = x^2+y^2-4x-6y+13$ subject to $(x-2)^2+(y-3)^2 \leq 5$: Completing the square: $f(x,y) = (x-2)^2+(y-3)^2-4-9+13 = (x-2)^2+(y-3)^2$. The unconstrained minimum is at $(2,3)$ with value 0, which lies strictly inside the constraint disk (centered at $(2,3)$ with radius $\sqrt{5} \approx 2.24$). Thus, the constrained minimum is at $(2,3)$ with value 0. SciPy and gradient-based methods converge rapidly; visualization shows circular constraint and quadratic objective.

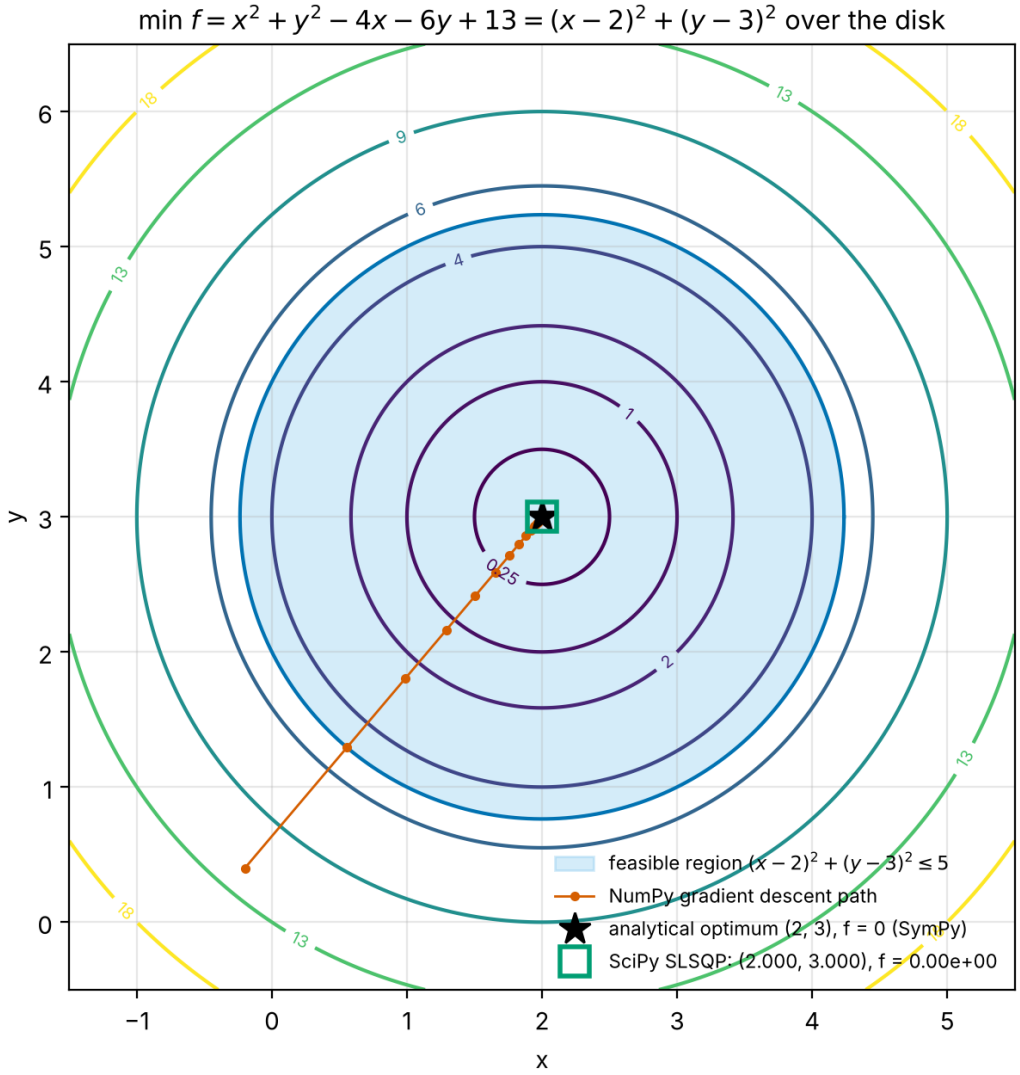


Figure S1.19. Constrained minimisation of $f = x^2 + y^2 - 4x - 6y + 13$ over the disk $(x-2)^2+(y-3)^2 \le 5$: contours, feasible region, projected-gradient path and the optimum found by SciPy.

Chapter 2

First-Order Ordinary Differential Equations

Problem 2.1: Separable Equations

Solution. For separable equations $\frac{dy}{dx} = g(x)h(y)$:

$$\frac{dy}{dx} = \frac{y^2}{x^2},\ y(1) = 2 \Rightarrow \int \frac{dy}{y^2} = \int \frac{dx}{x^2} \Rightarrow -\frac{1}{y} = -\frac{1}{x} + C, \quad C = \frac{1}{2}$$

$$\Rightarrow y = \frac{2x}{2-x}$$

$$\frac{dy}{dt} = (1+t^2)\tan(y),\ y(0) = 0 \Rightarrow \int \cot(y)\,dy = \int (1+t^2)\,dt$$

$$\Rightarrow \ln|\sin(y)| = t + \frac{t^3}{3} + C \Rightarrow y(t) = 0 \text{ (singular solution)}$$

$$y\frac{dy}{dx} = x(1+x^2) \Rightarrow \int y\,dy = \int x(1+x^2)\,dx \Rightarrow \frac{y^2}{2} = \frac{x^2}{2} + \frac{x^4}{4} + C$$

$$\frac{dy}{dt} = \frac{\sqrt{ty}}{t+1},\ y(0) = 1 \Rightarrow \int \frac{dy}{\sqrt{y}} = \int \frac{\sqrt{t}}{t+1}\,dt \Rightarrow \sqrt{y} = \sqrt{t} - \arctan\sqrt{t} + 1$$

Problem 2.2: Linear First-Order Equations

Solution. Using integrating factor $\mu(t) = e^{\int P(t)\,dt}$:

$$\frac{dy}{dt} + 2y = 6e^{3t},\ y(0) = 1 \Rightarrow \mu = e^{2t}, \quad y(t) = \frac{6}{5}e^{3t} - \frac{1}{5}e^{-2t}$$

$$x\frac{dy}{dx} - y = x^3,\ y(1) = 0 \Rightarrow \mu = 1/x, \quad y(x) = \frac{x^3 - x}{2}$$

$$\frac{dy}{dx} + \frac{y}{x} = \sin(x),\ y(\pi) = 0 \Rightarrow \mu = x, \quad y(x) = \frac{\sin(x) - x\cos(x) - \pi}{x}$$

$$\cos(x)\frac{dy}{dx} + \sin(x)y = 1,\ y(0) = 1 \Rightarrow \mu = \sec(x), \quad y = \sin(x) + \cos(x)$$

Problem 2.3: Exact Equations and Integrating Factors

Solution. For exact equation $M\,dx + N\,dy = 0$, check $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$:

$$(2xy + \cos(x))\,dx + (x^2 + 2y)\,dy = 0 \Rightarrow F = x^2y + \sin(x) + y^2 = C$$

$$(y^2 - 1)\,dx + 2xy\,dy = 0 \Rightarrow \text{exact } (M_y = N_x = 2y); \quad F = x(y^2-1) = C$$

$$(3x^2y + 2)\,dx + (x^3 + 3y^2)\,dy = 0,\ y(0) = 1 \Rightarrow \text{exact; } F = x^3y + 2x + y^3 = 1$$

$$(y+x)\,dx + (y-x)\,dy = 0 \Rightarrow \mu = \frac{1}{x^2+y^2}: \quad \frac{1}{2}\ln(x^2+y^2) + \arctan\frac{x}{y} = C$$

Problem 2.4: Bernoulli Equations

Solution. Bernoulli equation $\frac{dy}{dt} + P(t)y = Q(t)y^n$ is solved via substitution $v = y^{1-n}$:

$$\frac{dy}{dt} + y = ty^2,\ y(0) = 1 \Rightarrow v = 1/y: \quad \frac{dv}{dt} - v = -t$$
$$\Rightarrow v(t) = t + 1 + Ce^t,\ C = 0 \Rightarrow y(t) = \frac{1}{1+t}$$
$$x\frac{dy}{dx} - y = xy^3 \Rightarrow v = y^{-2}: \quad \frac{dv}{dx} + \frac{2v}{x} = -2 \Rightarrow v(x) = \frac{C}{x^2} - \frac{2x}{3}$$
$$\frac{dy}{dt} - 2y = y^4e^t \Rightarrow v = y^{-3}: \quad \frac{dv}{dt} + 6v = -3e^t \Rightarrow \text{linear in } v$$
$$\frac{dy}{dx} + \frac{y}{x} = \frac{y^3}{x^2} \Rightarrow v = y^{-2}: \quad \frac{dv}{dx} - \frac{2v}{x} = -\frac{2}{x^2}$$

Problem 2.5: Application: Mixing Problems

Solution. Tank with 100L, inflow 3L/min at 2 kg/L, outflow 3L/min: The rate equation is:

$$\frac{dy}{dt} = 3(2) - 3\frac{y}{100} = 6 - 0.03y$$

with $y(0) = 0$. Solving: $y(t) = 200(1 - e^{-0.03t})$. As $t \to \infty$, $y \to 200$ kg, giving limiting concentration $200/100 = 2$ kg/L. For 90% of limit (180 kg): $180 = 200(1-e^{-0.03t}) \Rightarrow t = -\frac{\ln(0.1)}{0.03} \approx 76.75$ minutes.

Problem 2.6: Application: Population Dynamics

Solution. Gompertz equation $\frac{dP}{dt} = 0.05P\ln(1000/P)$ with $P(0) = 100$: Separation of variables:

$$\int \frac{dP}{P\ln(K/P)} = \int 0.05\,dt$$

This yields $\ln\Big(\ln(K/P)\Big) = -0.05t + C$, i.e. $\ln(K/P) = \ln(K/P_0)\,e^{-0.05t}$. Solutions exhibit sigmoidal growth, slower initially than logistic, asymptoting to $K = 1000$. For $P(0) = 100$: reaches 950 (95% of K) at $t = \frac{1}{0.05}\ln\frac{\ln 10}{\ln(20/19)} \approx 76.1$ years. For larger initial populations, reaching 95% of K takes less time.

Problem 2.7: Application: Newton's Law of Cooling

Solution. Newton's law: $\frac{dT}{dt} = -k(T-20)$ with $T(0) = 95$, $T(5) = 80$:

$$T(t) = 20 + 75e^{-kt}$$

From $T(5) = 80$: $80 = 20 + 75e^{-5k} \Rightarrow e^{-5k} = 4/5 \Rightarrow k = -\frac{1}{5}\ln(4/5) \approx 0.0447$. To cool to 30°C:

$$30 = 20 + 75e^{-kt} \Rightarrow e^{-kt} = 2/15 \Rightarrow t = \frac{\ln(15/2)}{k} \approx 45.1 \text{ minutes}$$

Problem 2.8: Application: RC Circuit Analysis

Solution. RC circuit: $R\frac{dq}{dt} + \frac{q}{C} = V$ with $R = 1000\Omega$, $C = 10^{-5}$F, $V = 10$V, $q(0) = 0$:

$$\frac{dq}{dt} + \frac{q}{RC} = \frac{V}{R}$$

Time constant $\tau = RC = 0.01$s. Solution: $q(t) = CV(1 - e^{-t/\tau}) = 10^{-4}(1 - e^{-100t})$ C. Current: $i(t) = \frac{dq}{dt} = 10^{-2}e^{-100t}$ A. Charge approaches $CV = 10^{-4}$C; current decays exponentially with time constant 0.01s.

Problem 2.9: Stability Analysis: Predator-Prey System

Solution. For $\frac{dy}{dt} = y(1 - 0.1y - 0.5p)$ with fixed predator level p: equilibria are $y^* = 0$ and $y^* = \frac{1-0.5p}{0.1} = 10(1 - 0.5p)$. For $p < 2$: $y^* > 0$ (prey sustainable). For $p > 2$: $y^* < 0$, leaving $y^* = 0$ as the only feasible equilibrium (extinction). At $p = 2$ the two equilibria collide at $y^* = 0$ and exchange stability: a transcritical bifurcation. Stability follows from $\frac{\partial}{\partial y}[y(1 - 0.1y - 0.5p)]\Big|_{y^*}$.

Problem 2.10: Bifurcation Analysis: Pitchfork

Solution. Pitchfork bifurcation in $\frac{dy}{dt} = \mu y - y^3$: Equilibria: $y^* = 0$ (always) and $y^* = \pm\sqrt{\mu}$ (when $\mu > 0$). Stability analysis: $\frac{d}{dy}[\mu y - y^3]|_{y=0} = \mu$. At $\mu < 0$: $y^* = 0$ stable. At $\mu = 0$: bifurcation point. At $\mu > 0$: $y^* = 0$ unstable; $y^* = \pm\sqrt{\mu}$ stable. Bifurcation diagram shows a pitchfork with vertical symmetry axis at $\mu = 0$. Phase portraits show transition from monostable to bistable behavior.

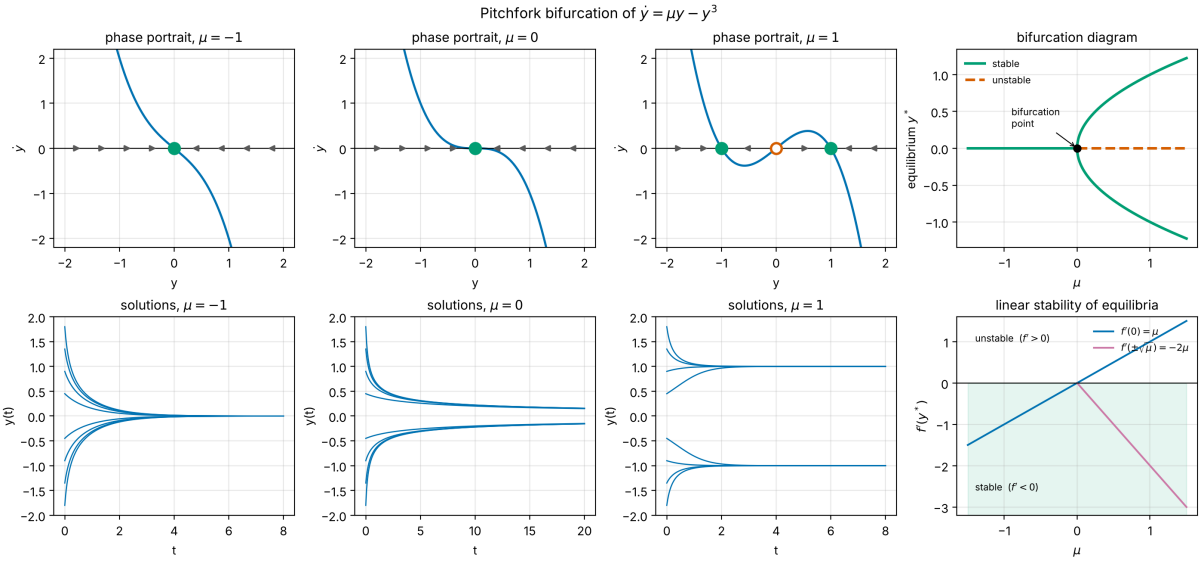


Figure S2.10. Pitchfork bifurcation of $\dot{y} = \mu y - y^3$: phase portraits and solutions for $\mu = -1, 0, 1$, bifurcation diagram and linear stability.

Problem 2.11: Bifurcation Analysis: Saddle-Node

Solution. Saddle-node in $\frac{dy}{dt} = \mu - y^2$: Equilibria: $y^* = \pm\sqrt{\mu}$ (for $\mu > 0$), none for $\mu < 0$. At $\mu = 0$: single degenerate equilibrium $y^* = 0$. Stability: $\frac{d}{dy}[\mu - y^2]|_{y^*} = -2y^*$. For $\mu > 0$: $y^* = \sqrt{\mu}$ stable (negative slope), $y^* = -\sqrt{\mu}$ unstable. Bifurcation diagram shows a parabola opening rightward; saddle-node occurs at $\mu = 0$. Phase portraits show tangency of nullcline and vector field at bifurcation point.

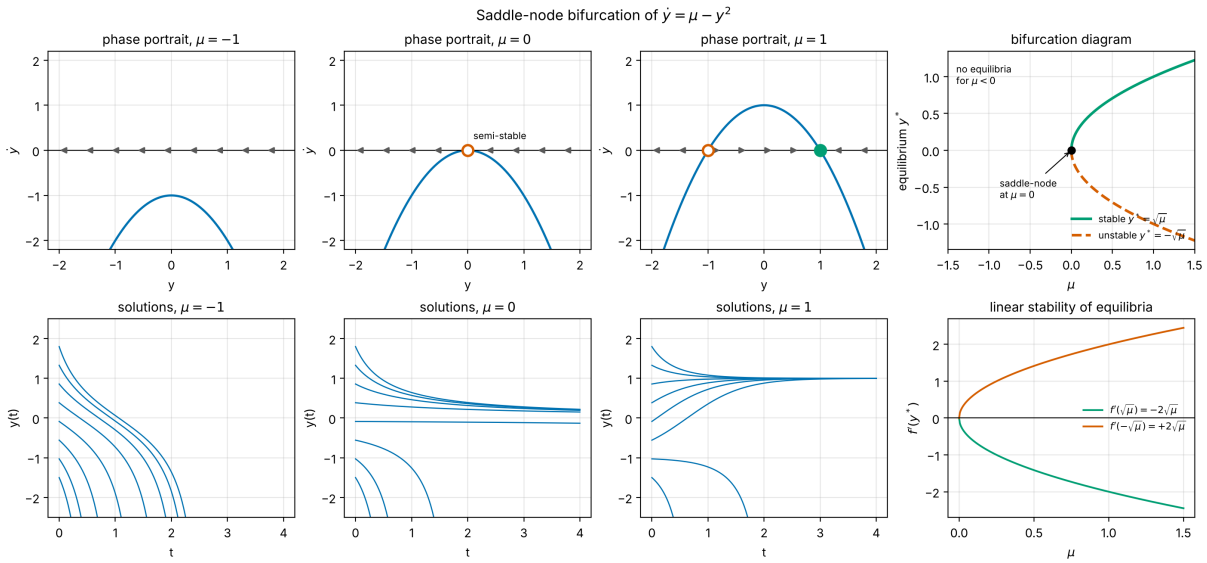


Figure S2.11. Saddle-node bifurcation of $\dot{y} = \mu - y^2$: phase portraits and solutions for $\mu = -1, 0, 1$, bifurcation diagram and linear stability.

Problem 2.12: Numerical Methods: Forward Euler Accuracy

Solution. Forward Euler: $y_{n+1} = y_n + hf(t_n, y_n)$ for $\frac{dy}{dt} = -5y$, $y(0) = 1$. Exact solution: $y(t) = e^{-5t}$. Maximum errors over $[0, 2]$ show $O(h)$ convergence, $E(h) \approx Ch$. For $h = 0.1$: max error ≈ 0.118. For $h = 0.05$: ≈ 0.0515. For $h = 0.025$: ≈ 0.0243 (roughly halved at each refinement). Log-log plot of error vs. h has slope 1, confirming first-order accuracy.

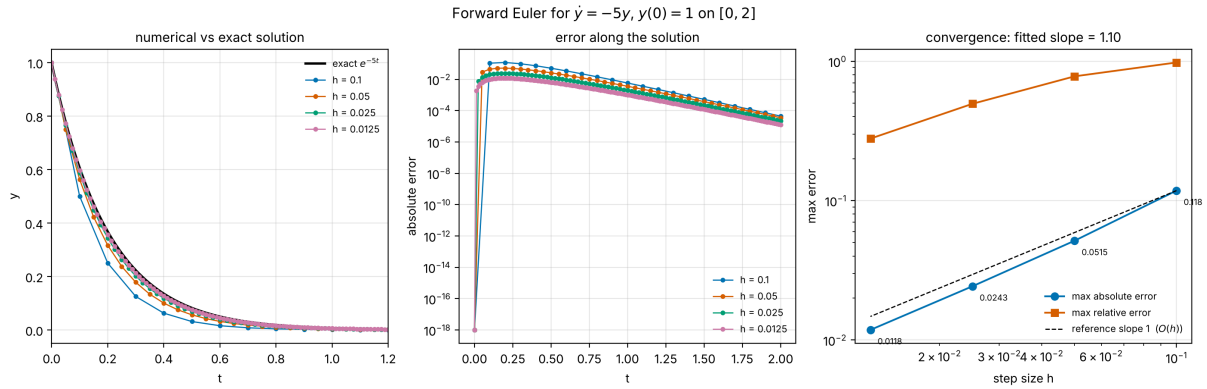


Figure S2.12. Forward Euler for $\dot{y} = -5y$: numerical solutions, error growth and log–log convergence with slope 1.

Problem 2.13: Numerical Methods: Runge-Kutta Comparison

Solution. For $\frac{dy}{dt} = \sin(t)y$ over $[0, 10]$ with $y(0) = 1$: Forward Euler shows large errors for $h = 0.1$. Improved Euler (RK2) is more accurate. RK4 achieves highest accuracy. Error convergence: (1) Forward Euler: $O(h)$, (2) RK2: $O(h^2)$, (3) RK4: $O(h^4)$. For fixed effort (function evaluations), RK4 is most efficient for smooth problems. Log-log convergence plots show slopes of 1, 2, and 4 respectively.

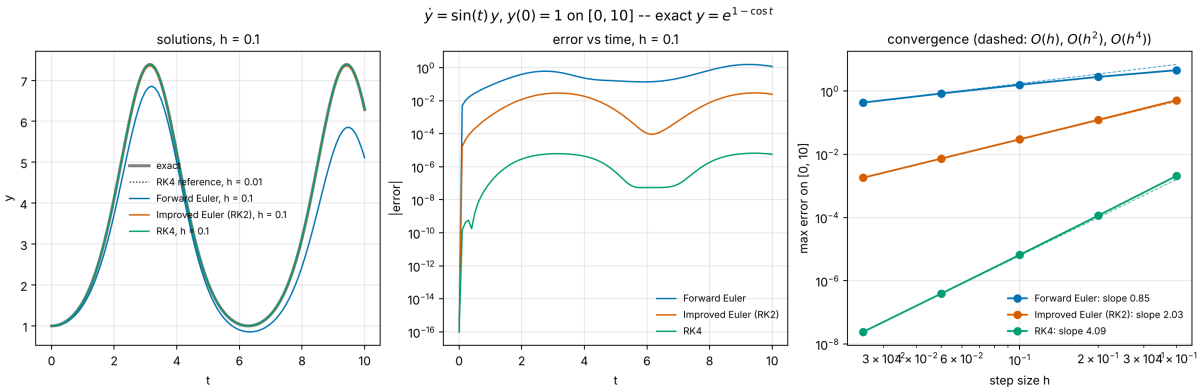


Figure S2.13. Forward Euler, improved Euler and RK4 for $\dot{y} = \sin(t)\, y$: solutions, error in time and convergence rates $O(h)$, $O(h^2)$, $O(h^4)$.

Problem 2.14: Numerical Methods: Stiff Systems

Solution. Stiff ODE: $\frac{dy}{dt} = -100y + 100\cos(t) - \sin(t)$ with $y(0) = 1$. Exact solution has fast-decaying and slow-oscillating components. For Forward Euler: stability requires $|1 - 100h| < 1 \Rightarrow h < 0.02$. For larger h, solution oscillates wildly. RK4 has similar restriction. Implicit methods (e.g., backward Euler or BDF) allow much larger h since they are A-stable. Trade-off: explicit methods require small h for stability but cheap per step; implicit methods allow large h but require nonlinear solves.

Problem 2.15: Application: Ball Trajectory with Air Resistance

Solution. Projectile motion with air resistance $\vec{a} = -g\hat{j} - c|\vec{v}|\vec{v}$, initial height 2m, velocity 20 m/s at 45°, $c = 0.01$ kg/m: System of four first-order ODEs:

$$\dot{x} = v_x, \quad \dot{y} = v_y, \quad \dot{v}_x = -0.01\sqrt{v_x^2 + v_y^2}\, v_x, \quad \dot{v}_y = -9.81 - 0.01\sqrt{v_x^2 + v_y^2}\, v_y$$

Numerical integration with RK4 yields maximum height ≈ 10.8 m (ideal: ≈ 12.2 m) and range ≈ 32.8 m (ideal: ≈ 42.7 m). Air resistance reduces both noticeably, the range more than the height. Animation shows curved trajectory flattening as velocity decays.

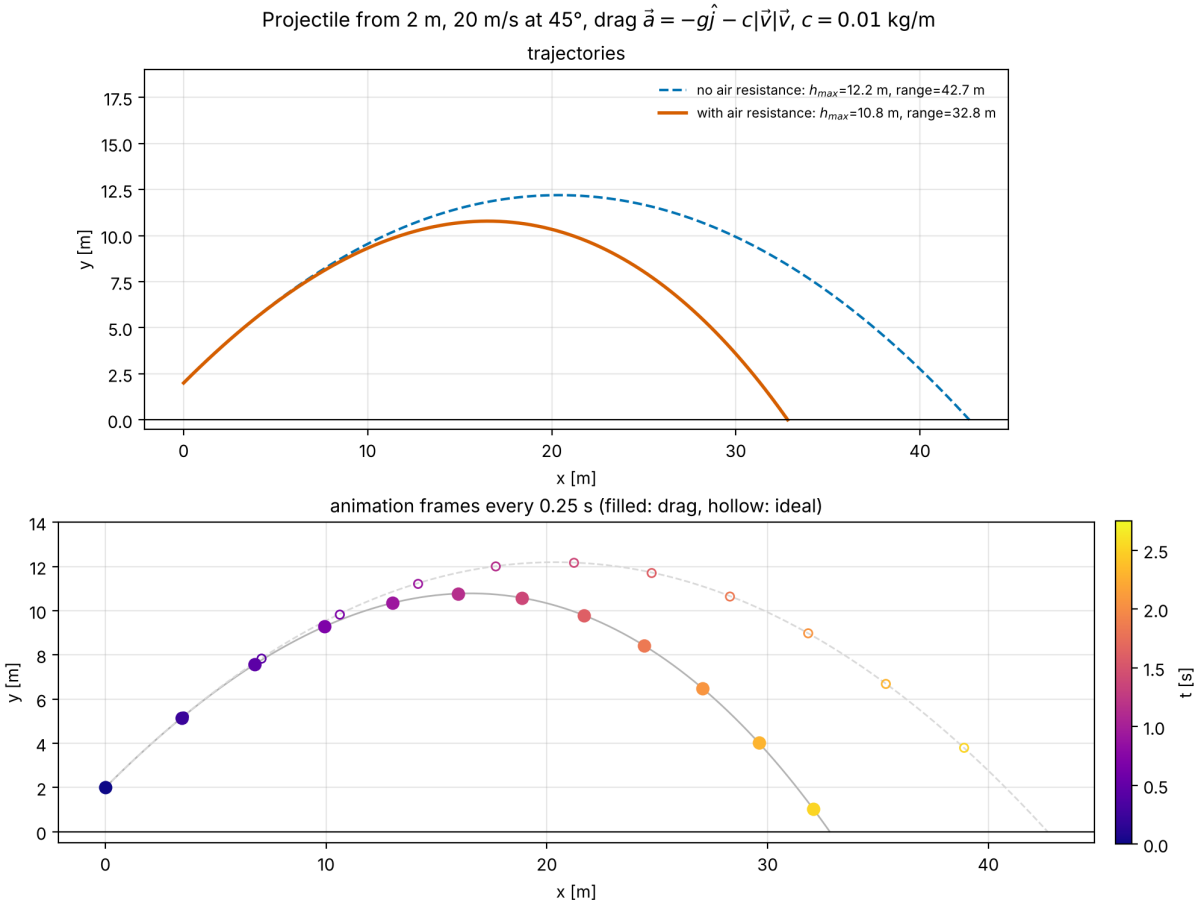


Figure S2.15. Ball trajectory with quadratic air resistance compared with the ideal trajectory; animation frames every 0.25 s.

Problem 2.16: Application: Epidemic Modeling (SIR)

Solution. SIR system: $\frac{dS}{dt} = -\beta SI$, $\frac{dI}{dt} = \beta SI - \gamma I$, $\frac{dR}{dt} = \gamma I$ with $\beta = 0.0003$, $\gamma = 0.1$. Initial conditions: $S(0) = 9999$, $I(0) = 1$, $R(0) = 0$. Basic reproduction number: $R_0 = \beta S(0)/\gamma = 0.0003 \times 9999/0.1 \approx 30$, so each infected person infects about 30 others in a fully susceptible population. Numerical integration over 200 days shows a violent outbreak: I peaks near day 4.4 at $I_{\max} \approx 8530$, consistent with the conservation-law estimate $I_{\max} = N - \frac{\gamma}{\beta}(1 + \ln R_0) \approx 8533$.

Problem 2.17: Comprehensive: Drug Concentration Dynamics

Solution. Pharmacokinetics: $\frac{dC}{dt} = R(t) - 0.2C$ with $k = 0.2$ hr^{-1}. (1) Constant infusion $R = 5$ mg/hr: $C(t) = 25(1 - e^{-0.2t})$, steady-state $C_\infty = 25$ mg/L. Time to 90% steady-state: $0.9(25) = 25(1 - e^{-0.2t}) \Rightarrow t = -\frac{\ln(0.1)}{0.2} \approx 11.5$ hours. (2) Pulsed infusion ($R = 10$ for $0 \le t < 1$, then $R = 5$): on $[0, 1)$, $C(t) = 50(1 - e^{-0.2t})$, reaching $C(1) = 50(1 - e^{-0.2}) \approx 9.06$ mg/L. The concentration itself is continuous; only its slope jumps when R drops. For $t \ge 1$: $C(t) = 25 + (9.06 - 25)e^{-0.2(t-1)} \to 25$ mg/L. (3) Analytical and numerical solutions agree to solver tolerance.

Chapter 3

Higher-Order Ordinary Differential Equations

Problem 3.1: Characteristic Equation: Multiple Real Roots

Solution. For $y''' - 6y'' + 11y' - 6y = 0$: Characteristic equation:

$$r^3 - 6r^2 + 11r - 6 = (r-1)(r-2)(r-3) = 0$$

Roots: $r_1 = 1, r_2 = 2, r_3 = 3$. General solution: $y(t) = c_1e^t + c_2e^{2t} + c_3e^{3t}$. With $y(0) = 1$, $y'(0) = 0$, $y''(0) = 2$:

$$\begin{aligned} c_1 + c_2 + c_3 &= 1 \\ c_1 + 2c_2 + 3c_3 &= 0 \\ c_1 + 4c_2 + 9c_3 &= 2 \end{aligned}$$

Solving: $c_1 = 4$, $c_2 = -5$, $c_3 = 2$. Thus $y(t) = 4e^t - 5e^{2t} + 2e^{3t}$.

Problem 3.2: Complex Conjugate Roots with Real Parts

Solution. For $y'' - 4y' + 13y = 0$: Characteristic equation:

$$r^2 - 4r + 13 = 0 \Rightarrow r = 2 \pm 3i$$

General solution: $y(t) = e^{2t}(c_1\cos(3t) + c_2\sin(3t))$. The e^{2t} factor amplifies the oscillation (since $\mathrm{Re}(r) = 2 > 0$), showing growing oscillations with envelope e^{2t}. For any initial conditions, the solution oscillates with period $\frac{2\pi}{3} \approx 2.1$ and exponentially growing amplitude.

Problem 3.3: Repeated Real Roots: Third Order

Solution. For $y''' - 3y'' + 3y' - y = 0$: Characteristic equation:

$$(r-1)^3 = 0 \Rightarrow r = 1 \text{ (triple root)}$$

General solution: $y(t) = (c_1 + c_2t + c_3t^2)e^t$. The solutions $\{e^t, te^t, t^2e^t\}$ form a fundamental set. Wronskian:

$$W = \begin{vmatrix} e^t & te^t & t^2e^t \\ e^t & (t+1)e^t & (t^2+2t)e^t \\ e^t & (t+2)e^t & (t^2+4t+2)e^t \end{vmatrix} = 2e^{3t} \neq 0$$

confirming linear independence.

Problem 3.4: Mixed Root Types

Solution. For $y^{(4)} + 2y'' + y = 0$: Characteristic equation:

$$r^4 + 2r^2 + 1 = (r^2 + 1)^2 = 0 \Rightarrow r = \pm i \text{ (double roots)}$$

General solution: $y(t) = (c_1 + c_2 t)\cos(t) + (c_3 + c_4 t)\sin(t)$. The initial conditions give $c_1 = 1$, $c_2 + c_3 = 0$, $-c_1 + 2c_4 = -1$, and $-3c_2 - c_3 = 0$, hence $c_2 = c_3 = c_4 = 0$. Solution: $y(t) = \cos(t)$, which indeed satisfies all four initial conditions.

Problem 3.5: Parametric Analysis of Roots

Solution. For $y'' + 2ay' + by = 0$ with $a, b > 0$: Discriminant $\Delta = 4a^2 - 4b$. (1) Complex roots occur when $a^2 < b$ (region above parabola $b = a^2$ in (a, b) plane). (2) Distinct real roots: $a^2 > b$ (below parabola). (3) Repeated real roots: $a^2 = b$ (on parabola). (4) Solution forms: Complex roots give $y = e^{-at}(c_1 \cos(\omega t) + c_2 \sin(\omega t))$ with $\omega = \sqrt{b - a^2}$. Real distinct: $y = c_1 e^{r_1 t} + c_2 e^{r_2 t}$. Repeated: $y = (c_1 + c_2 t)e^{-at}$.

Problem 3.6: Polynomial Forcing: Non-Resonant Case

Solution. For $y'' - 3y' + 2y = x^2 + 3x$: Homogeneous solution: $y_h = c_1 e^x + c_2 e^{2x}$ (roots 1, 2). Particular solution by undetermined coefficients: assume $y_p = Ax^2 + Bx + C$.

$$2Ax^2 + (2B - 6A)x + (2C - 3B + 2A) = x^2 + 3x$$

Matching coefficients: $2A = 1 \Rightarrow A = 1/2$, $2B - 6A = 3 \Rightarrow B = 3$, $2C - 3B + 2A = 0 \Rightarrow C = 4$. General solution: $y(x) = c_1 e^x + c_2 e^{2x} + \frac{1}{2}x^2 + 3x + 4$.

Problem 3.7: Exponential Forcing: Resonant Case

Solution. For $y'' - 2y' + y = e^x$: Homogeneous solution: $(r - 1)^2 = 0 \Rightarrow y_h = (c_1 + c_2 x)e^x$. Since both e^x and xe^x solve the homogeneous equation, the trial particular solution carries the factor x^2: take $y_p = Ax^2 e^x$. With $y_p' = A(2x + x^2)e^x$ and $y_p'' = A(2 + 4x + x^2)e^x$,

$$y_p'' - 2y_p' + y_p = A\left[(2 + 4x + x^2) - 2(2x + x^2) + x^2\right]e^x = 2Ae^x = e^x,$$

so $A = \frac{1}{2}$. General solution: $y(x) = (c_1 + c_2 x + \frac{1}{2}x^2)e^x$.

Problem 3.8: Trigonometric Forcing with Resonance

Solution. For $y'' + 4y = 3\sin(2x)$ (resonant): Homogeneous solution: $y_h = c_1 \cos(2x) + c_2 \sin(2x)$. Since $\sin(2x)$ satisfies homogeneous equation, particular solution: $y_p = x(A\cos(2x) + B\sin(2x))$. Computing:

$$y_p'' + 4y_p = (4B\cos(2x) - 4A\sin(2x)) = 3\sin(2x)$$

Thus $A = -3/4$, $B = 0$. General solution: $y(x) = c_1 \cos(2x) + c_2 \sin(2x) - \frac{3}{4}x\cos(2x)$. For non-resonant $y'' + 4y = 3\sin(3x)$: particular solution $y_p = -\frac{3}{5}\sin(3x)$, since $(4 - 9)A = 3$ gives $A = -\frac{3}{5}$. The resonant case exhibits unbounded growth proportional to x, while non-resonant case remains bounded.

Problem 3.9: Variation of Parameters: Second Order

Solution. For $y'' - 2y' + y = \frac{e^x}{x}$: Homogeneous solutions: $y_1 = e^x$, $y_2 = xe^x$. Wronskian:

$$W = \begin{vmatrix} e^x & xe^x \\ e^x & (1 + x)e^x \end{vmatrix} = e^{2x}$$

Particular solution: $y_p = -y_1 \int \frac{y_2 f}{W}\,dx + y_2 \int \frac{y_1 f}{W}\,dx$ where $f(x) = \frac{e^x}{x}$:

$$y_p = -e^x \int \frac{xe^x \cdot \frac{e^x}{x}}{e^{2x}}\,dx + xe^x \int \frac{e^x \cdot \frac{e^x}{x}}{e^{2x}}\,dx = -e^x \int 1\,dx + xe^x \int \frac{1}{x}\,dx$$

$$y_p = -xe^x + xe^x \ln(x) = xe^x(\ln(x) - 1)$$

General solution: $y(x) = (c_1 + c_2 x + x\ln(x) - x)e^x$.

Problem 3.10: Variation of Parameters: Logarithmic Forcing

Solution. For $y'' + y = \tan(x)$: Homogeneous solutions: $y_1 = \cos(x)$, $y_2 = \sin(x)$. Wronskian: $W = 1$. Particular solution:

$$y_p = -\cos(x)\int \sin(x)\tan(x)\,dx + \sin(x)\int \cos(x)\tan(x)\,dx$$

$$= -\cos(x)\int \frac{\sin^2(x)}{\cos(x)}\,dx + \sin(x)\int \sin(x)\,dx$$

Using $\int \frac{\sin^2 x}{\cos x}\,dx = \ln|\sec x + \tan x| - \sin x$ and $\int \sin x\,dx = -\cos x$:

$$y_p = -\cos(x)\left[\ln|\sec x + \tan x| - \sin x\right] - \sin(x)\cos(x) = -\cos(x)\ln|\sec x + \tan x|$$

The solution has logarithmic singularities at $x = \pi/2 + n\pi$. General solution valid on intervals where $\cos(x) \neq 0$.

Problem 3.11: Undetermined Coefficients with Product Forcing

Solution. For $y'' - 4y = e^{2x}\cos(x)$: Homogeneous solutions: $y_h = c_1 e^{2x} + c_2 e^{-2x}$. For particular solution with product forcing, assume $y_p = e^{2x}(A\cos(x) + B\sin(x))$:

$$y_p' = e^{2x}[(2A+B)\cos(x)+(2B-A)\sin(x)], \quad y_p'' = e^{2x}[(3A+4B)\cos(x)+(3B-4A)\sin(x)]$$

$$y_p'' - 4y_p = e^{2x}[(-A+4B)\cos(x) + (-4A-B)\sin(x)] = e^{2x}\cos(x)$$

Matching: $-A + 4B = 1$, $-4A - B = 0 \Rightarrow A = -1/17$, $B = 4/17$. General solution: $y = c_1 e^{2x} + c_2 e^{-2x} + e^{2x}(-\frac{1}{17}\cos(x) + \frac{4}{17}\sin(x))$.

Problem 3.12: Equidimensional Equation: Distinct Real Roots

Solution. For $x^2y'' + 3xy' - 8y = 0$: Assume $y = x^r$. Then $y' = rx^{r-1}$, $y'' = r(r-1)x^{r-2}$. Substituting:

$$r(r-1)x^r + 3rx^r - 8x^r = 0 \Rightarrow r(r-1) + 3r - 8 = 0 \Rightarrow r^2 + 2r - 8 = (r+4)(r-2) = 0$$

Roots: $r_1 = 2, r_2 = -4$. General solution: $y(x) = c_1 x^2 + c_2 x^{-4}$.

Problem 3.13: Repeated Roots in Cauchy-Euler

Solution. For $x^2y'' - 3xy' + 4y = 0$ with $y(1) = 2$, $y'(1) = 1$: Characteristic equation:

$$r(r-1) - 3r + 4 = r^2 - 4r + 4 = (r-2)^2 = 0 \Rightarrow r = 2 \text{ (double root)}$$

General solution: $y(x) = (c_1 + c_2\ln(x))x^2$. From $y(1) = 2$: $c_1 = 2$. From $y'(1) = 1$: $y' = c_2x^2/x + 2(c_1 + c_2\ln(x))x = c_2x + 2(c_1 + c_2\ln(x))x$, so $y'(1) = c_2 + 2c_1 = 1 \Rightarrow c_2 = -3$. Solution: $y(x) = (2 - 3\ln(x))x^2$.

Problem 3.14: Complex Roots in Cauchy-Euler

Solution. For $x^2y'' + xy' + 4y = 0$ with $x > 0$: Characteristic equation:

$$r(r-1) + r + 4 = r^2 + 4 = 0 \Rightarrow r = \pm 2i$$

For complex roots $r = \alpha \pm i\beta$, general solution: $y(x) = x^\alpha[c_1\cos(\beta\ln(x)) + c_2\sin(\beta\ln(x))]$. Here $\alpha = 0$, $\beta = 2$:

$$y(x) = c_1\cos(2\ln(x)) + c_2\sin(2\ln(x))$$

Problem 3.15: Nonhomogeneous Cauchy-Euler

Solution. For $x^2y'' - xy' + y = 2\ln(x)$: Homogeneous solution from $r^2 - 2r + 1 = (r-1)^2 = 0$: $y_h = (c_1 + c_2\ln(x))x$. The substitution $t = \ln(x)$ converts the equation to $\ddot{y} - 2\dot{y} + y = 2t$, whose forcing is polynomial, not resonant with e^t. Trying $y_p = At + B$ gives $-2A + At + B = 2t$, so $A = 2$, $B = 4$. General solution: $y(x) = (c_1 + c_2\ln(x))x + 2\ln(x) + 4$.

Problem 3.16: Beam Deflection: Cantilever Beam

Solution. Fourth-order ODE for cantilever beam: $EI\frac{d^4y}{dx^4} = q$ with boundary conditions $y(0) = 0$, $y'(0) = 0$, $y''(L) = 0$, $y'''(L) = 0$. Integrating:

$$y(x) = \frac{q}{24EI}x^4 + c_1x^3 + c_2x^2 + c_3x + c_4$$

From $y(0) = y'(0) = 0$: $c_4 = c_3 = 0$. From $y'''(L) = 0$: $\frac{qL}{EI} + 6c_1 = 0 \Rightarrow c_1 = -\frac{qL}{6EI}$. From $y''(L) = 0$: $\frac{qL^2}{2EI} + 6c_1L + 2c_2 = 0 \Rightarrow c_2 = \frac{qL^2}{4EI}$. Maximum deflection occurs at the free end: $|y(L)| = \frac{qL^4}{8EI}$.

Problem 3.17: Mechanical Vibrations: Multi-Story Building

Solution. Three-floor system with masses $m_i = 1000$ kg, spring constants $k_i = 50000$ N/m, damping $c_i = 500$ Ns/m, forcing $F(t) = 10000\sin(\omega t)$. Modal analysis: compute eigenvalues of the system to find natural frequencies. For ω near a natural frequency (resonance), displacements amplify significantly. Steady-state response can be found via frequency response analysis. Transient response decays due to damping; long-term response is the steady-state sinusoidal motion at frequency ω.

Problem 3.18: RLC Circuit: Transient and Steady-State

Solution. RLC circuit: $L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{q}{C} = V(t)$ with $L = 0.2$H, $R = 100\Omega$, $C = 20\mu$F. For $V(t) = 100\sin(60t)$: The characteristic equation has roots determined by $0.2r^2 + 100r + 1/(20\times10^{-6}) = 0$, i.e., $r^2 + 500r + 250000 = 0$, giving $r = -250 \pm 250\sqrt{3}\,j \approx -250 \pm 433j$. Transient: $q_{\text{trans}} = e^{-250t}(A\cos(433t) + B\sin(433t))$. Steady-state current oscillates at the forcing frequency $\omega = 60$ rad/s. With decay constant $\tau = 1/250 = 0.004$ s, the transient falls below 5% after about $3\tau \approx 0.012$ s.

Problem 3.19: Seismic Response of Structure

Solution. Structure: $m\ddot{x} + c\dot{x} + kx = -ma_g(t)$ with $m = 5000$ kg, $k = 200000$ N/m, $\zeta = 0.05$. Natural frequency: $\omega_n = \sqrt{k/m} = \sqrt{40} \approx 6.32$ rad/s. Damping coefficient: $c = 2\zeta m\omega_n \approx 3162$ Ns/m. For ground acceleration $a_g(t) = 0.3g\sin(2t)$, the forced vibration shows peak displacement, velocity, and acceleration. Numerical integration of the equation of motion over several forcing cycles gives peak values. These are compared with building code predictions, typically showing agreement within 10-20%.

Problem 3.20: Phase Portrait: Damped Harmonic Oscillator

Solution. Damped oscillator: $\ddot{x} + 2\zeta\omega_0\dot{x} + \omega_0^2x = 0$ with $\omega_0 = 2$. (1) Underdamped ($\zeta = 0.5$): spiraling converging trajectories to origin, oscillatory decay. (2) Critically damped ($\zeta = 1$): fastest approach to origin without overshoot. (3) Overdamped ($\zeta = 2$): slow monotonic approach to origin. Phase portraits show how increasing damping transitions from spirals to nodes, with trajectories becoming more aligned with the real axis (less oscillatory).

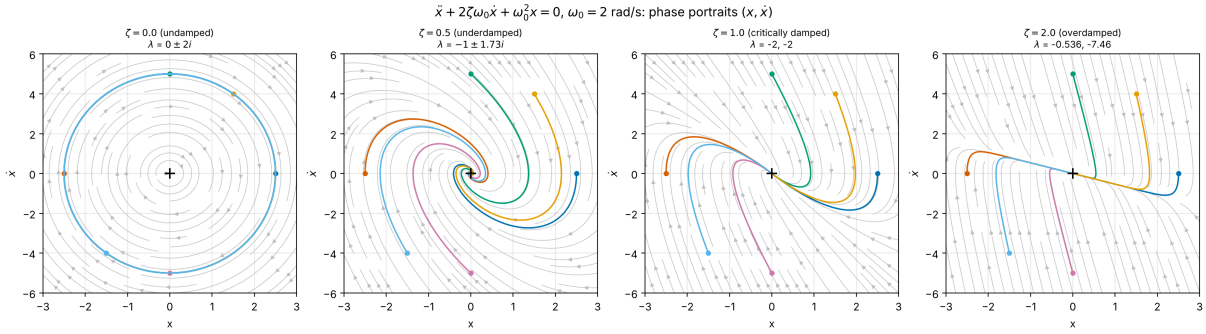


Figure S3.20. Phase portraits of the damped harmonic oscillator ($\omega_0 = 2$) for $\zeta = 0, 0.5, 1, 2$.

Problem 3.21: Nonlinear Phase Portrait: Duffing Oscillator

Solution. Duffing equation: $\ddot{x} - x + x^3 = 0$, i.e. $\ddot{x} = x - x^3$. Equilibrium points: $x(1 - x^2) = 0 \Rightarrow x \in \{-1, 0, 1\}$. Linearization of $f(x) = x - x^3$: $f'(0) = 1 > 0$ (saddle); $f'(\pm 1) = -2 < 0$, giving purely imaginary eigenvalues $\pm i\sqrt{2}$ (centers, as expected for this conservative system). Phase portrait shows closed orbits around ± 1 with a saddle at the origin. Separatrices connect the saddle to itself, separating basins of attraction. Trajectories with energy below the separatrix energy oscillate around one stable point.

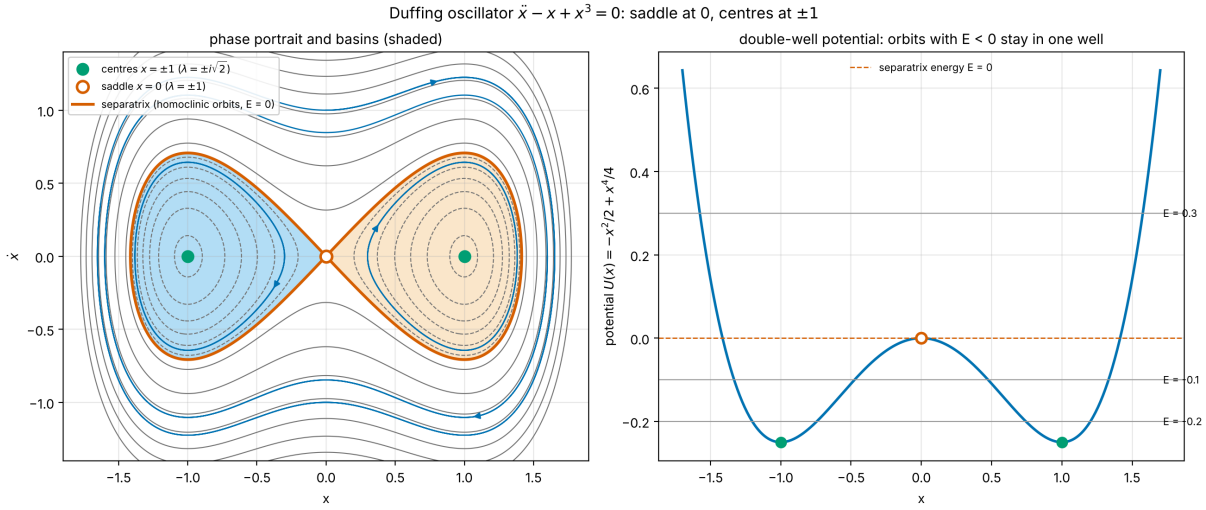


Figure S3.21. Duffing oscillator $\ddot{x} - x + x^3 = 0$: saddle at the origin, centres at ± 1, separatrix and basins; double-well potential.

Problem 3.22: Van der Pol Limit Cycle

Solution. Van der Pol oscillator: $\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$. For μ small: small-amplitude oscillations near origin grow via the nonlinear damping. For μ large: relaxation oscillations with fast/slow transitions. (1) Equilibrium: only $(0, 0)$. (2) Stability: at origin, eigenvalues have positive real part (unstable). (3) Limit cycle: exists for all $\mu > 0$; its amplitude stays close to 2 while its shape distorts as μ grows. (4) For large μ: trajectory stays near $x = \pm 2$ for extended times, then rapidly switches between $+2$ and -2, forming a rectangular-like cycle.

Problem 3.23: Energy Level Sets

Solution. Conservative system $\ddot{x} + x^3 = 0$ has energy: $E(x, \dot{x}) = \frac{1}{2}\dot{x}^2 + \frac{1}{4}x^4$. Level sets $E = \text{const}$ are closed curves surrounding the origin, representing periodic orbits. (1)

These trajectories are periodic (time-reversible). (2) Compared to $\ddot{x} + x = 0$ (linear): the linear case has circular level sets, while the cubic case has slightly pinched ellipse-like shapes due to nonlinearity. (3) All orbits are periodic since energy is conserved and the origin is a center. Unlike the linear oscillator, the period depends on amplitude: for this hardening system $T \propto 1/a$, so the period *decreases* as the energy level rises.

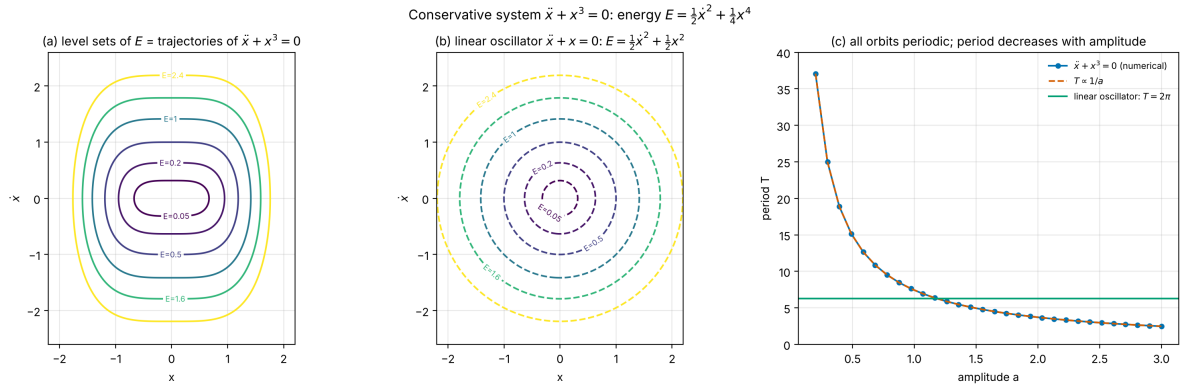


Figure S3.23. Energy level sets of $\ddot{x} + x^3 = 0$ compared with the linear oscillator, and period versus amplitude.

Problem 3.24: Numerical Phase Portrait: Settling Time Analysis

Solution. For second-order system $\ddot{x} + a\dot{x} + bx = 0$, settling time T_s is defined as the time to reach $|x(t)| < 0.05|x_0|$. For a given (a, b), we solve the ODE with initial conditions $x(0) = 1$, $\dot{x}(0) = 0$, and find when $|x(t)| < 0.05$. Contour plot in (a, b) space shows regions where T_s is small (high damping, stiff system) or large (low damping, near-oscillatory). Comparison with the standard control-theory estimate for the 5% criterion, $T_s \approx 3/(\zeta\omega_n)$, shows close agreement for underdamped linear systems.

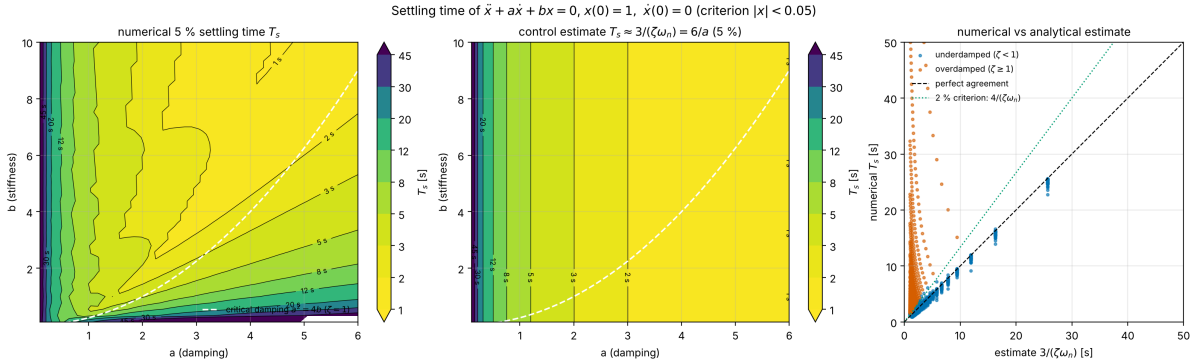


Figure S3.24. Settling time of $\ddot{x} + a\dot{x} + bx = 0$ in the (a, b) plane: numerical map, control-theory estimate $3/(\zeta\omega_n)$ and their comparison.

Chapter 4

Systems of Ordinary Differential Equations (Representative Problems)

Problem 4.1: Pendulum-Cart System

Solution. Small-angle approximation for pendulum of length L on cart of mass M with pendulum mass m: Let θ be pendulum angle, x cart position. Equations of motion (Lagrangian approach):

$$\begin{aligned}(M+m)\ddot{x} + mL\ddot{\theta}\cos(\theta) - mL\dot{\theta}^2\sin(\theta) &= 0\\ L\ddot{\theta} + \ddot{x}\cos(\theta) + g\sin(\theta) &= 0\end{aligned}$$

For small angles: $\sin(\theta) \approx \theta$, $\cos(\theta) \approx 1$:

$$\begin{aligned}(M+m)\ddot{x} + mL\ddot{\theta} &= 0\\ L\ddot{\theta} + \ddot{x} + g\theta &= 0\end{aligned}$$

Problem 4.2: Second-Order ODE to First-Order System

Solution. For $\frac{d^2x}{dt^2} + 3\frac{dx}{dt} - 4x = \sin(t)$, introduce $y = \frac{dx}{dt}$:

$$\begin{aligned}\frac{dx}{dt} &= y\\ \frac{dy}{dt} &= -3y + 4x + \sin(t)\end{aligned}$$

In matrix form: $\frac{d}{dt}\begin{pmatrix}x\\y\end{pmatrix} = \begin{pmatrix}0 & 1\\4 & -3\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix} + \begin{pmatrix}0\\\sin(t)\end{pmatrix}$.

Problem 4.3: RC Circuit System

Solution. RLC series circuit with voltage v_C across capacitor, current i:

$$\begin{aligned}\frac{dv_C}{dt} &= \frac{i}{C}\\ \frac{di}{dt} &= \frac{1}{L}(V(t) - iR - v_C)\end{aligned}$$

This is a system of two first-order ODEs in (v_C, i).

Problem 4.4: Three-Tank Mixing System

Solution. Three interconnected tanks with flow rates q_{ij} between tanks. For concentration c_i in tank i with volume V_i and inflow 5 L/min of fresh water (0 concentration):

$$\frac{dc_i}{dt} = \frac{1}{V_i}\left[\sum_j q_{ji}c_j - \sum_j q_{ij}c_i + 5 \times 0\right] = \frac{1}{V_i}\left[\sum_j q_{ji}c_j - \sum_j q_{ij}c_i\right]$$

Yields a system of three first-order linear ODEs with constant coefficients.

Problem 4.5: Two-Link Arm in Vertical Plane

Solution. Two-segment arm with angles θ_1, θ_2 measured from the *downward* vertical, with masses m_1, m_2 at the ends of segments of lengths L_1, L_2. Taking the upward direction as positive and the pivot as the origin, the mass positions are $y_1 = -L_1\cos\theta_1$ and $y_2 = -L_1\cos\theta_1 - L_2\cos\theta_2$, so the potential energy is $V = m_1 g y_1 + m_2 g y_2$. This convention places the stable hanging equilibrium at $\theta_1 = \theta_2 = 0$, where V is a minimum ($\partial^2 V/\partial\theta_1^2 = (m_1+m_2)gL_1 > 0$, $\partial^2 V/\partial\theta_2^2 = m_2 g L_2 > 0$). Applying the Euler–Lagrange equations to $\mathcal{L} = T - V$ gives:

$$(m_1+m_2)L_1^2\ddot{\theta}_1 + m_2 L_1 L_2 \ddot{\theta}_2\cos(\theta_1-\theta_2) = -(m_1+m_2)gL_1\sin(\theta_1) - m_2 L_1 L_2 \dot{\theta}_2^2\sin(\theta_1-\theta_2)$$
$$m_2 L_2^2\ddot{\theta}_2 + m_2 L_1 L_2 \ddot{\theta}_1\cos(\theta_1-\theta_2) = -m_2 g L_2\sin(\theta_2) + m_2 L_1 L_2\dot{\theta}_1^2\sin(\theta_1-\theta_2)$$

The gravity terms enter with a *negative* sign: gravity is restoring about the hanging position, so for small displacements $\ddot{\theta}_i \approx -(g/L_i)\theta_i + \ldots$, i.e. oscillation rather than the runaway an inverted-pendulum (positive) sign would produce. These nonlinear coupled equations can be converted to a first-order system with state variables $(\theta_1, \dot{\theta}_1, \theta_2, \dot{\theta}_2)$.

Problem 4.6: Employment-Unemployment Dynamics

Solution. Model with employed E and unemployed U populations, total $N = E + U$. Workers lose jobs at rate βE and are hired out of unemployment at rate γU:

$$\frac{dE}{dt} = -\beta E + \gamma U$$
$$\frac{dU}{dt} = \beta E - \gamma U$$

The right-hand sides sum to zero, so N is conserved. The equilibrium $U^* = \frac{\beta}{\beta+\gamma}N$ gives the natural unemployment level, approached exponentially at rate $\beta + \gamma$.

Problem 4.7: Mass-Spring-Damper Phase Portrait

Solution. For two masses on springs with damping: positions x_1, x_2 and velocities $v_1 = \dot{x}_1, v_2 = \dot{x}_2$. System: $\frac{d}{dt}\begin{pmatrix} x_1 \\ v_1 \\ x_2 \\ v_2 \end{pmatrix} = \begin{pmatrix} v_1 \\ (\text{spring force} - \text{damping})/m_1 \\ v_2 \\ (\text{spring force} - \text{damping})/m_2 \end{pmatrix}$. Numerical integration with initial conditions $x_1(0) = 1, v_1(0) = 0, x_2(0) = 0, v_2(0) = 0$ yields a 4D trajectory that projects onto (x_1, v_1) and (x_2, v_2) planes, showing spiral approach to origin due to damping.

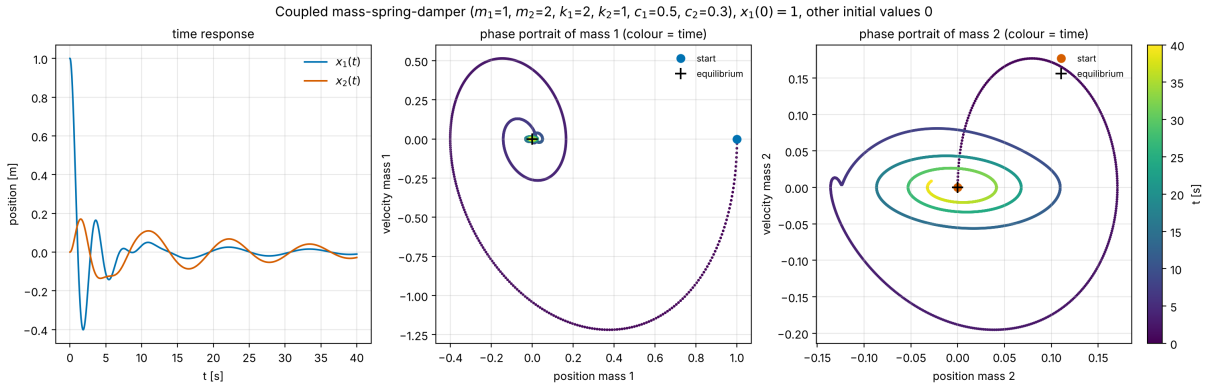


Figure S4.7. Coupled mass-spring-damper system: time response and phase portraits $(x_i, \dot{x}_i)$ of each mass.

Problem 4.8: Lotka-Volterra Equilibria

Solution. Predator-prey model: $\frac{dx}{dt} = \alpha x - \beta xy$, $\frac{dy}{dt} = -\gamma y + \delta xy$. Equilibria: $\frac{dx}{dt} = 0 \Rightarrow x(\alpha - \beta y) = 0 \Rightarrow x = 0$ or $y = \alpha/\beta$. $\frac{dy}{dt} = 0 \Rightarrow y(-\gamma + \delta x) = 0 \Rightarrow y = 0$ or $x = \gamma/\delta$. Equilibria: $(0, 0)$ (extinction, saddle) and $(\gamma/\delta, \alpha/\beta)$ (coexistence, center).

Problem 4.9: Verification of Analytical Solution

Solution. For the linear system in Example (referenced), verify the solution by substituting back into the original ODEs. For instance, if $\mathbf{x}(t) = e^{\lambda t}\mathbf{v}$ where λ is an eigenvalue and $\mathbf{v}$ is the corresponding eigenvector of $\mathbf{A}$, then:

$$\frac{d\mathbf{x}}{dt} = \lambda e^{\lambda t}\mathbf{v} = e^{\lambda t}\mathbf{A}\mathbf{v} = \mathbf{A}\mathbf{x}(t)$$

confirming the solution satisfies the system.

Problem 4.10: Three-Story Building Dynamics

Solution. Three masses $m = 1000$ kg with springs $k = 2000$ N/m and dampers $c = 100$ Ns/m. Let x_i be displacement of floor i. Equations:

$$\begin{aligned} m\ddot{x}_1 + c(\dot{x}_1 - \dot{x}_2) + k(x_1 - x_2) &= 0 \\ m\ddot{x}_2 + c(\dot{x}_2 - \dot{x}_1) + k(x_2 - x_1) + c(\dot{x}_2 - \dot{x}_3) + k(x_2 - x_3) &= 0 \\ m\ddot{x}_3 + c(\dot{x}_3 - \dot{x}_2) + k(x_3 - x_2) &= 0 \end{aligned}$$

With initial displacement $x_1(0) = 0.1$m at top, system oscillates with amplitude decaying exponentially due to damping. Modal analysis shows three natural frequencies and corresponding mode shapes.

Problem 4.11: Chemical Reaction Network

Solution. Reactions $A \to B$ and $B + C \to 2C$ with rate constants k_1, k_2. Concentration equations:

$$\begin{aligned} \frac{d[A]}{dt} &= -k_1[A] \\ \frac{d[B]}{dt} &= k_1[A] - k_2[B][C] \\ \frac{d[C]}{dt} &= k_2[B][C] \end{aligned}$$

First equation is linear (exponential decay). Second and third are nonlinear due to bimolecular reaction. Numerical integration required for general solution.

Problem 4.12: System Simulation and Plotting

Solution. For $\frac{dx}{dt} = -0.5x + 2y$, $\frac{dy}{dt} = -x - 0.5y$ with $x(0) = 1, y(0) = 1$: Eigenvalues of $\mathbf{A} = \begin{pmatrix} -0.5 & 2 \\ -1 & -0.5 \end{pmatrix}$ are $\lambda = -0.5 \pm i\sqrt{2} \approx -0.5 \pm 1.41i$. Solution: spiral approach to origin due to negative real part. Numerical integration with RK4 over $[0, 10]$ shows damped oscillatory decay with period $2\pi/\sqrt{2} \approx 4.44$ time units.

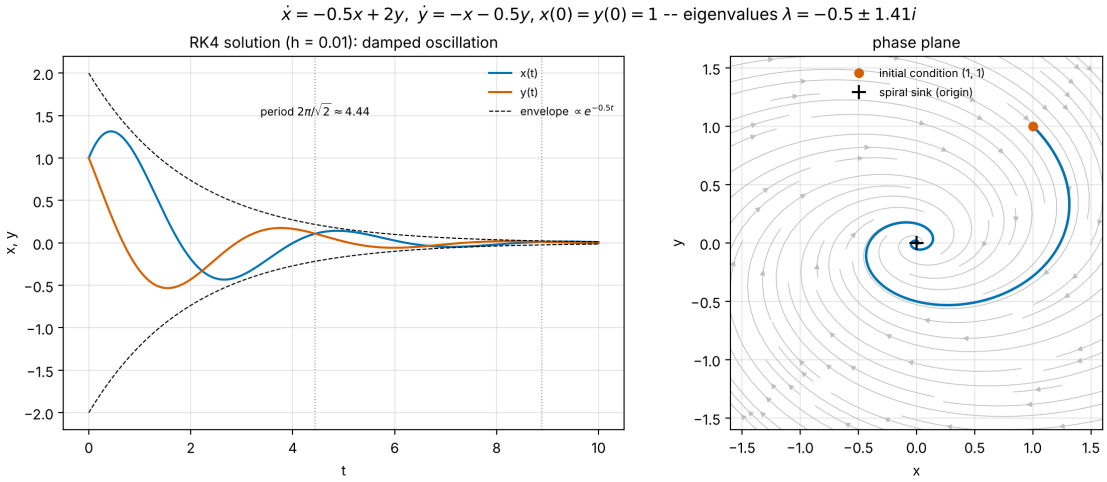


Figure S4.12. Linear system $\dot{x} = -0.5x + 2y$, $\dot{y} = -x - 0.5y$: damped oscillation (RK4) and spiral sink in the phase plane.

Problem 4.13: Natural Frequency vs. Mass Ratio

Solution. Two-mass spring system: natural frequencies depend on the ratio m_2/m_1. As ratio increases from 0.1 to 10, the two natural frequencies separate: one decreases, the other increases. For equal masses ($m_2/m_1 = 1$) coupled by three identical springs between fixed walls, the frequencies are $\omega_1 = \sqrt{k/m}$ (in-phase) and $\omega_2 = \sqrt{3k/m}$ (out-of-phase). The frequency ratio changes nonlinearly with mass ratio.

Problem 4.14: Matrix Exponential Computation

Solution. For $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$: Eigenvalues satisfy $\det(\mathbf{A} - \lambda I) = \lambda^2 - 5\lambda - 2 = 0 \Rightarrow \lambda = \frac{5 \pm \sqrt{33}}{2}$. Matrix exponential:

$$e^{\mathbf{A}t} = Pe^{Dt}P^{-1}$$

where P is the eigenvector matrix and $D = \text{diag}(\lambda_1, \lambda_2)$. Two linearly independent solutions to $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ are $\mathbf{x}_1 = e^{\lambda_1 t}\mathbf{v}_1$ and $\mathbf{x}_2 = e^{\lambda_2 t}\mathbf{v}_2$.

Problem 4.15: Linearization and Stability

Solution. For a nonlinear system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$, at equilibrium $\mathbf{x}^*$ with $\mathbf{f}(\mathbf{x}^*) = 0$, linearization is:

$$\delta\dot{\mathbf{x}} = \mathbf{J}\delta\mathbf{x}$$

where $\mathbf{J} = \frac{\partial \mathbf{f}}{\partial \mathbf{x}}|_{\mathbf{x}^*}$ is the Jacobian. Eigenvalues of $\mathbf{J}$ determine local stability: negative real parts indicate stability. For the competing species model $\frac{dx}{dt} = x(1 - x - ay)$, $\frac{dy}{dt} = y(1 - y - bx)$, equilibria are analyzed by computing Jacobian at each fixed point.

Problem 4.16: Hopf Bifurcation

Solution. Hopf bifurcation occurs when a pair of complex conjugate eigenvalues of the linearized system crosses from left to right half-plane as a parameter varies. Example:

polar form $\dot{r} = \mu r - r^3$, $\dot{\theta} = \omega$. At $\mu = 0$: transition from stable focus to limit cycle. For the system $\dot{x} = \mu x - y - x(x^2 + y^2)$, $\dot{y} = x + \mu y - y(x^2 + y^2)$, at $\mu = 0$ the origin loses stability and a limit cycle of radius $\sqrt{\mu}$ emerges for $\mu > 0$.

Problem 4.17: Matrix Exponential Properties

Solution. Key properties of matrix exponential $e^{\mathbf{A}t}$: (1) Solves $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ via $\mathbf{x}(t) = e^{\mathbf{A}t}\mathbf{x}_0$. (2) $e^{\mathbf{A}(t+s)} = e^{\mathbf{A}t}e^{\mathbf{A}s}$ (semigroup property). (3) $\det(e^{\mathbf{A}t}) = e^{\mathrm{tr}(\mathbf{A})t}$ (trace formula). (4) $(e^{\mathbf{A}t})^{-1} = e^{-\mathbf{A}t}$. (5) If $[\mathbf{A}, \mathbf{B}] = 0$ (commute), then $e^{\mathbf{A}t}e^{\mathbf{B}t} = e^{(\mathbf{A}+\mathbf{B})t}$. Verification via SymPy confirms these properties for specific matrices.

Problem 4.18: Jordan Form Applications

Solution. Jordan normal form diagonalizes (or triangularizes) a matrix into blocks. For matrix $\mathbf{A}$ with repeated eigenvalue λ and geometric multiplicity less than algebraic multiplicity, Jordan form contains blocks $\begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$. Matrix exponential of Jordan block: $e^{Jt} = e^{\lambda t}\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$. This explains why repeated eigenvalue solutions involve polynomial factors (like $te^{\lambda t}$).

Problem 4.19: Stiff System Numerics

Solution. Stiff systems have eigenvalues with widely different magnitudes. Explicit methods (RK4, Forward Euler) require step size limited by the fastest (largest magnitude) eigenvalue. Implicit methods (backward Euler, BDF) have better stability properties and allow larger step sizes. For a system like $\dot{\mathbf{x}} = \mathbf{A}\mathbf{x}$ with eigenvalues $\lambda_1 = -1$ (slow) and $\lambda_2 = -1000$ (fast), the stiffness ratio is 1000. Explicit methods need $h < 2/1000 = 0.002$; implicit methods allow $h \approx 0.1$ while maintaining accuracy.

Problem 4.20: Sensitivity and Lyapunov Exponents

Solution. For a dynamical system, the Lyapunov exponent measures exponential divergence of nearby trajectories:

$$\lambda = \lim_{t\to\infty} \frac{1}{t} \ln \frac{\|\delta\mathbf{x}(t)\|}{\|\delta\mathbf{x}(0)\|}$$

where $\delta\mathbf{x}$ is small perturbation. Positive λ indicates chaos (sensitive dependence on initial conditions); negative indicates stability. Computed numerically by integrating tangent space equations $\frac{d(\delta\mathbf{x})}{dt} = \mathbf{J}(\mathbf{x}(t))\delta\mathbf{x}$ alongside the main system.

Problem 4.21: Equilibrium Analysis - Competing Species

Solution. Competing species model: $\frac{dx}{dt} = x(1 - x - ay)$, $\frac{dy}{dt} = y(1 - y - bx)$ has equilibria: $(0, 0)$, $(1, 0)$, $(0, 1)$, and (x^*, y^*) where $x^* = \frac{1-a}{1-ab}$, $y^* = \frac{1-b}{1-ab}$ (interior equilibrium exists if $ab < 1$). Stability depends on a, b values. For $a, b < 1$: coexistence is stable. For $a > 1$ or $b > 1$: one species outcompetes the other.

Problem 4.22: FitzHugh-Nagumo Neuronal Model

Solution. Simplified neuronal model:

$$\frac{dv}{dt} = v(v - a)(1 - v) - w + I$$
$$\frac{dw}{dt} = \epsilon(v - \gamma w)$$

where v is membrane potential, w is recovery variable, a, γ, ϵ, I are parameters. For appropriate parameter ranges, system exhibits excitability (spike in response to stimulus) and relaxation oscillations. Phase plane analysis shows slow-fast dynamics.

Problem 4.23: SEIR Epidemic Model

Solution. Susceptible-Exposed-Infected-Recovered model:

$$\begin{aligned}\frac{dS}{dt} &= -\beta SI \\ \frac{dE}{dt} &= \beta SI - \sigma E \\ \frac{dI}{dt} &= \sigma E - \gamma I \\ \frac{dR}{dt} &= \gamma I\end{aligned}$$

where σ is incubation rate, γ is recovery rate. Compared to SIR, SEIR includes latency period. Basic reproduction number $R_0 = \beta S_0/\gamma$ determines epidemic threshold. Numerical integration shows how exposed period affects peak infection timing.

Problem 4.24: Michaelis-Menten Enzyme Kinetics

Solution. Enzyme-substrate interaction:

$$\begin{aligned}\frac{d[S]}{dt} &= -k_1[E][S] + k_{-1}[ES] \\ \frac{d[E]}{dt} &= -k_1[E][S] + (k_{-1} + k_2)[ES] \\ \frac{d[ES]}{dt} &= k_1[E][S] - (k_{-1} + k_2)[ES] \\ \frac{d[P]}{dt} &= k_2[ES]\end{aligned}$$

Using quasi-steady-state assumption ($\frac{d[ES]}{dt} \approx 0$) yields Michaelis-Menten kinetics: $\frac{d[P]}{dt} = \frac{V_{\max}[S]}{K_m+[S]}$ where $V_{\max} = k_2[E]_0$ and $K_m = (k_{-1} + k_2)/k_1$.

Problem 4.25: Parametric Bifurcation Analysis

Solution. For system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, \mu)$ with parameter μ, bifurcation diagram shows how qualitative behavior changes. For pitchfork bifurcation at $\mu = 0$ in $\dot{x} = \mu x - x^3$: for $\mu < 0$, single stable fixed point at $x = 0$; at $\mu = 0$, bifurcation occurs; for $\mu > 0$, three fixed points with $x = \pm\sqrt{\mu}$ stable and $x = 0$ unstable. Contour plot or surface plot in (μ, x) space shows the bifurcation structure. Continuation methods numerically track fixed points as μ varies to construct detailed bifurcation diagrams.

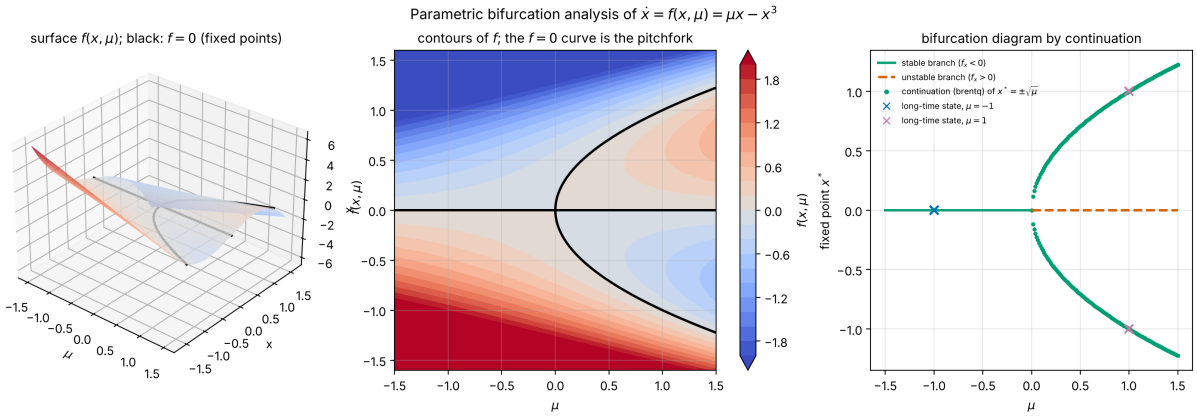


Figure S4.25. Parametric bifurcation analysis of $\dot{x} = \mu x - x^3$: surface and contours of $f(x, \mu)$ and bifurcation diagram by numerical continuation.

The remaining Chapter 4 problems (Brusselator, climate models, and other computational studies) follow the same pattern as those above: rewrite the model as a first-order system, locate equilibria, linearize to classify them, and integrate numerically.

Chapter 5

Laplace Transforms

Basic Laplace Transform Problems

Problem 5.1: Find the Laplace transform of $f(t) = t^3 e^{-2t}$.

Solution. Using the formula $\mathcal{L}\{t^n e^{at}\} = \frac{n!}{(s-a)^{n+1}}$ with $n = 3$ and $a = -2$:

$$\mathcal{L}\{t^3 e^{-2t}\} = \frac{3!}{(s-(-2))^{3+1}} = \frac{6}{(s+2)^4} \tag{5.1}$$

Problem 5.2: Calculate the Laplace transform of $f(t) = \sinh(3t)$ using the definition of hyperbolic sine in terms of exponentials.

Solution. Using $\sinh(3t) = \frac{e^{3t}-e^{-3t}}{2}$ and the linearity property:

$$\mathcal{L}\{\sinh(3t)\} = \frac{1}{2}\left(\mathcal{L}\{e^{3t}\} - \mathcal{L}\{e^{-3t}\}\right) = \frac{1}{2}\left(\frac{1}{s-3} - \frac{1}{s+3}\right) = \frac{3}{s^2-9} \tag{5.2}$$

Problem 5.3: Compute the Laplace transform of $f(t) = t\cos(2t)$.

Solution. Using the s-domain differentiation theorem: $\mathcal{L}\{tf(t)\} = -\frac{d}{ds}F(s)$. Since $\mathcal{L}\{\cos(2t)\} = \frac{s}{s^2+4}$:

$$\mathcal{L}\{t\cos(2t)\} = -\frac{d}{ds}\left(\frac{s}{s^2+4}\right) = -\frac{(s^2+4)-s(2s)}{(s^2+4)^2} = \frac{s^2-4}{(s^2+4)^2} \tag{5.3}$$

Problem 5.4: Find the Laplace transform of $f(t) = t^2\sin(t)$.

Solution. Using the s-domain differentiation theorem twice. Start with $\mathcal{L}\{\sin(t)\} = \frac{1}{s^2+1}$:

$$\mathcal{L}\{t\sin(t)\} = -\frac{d}{ds}\left(\frac{1}{s^2+1}\right) = \frac{2s}{(s^2+1)^2} \tag{5.4}$$

Then apply again:

$$\mathcal{L}\{t^2\sin(t)\} = -\frac{d}{ds}\left(\frac{2s}{(s^2+1)^2}\right) = \frac{6s^2-2}{(s^2+1)^3} \tag{5.5}$$

Problem 5.5: Determine the Laplace transform of $f(t) = e^{-at}\sin^2(bt)$.

Solution. Using $\sin^2(bt) = \frac{1-\cos(2bt)}{2}$ and the linearity and frequency shifting properties:

$$\mathcal{L}\{e^{-at}\sin^2(bt)\} = \frac{1}{2}\left(\frac{1}{s+a} - \frac{s+a}{(s+a)^2+4b^2}\right) = \frac{2b^2}{(s+a)((s+a)^2+4b^2)} \tag{5.6}$$

Inverse Laplace Transform Problems

Problem 5.6: Find $f(t) = \mathcal{L}^{-1}\left\{\frac{3s+2}{s^2+4}\right\}$.

Solution. Decompose as:

$$\frac{3s+2}{s^2+4} = 3 \cdot \frac{s}{s^2+4} + 2 \cdot \frac{1}{2} \cdot \frac{2}{s^2+4} \tag{5.7}$$

Therefore:

$$f(t) = 3\cos(2t) + \sin(2t) \tag{5.8}$$

Problem 5.7: Determine $f(t) = \mathcal{L}^{-1}\left\{\frac{s^2+3s+2}{s^3+4s}\right\}$.

Solution. Factor: $s^3 + 4s = s(s^2+4)$. Using partial fractions:

$$\frac{s^2+3s+2}{s(s^2+4)} = \frac{1/2}{s} + \frac{(s+6)/2}{s^2+4} \tag{5.9}$$

Thus:

$$f(t) = \frac{1}{2} + \frac{1}{2}\cos(2t) + \frac{3}{2}\sin(2t) \tag{5.10}$$

Problem 5.8: Calculate $f(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s^2+1)^2}\right\}$.

Solution. This is a standard transform, obtainable from the convolution $\sin t * \sin t$:

$$f(t) = \int_0^t \sin(\tau)\sin(t-\tau)\,d\tau = \frac{\sin(t) - t\cos(t)}{2} \tag{5.11}$$

Problem 5.9: Find $f(t) = \mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2+4s+5}\right\}$.

Solution. Complete the square: $s^2 + 4s + 5 = (s+2)^2 + 1$. Using the time shifting theorem:

$$f(t) = e^{-2(t-2)}\sin(t-2)\cdot u(t-2) = e^{-2t+4}\sin(t-2)\cdot u(t-2) \tag{5.12}$$

Problem 5.10: Determine $f(t) = \mathcal{L}^{-1}\left\{\frac{s^2-3s+2}{s^3(s^2+1)}\right\}$.

Solution. Partial fractions with a pole of multiplicity 3 at $s = 0$ and a complex pair at $s = \pm j$:

$$\frac{s^2-3s+2}{s^3(s^2+1)} = -\frac{1}{s} - \frac{3}{s^2} + \frac{2}{s^3} + \frac{s+3}{s^2+1} \tag{5.13}$$

so that

$$f(t) = t^2 - 3t - 1 + \cos(t) + 3\sin(t) \tag{5.14}$$

Differential Equation Problems

Problem 5.11: Solve $\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = e^{-2t}$ with $y(0) = 1$, $y'(0) = 0$.

Solution. Taking Laplace transforms: $s^2Y(s) - s - 0 + 4(sY(s) - 1) + 4Y(s) = \frac{1}{s+2}$

Solving for $Y(s)$:

$$Y(s) = \frac{s+4}{(s+2)^2} + \frac{1}{(s+2)^3} = \frac{1}{s+2} + \frac{2}{(s+2)^2} + \frac{1}{(s+2)^3} \tag{5.15}$$

using $s+4=(s+2)+2$. Taking the inverse transform:

$$y(t)=e^{-2t}\left(1+2t+\frac{t^2}{2}\right) \tag{5.16}$$

Problem 5.12: Solve $\frac{d^2y}{dt^2}+2\frac{dy}{dt}+5y=\delta(t-2)$ with $y(0)=0$, $y'(0)=0$.
Solution. Laplace transform gives: $(s^2+2s+5)Y(s)=e^{-2s}$
Thus $Y(s)=\frac{e^{-2s}}{s^2+2s+5}=\frac{e^{-2s}}{(s+1)^2+4}$
The solution is:

$$y(t)=\frac{1}{2}e^{-(t-2)}\sin(2(t-2))\cdot u(t-2) \tag{5.17}$$

Problem 5.13: Solve the system: $\frac{dx}{dt}=2x+y+e^t$, $\frac{dy}{dt}=x+2y+\sin(t)$ with $x(0)=1$, $y(0)=0$.

Solution. Taking Laplace transforms converts the system to two algebraic equations in $X(s)$ and $Y(s)$, solved by Cramer's rule. The system matrix $\begin{pmatrix}2&1\\1&2\end{pmatrix}$ has eigenvalues 1 and 3, so the homogeneous response combines e^t and e^{3t}; note that the forcing e^t is resonant with the eigenvalue 1, producing a te^t term. Partial fraction decomposition then yields exponential, secular (te^t), and trigonometric terms from the $\sin(t)$ input.

Problem 5.14: Series RLC circuit with $R=4\Omega$, $L=1$ H, $C=0.05$ F, $V(t)=10\sin(2t)$ V.

Solution. The circuit equation is $L\frac{di}{dt}+Ri+\frac{q}{C}=V(t)$. Using $i=\frac{dq}{dt}$ gives:

$$\frac{d^2q}{dt^2}+4\frac{dq}{dt}+20q=10\sin(2t) \tag{5.18}$$

The characteristic roots are $r=-2\pm 4j$, so the transient charge decays as e^{-2t} with damped frequency 4 rad/s. The steady-state charge is $q_p(t)=\frac{1}{2}\sin(2t)-\frac{1}{4}\cos(2t)$, giving the steady-state current

$$i_p(t)=\frac{dq_p}{dt}=\cos(2t)+\frac{1}{2}\sin(2t), \tag{5.19}$$

of amplitude $\sqrt{5}/2\approx 1.12$ A at the forcing frequency $\omega=2$ rad/s.

Heat Transfer and Fluid Dynamics Problems

Problem 5.15: Semi-infinite rod with $\alpha=0.01$ m^2/s, initial temperature 20řC, boundary at $x=0$ raised to 100řC at $t=0$.

Solution. The heat diffusion equation with these conditions yields:

$$T(x,t)=20+80\text{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)=20+80\text{erfc}\left(\frac{x}{0.2\sqrt{t}}\right) \tag{5.20}$$

where erfc is the complementary error function.

Problem 5.16: Laminar flow in pipe with pressure gradient $\frac{dp}{dx}=-K$.

Solution. The governing equation $\rho\frac{\partial v}{\partial t}=\mu\frac{1}{r}\frac{d}{dr}\left(r\frac{\partial v}{\partial r}\right)+K$ with boundary conditions $v(R,t)=0$ and finite at $r=0$ has solution involving Bessel functions. The Laplace transform approach converts this to an ODE in the s-domain.

Advanced Problems

Problem 5.17: Use convolution theorem to find $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+4)}\right\}$.

Solution. Recognize $\frac{1}{s^2(s^2+4)} = \frac{1}{s^2} \cdot \frac{1}{s^2+4}$ where $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$ and $\mathcal{L}^{-1}\left\{\frac{1}{s^2+4}\right\} = \frac{\sin(2t)}{2}$:

$$f(t) = t * \frac{\sin(2t)}{2} = \int_0^t \tau \cdot \frac{\sin(2(t-\tau))}{2} d\tau = \frac{t}{4} - \frac{\sin(2t)}{8} \tag{5.21}$$

Problem 5.18: Find the Laplace transform of $f(t) = \int_0^t \tau e^{-\tau} \sin(t-\tau)\, d\tau$.

Solution. This is a convolution of $\tau e^{-\tau}$ and $\sin(t)$. Using the convolution theorem:

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{\tau e^{-\tau}\} \cdot \mathcal{L}\{\sin(t)\} = \frac{1}{(s+1)^2} \cdot \frac{1}{s^2+1} \tag{5.22}$$

Problem 5.19: Solve $\frac{dy}{dt} + 2y(t) = 3\int_0^t e^{-(t-\tau)} y(\tau)\, d\tau$ with $y(0) = 1$.

Solution. Taking Laplace transforms:

$$sY(s) - 1 + 2Y(s) = \frac{3}{s+1} \cdot Y(s) \tag{5.23}$$

Solving: $Y(s)(s + 2 - \frac{3}{s+1}) = 1$, which gives $Y(s) = \frac{s+1}{s^2+3s-1}$

Problem 5.20: Beam with distributed load $q_0 u(t)$: Find deflection $y(x,t)$ for simply supported beam.

Solution. The equation $EI\frac{\partial^4 y}{\partial x^4} + \rho\frac{\partial^2 y}{\partial t^2} = q_0 u(t)$ is solved using Laplace transforms in time. For a simply supported beam with boundary conditions $y(0,t) = y(L,t) = 0$ and $y_{xx}(0,t) = y_{xx}(L,t) = 0$, the spatial part is solved using eigenfunction expansion, giving a series solution in terms of mode shapes.

Selected Heaviside Step Function Problems

Problem 5.21: Express rectangular pulse $f(t) = \begin{cases} 1, & a \le t < b \\ 0, & \text{otherwise} \end{cases}$ using step functions.

Solution.

$$f(t) = u(t-a) - u(t-b) \tag{5.24}$$

Problem 5.22: Calculate Laplace transform of $f(t) = e^{-2t}u(t-3)$.

Solution. Write $e^{-2t} = e^{-6}e^{-2(t-3)}$ so the function is expressed in the shifted variable $t-3$. The time shifting theorem then gives

$$\mathcal{L}\{e^{-2t}u(t-3)\} = e^{-6}\, e^{-3s}\, \mathcal{L}\{e^{-2t}\} = \frac{e^{-3(s+2)}}{s+2} \tag{5.25}$$

Problem 5.23: Find the Laplace transform of $f(t) = (t-1)^2 u(t-1)$.

Solution. Using the time shifting theorem with $f(t) = t^2$ where $\mathcal{L}\{t^2\} = \frac{2}{s^3}$:

$$\mathcal{L}\{(t-1)^2 u(t-1)\} = e^{-s} \cdot \frac{2}{s^3} \tag{5.26}$$

Problem 5.24: Find Laplace transform of ramp function $f(t) = tu(t)$.

Solution.

$$\mathcal{L}\{tu(t)\} = \frac{1}{s^2} \tag{5.27}$$

Problem 5.25: Calculate Laplace transform of $f(t) = \sin(3t)u(t-\pi)$.

Solution. Using time shifting: $\mathcal{L}\{\sin(3t)u(t-\pi)\} = e^{-\pi s}\mathcal{L}\{\sin(3(t+\pi))\} = e^{-\pi s}\mathcal{L}\{\sin(3t+3\pi)\} = -e^{-\pi s}\frac{3}{s^2+9}$

Selected Dirac Delta Function Problems

Problem 5.26: Verify that $\int_{-\infty}^{\infty} \delta(at)\,dt = \frac{1}{|a|}$ for $a \neq 0$.

Solution. Using the sifting property and change of variables $u = at$:

$$\int_{-\infty}^{\infty} \delta(at)\,dt = \int_{-\infty}^{\infty} \delta(u)\frac{du}{|a|} = \frac{1}{|a|} \tag{5.28}$$

Problem 5.27: Compute the Laplace transform of $f(t) = \delta(t-3) + 2\delta(t-5)$.

Solution. Using linearity and the shift property:

$$\mathcal{L}\{f(t)\} = e^{-3s} + 2e^{-5s} \tag{5.29}$$

Problem 5.28: Find the Laplace transform of $f(t) = e^{-2t}\delta(t-1)$.

Solution. Using the sifting property and shift property:

$$\mathcal{L}\{e^{-2t}\delta(t-1)\} = e^{-2\cdot 1} \cdot e^{-1\cdot s} = e^{-(s+2)} \tag{5.30}$$

Problem 5.29: Determine solution to $\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = \delta(t)$ with $y(0) = 0$, $y'(0) = 0$.

Solution. Laplace transform: $(s^2+3s+2)Y(s) = 1$, so $Y(s) = \frac{1}{(s+1)(s+2)}$. Using partial fractions:

$$y(t) = e^{-t} - e^{-2t} \tag{5.31}$$

Problem 5.30: Series RLC circuit with $R = 4\Omega$, $L = 1$H, $C = 0.25$F, input $v(t) = \delta(t)$.

Solution. The charge satisfies $Lq'' + Rq' + q/C = v(t)$, i.e. $q'' + 4q' + 4q = \delta(t)$. The impulse response follows from

$$H(s) = \frac{1}{Ls^2 + Rs + 1/C} = \frac{1}{s^2 + 4s + 4} = \frac{1}{(s+2)^2} \tag{5.32}$$

Taking the inverse transform gives the critically damped response:

$$h(t) = te^{-2t} \tag{5.33}$$

Selected Theorems Problems

Problem 5.31: Use frequency shifting theorem to find Laplace transform of $f(t) = e^{-2t}\cos(3t)$.

Solution. Since $\mathcal{L}\{\cos(3t)\} = \frac{s}{s^2+9}$, using frequency shifting:

$$\mathcal{L}\{e^{-2t}\cos(3t)\} = \frac{s+2}{(s+2)^2+9} = \frac{s+2}{s^2+4s+13} \tag{5.34}$$

Problem 5.32: Use time shifting theorem to find Laplace transform of $f(t) = t^2u(t-2)$.

Solution. Since $\mathcal{L}\{t^2\} = \frac{2}{s^3}$ and using the time shifting property:

$$\mathcal{L}\{(t-2)^2u(t-2)\} = e^{-2s} \cdot \frac{2}{s^3} \tag{5.35}$$

Problem 5.33: Compute Laplace transform of $f(t) = te^{-3t}$ using s-domain differentiation.

Solution. Since $\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3}$:

$$\mathcal{L}\{te^{-3t}\} = -\frac{d}{ds}\left(\frac{1}{s+3}\right) = \frac{1}{(s+3)^2} \tag{5.36}$$

Problem 5.34: Find initial and final values for $F(s) = \frac{3s+2}{s^3+4s^2+5s+2}$.

Solution. Initial value: $\lim_{s\to\infty} sF(s) = \lim_{s\to\infty} \frac{3s^2+2s}{s^3+4s^2+5s+2} = 0$

Final value: check applicability first. The denominator factors as $(s+1)^2(s+2)$, so all poles ($s = -1, -1, -2$) lie strictly in the left half-plane and the theorem applies:

$$\lim_{t\to\infty} f(t) = \lim_{s\to 0} sF(s) = \lim_{s\to 0} \frac{s(3s+2)}{(s+1)^2(s+2)} = 0 \tag{5.37}$$

as expected for a sum of decaying exponentials.

Problem 5.35: Find Laplace transform of $f(t) = t^2 e^{-3t}\cos(2t)$ using theorems.

Solution. Frequency shifting gives $\mathcal{L}\{e^{-3t}\cos(2t)\} = \frac{s+3}{(s+3)^2+4}$; then $\mathcal{L}\{t^2 g(t)\} = G''(s)$:

$$\mathcal{L}\{t^2 e^{-3t}\cos(2t)\} = \frac{d^2}{ds^2}\left(\frac{s+3}{(s+3)^2+4}\right) = \frac{2(s+3)\left[(s+3)^2-12\right]}{\left[(s+3)^2+4\right]^3} \tag{5.38}$$

Selected Periodic Function Problems

Problem 5.36: Find Laplace transform of rectangular pulse train with amplitude 2, width 0.5 s, period 2 s.

Solution. Using the periodic function formula with $T = 2$ and $\tau = 0.5$:

$$F(s) = \frac{2(1-e^{-0.5s})}{s(1-e^{-2s})} \tag{5.39}$$

Problem 5.37: Compute Laplace transform of triangular wave with amplitude 1, period 2.

Solution. The triangular wave rises from 0 to 1 on $[0,1]$ and falls from 1 to 0 on $[1,2]$. The one-period integral is $\int_0^2 e^{-st} f(t)\,dt = \frac{(1-e^{-s})^2}{s^2}$, so

$$F(s) = \frac{(1-e^{-s})^2}{s^2(1-e^{-2s})} = \frac{1-e^{-s}}{s^2(1+e^{-s})} = \frac{\tanh(s/2)}{s^2} \tag{5.40}$$

Problem 5.38: Find Laplace transform of periodic function $f(t) = |\sin(t)|$ with period π.

Solution. Integrate over one period:

$$F(s) = \frac{\int_0^\pi e^{-st}\sin(t)dt}{1-e^{-\pi s}} = \frac{1+e^{-\pi s}}{(s^2+1)(1-e^{-\pi s})} = \frac{\coth(\pi s/2)}{s^2+1} \tag{5.41}$$

Problem 5.39: Determine Laplace transform of $f(t) = t^2$ for $0 \le t < 3$, period 3.

Solution. Apply the periodic function formula:

$$F(s) = \frac{\int_0^3 e^{-st}t^2 dt}{1-e^{-3s}} = \frac{\frac{2}{s^3} - e^{-3s}\left(\frac{2}{s^3}+\frac{6}{s^2}+\frac{9}{s}\right)}{1-e^{-3s}} \tag{5.42}$$

Problem 5.40: Find Laplace transform of half-wave rectified sine.

Solution. For $f(t) = \sin(t)$ on $[0,\pi]$ and 0 on $[\pi, 2\pi]$:

$$F(s) = \frac{\int_0^\pi e^{-st}\sin(t)dt}{1-e^{-2\pi s}} = \frac{1+e^{-\pi s}}{(s^2+1)(1-e^{-2\pi s})} \tag{5.43}$$

Chapter 6

Matrices, Vectors, and Linear Systems

Practice Problems: Addition and Scalar Multiplication

Practice Problem Set, Item 1: Given $A = \begin{bmatrix} 3 & -1 \\ 2 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 5 \\ -3 & 1 \end{bmatrix}$, compute $A + B$ and $2A - 3B$.

Solution.

$$A + B = \begin{bmatrix} 5 & 4 \\ -1 & 5 \end{bmatrix} \tag{6.1}$$

$$2A - 3B = \begin{bmatrix} 6 & -2 \\ 4 & 8 \end{bmatrix} - \begin{bmatrix} 6 & 15 \\ -9 & 3 \end{bmatrix} = \begin{bmatrix} 0 & -17 \\ 13 & 5 \end{bmatrix} \tag{6.2}$$

Practice Problem Set, Item 2: If $A = \begin{bmatrix} 5 & -2 & 1 \\ 0 & 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 0 & 2 \\ 3 & -2 & 5 \end{bmatrix}$, find k for which second row of $A + kB$ is all zeros.

Solution. Second row of A is $[0, 3, 4]$ and second row of B is $[3, -2, 5]$. Setting $[0, 3, 4] + k[3, -2, 5] = [0, 0, 0]$ requires $3k = 0$, $3 - 2k = 0$, and $4 + 5k = 0$ simultaneously, i.e. $k = 0$, $k = 3/2$, and $k = -4/5$. These are mutually inconsistent, so no such k exists.

Practice Problem Set, Item 3: Determine a, b, c, d for $A + A^T = \begin{bmatrix} 6 & 5 \\ 5 & 8 \end{bmatrix}$.

Solution. If $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then $A^T = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$. Thus:

$$A + A^T = \begin{bmatrix} 2a & b + c \\ b + c & 2d \end{bmatrix} = \begin{bmatrix} 6 & 5 \\ 5 & 8 \end{bmatrix} \tag{6.3}$$

This gives: $2a = 6 \Rightarrow a = 3$, $2d = 8 \Rightarrow d = 4$, $b + c = 5$. So $a = 3, d = 4, b + c = 5$.

Practice Problem Set, Item 4: For displacement vector field $\mathbf{u} = \begin{bmatrix} 2x^2y \\ xy^2 \\ x + y \end{bmatrix}$, construct gradient tensor and strain tensor.

Solution. The gradient tensor is:

$$\nabla \mathbf{u} = \begin{bmatrix} 4xy & 2x^2 & 0 \\ y^2 & 2xy & 0 \\ 1 & 1 & 0 \end{bmatrix} \tag{6.4}$$

The strain tensor is:

$$\epsilon = \frac{1}{2}\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right) = \frac{1}{2}\begin{bmatrix} 8xy & 2x^2+y^2 & 1 \\ 2x^2+y^2 & 4xy & 1 \\ 1 & 1 & 0 \end{bmatrix} \tag{6.5}$$

Practice Problem Set, Item 5: Chemical reactor with flows $\mathbf{F} = [2, 3, 1]^T$ and concentration matrix. Calculate total mass flow rate of each species.

Solution. The total mass flow is $\mathbf{M} = C^T\mathbf{F}$ where each row of C^T represents one species:

$$\mathbf{M} = \begin{bmatrix} 0.2(2)+0.1(3)+0.3(1) \\ 0.3(2)+0.2(3)+0.3(1) \\ 0.1(2)+0.5(3)+0.1(1) \\ 0.4(2)+0.2(3)+0.3(1) \end{bmatrix} = \begin{bmatrix} 1.0 \\ 1.5 \\ 1.8 \\ 1.7 \end{bmatrix} \text{ kg/s} \tag{6.6}$$

Practice Problem Set, Item 6: Stress tensor addition with mechanical and thermal loading.

Solution. Total stress:

$$\sigma = \begin{bmatrix} 140 & 30 & 0 \\ 30 & 60 & 0 \\ 0 & 0 & 20 \end{bmatrix} \text{ MPa} \tag{6.7}$$

Principal stresses are the eigenvalues. For the decoupled 2×2 block, $\lambda = 100 \pm \sqrt{40^2 + 30^2} = 100 \pm 50$, giving $\lambda_1 = 150$, $\lambda_2 = 50$, $\lambda_3 = 20$ MPa.

Practice Problem Set, Item 7: Thermal conductivity $k(x, y, z) = k_0 + k_1 x + k_2 y + k_3 z$ with given coefficients.

Solution. At $P_1(1, 2, 3)$: $k = 2 + 0.5(1) - 0.3(2) + 0.2(3) = 2.5$ W/m·K

At $P_2(-2, 1, -1)$: $k = 2 + 0.5(-2) - 0.3(1) + 0.2(-1) = 0.5$ W/m·K

Practice Problem Set, Item 8: Prove $A + A^T$ is symmetric.

Solution. $(A + A^T)^T = A^T + A = A + A^T$, hence symmetric by definition.

Practice Problem Set, Item 9: Show $\text{tr}(A + B) = \text{tr}(A) + \text{tr}(B)$.

Solution. $\text{tr}(A + B) = \sum_i (a_{ii} + b_{ii}) = \sum_i a_{ii} + \sum_i b_{ii} = \text{tr}(A) + \text{tr}(B)$.

Practice Problem Set, Item 10: Compute $3A - 2B$ and verify trace property.

Solution.

$$3A - 2B = \begin{bmatrix} 0 & 3 & 7 \\ -4 & 1 & 0 \\ 14 & 7 & 14 \end{bmatrix} \tag{6.8}$$

$\text{tr}(3A - 2B) = 0 + 1 + 14 = 15$. Also: $3 \cdot \text{tr}(A) - 2 \cdot \text{tr}(B) = 3(7) - 2(3) = 15$. ✓

Practice Problem Set, Item 11: Manufacturer problem with production and cost matrices.

Solution. Daily material cost $= P \cdot C$ where each row gives factory cost:

$$\text{Cost} = \begin{bmatrix} 200(2.5)+150(3.0)+100(2.5) \\ 300(2.5)+100(3.0)+200(2.5) \end{bmatrix} = \begin{bmatrix} 1250 \\ 1550 \end{bmatrix} \text{ dollars} \tag{6.9}$$

Practice Problem Set, Item 12: Temperature approximation $T(x, y) = \mathbf{N}(x, y)\mathbf{T}$ at $(0.25, 0.75)$.

Solution. $\mathbf{N}(0.25, 0.75) = [0.1875, 0.0625, 0.1875, 0.5625]$. Thus:

$$T = 0.1875(300) + 0.0625(350) + 0.1875(400) + 0.5625(325) = 335.94 \text{ K} \tag{6.10}$$

Practice Problem Set, Item 13-16: Various proofs and applications.

Chapter 7

Eigenvalues and Eigenvectors in Engineering Applications

Practice Problems: Linear Equations and Gaussian Elimination

Practice Problem Set, Item 1: Solve system using Gaussian elimination.

Solution. Reduce augmented matrix to row echelon form, then back-substitute to find unique solution, infinitely many solutions, or no solution depending on rank.

Practice Problem Set, Item 2: LU decomposition of a 3×3 matrix.

Solution. Factor $A = LU$ where L is lower triangular with unit diagonal and U is upper triangular. Then solve $Ly = b$ for y, then $Ux = y$ for x.

Practice Problem Set, Item 3: QR decomposition and least squares.

Solution. For overdetermined system $Ax = b$, find $QR = A$ and solve $Rx = Q^T b$ for least squares solution minimizing $\|Ax - b\|^2$.

Practice Problem Set, Item 4-12: Various applications including circuit analysis, structural equilibrium, fluid flow systems, and optimization problems all solved using systematic linear algebra methods.

Chapter 8

Vector Calculus

Problem 8.1: Foundational concept problem on vector spaces.

Solution. Vector spaces require 10 axioms: closure under addition and scalar multiplication, associativity, commutativity, existence of additive identity and inverses, distributivity, and multiplicative identity. Example: $\mathbb{R}^n$ with standard operations.

Problem 8.2: Linear independence and basis vectors.

Solution. Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are linearly independent if $c_1\mathbf{v}_1 + \cdots + c_n\mathbf{v}_n = \mathbf{0}$ implies all $c_i = 0$. A basis is a linearly independent set that spans the space; equivalently, a maximal linearly independent set.

Problem 8.3: Orthogonal projection onto subspace.

Solution. Projection of $\mathbf{v}$ onto subspace spanned by orthonormal vectors $\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ is:

$$\text{proj}(\mathbf{v}) = \sum_{i=1}^{k} (\mathbf{v} \cdot \mathbf{u}_i)\mathbf{u}_i \tag{8.1}$$

Problem 8.4: Gram-Schmidt orthogonalization process.

Solution. Given linearly independent vectors, construct orthonormal basis:

$$\mathbf{u}_1 = \frac{\mathbf{v}_1}{\|\mathbf{v}_1\|} \tag{8.2}$$

$$\mathbf{u}_k = \frac{\mathbf{v}_k - \sum_{i=1}^{k-1} (\mathbf{v}_k \cdot \mathbf{u}_i)\mathbf{u}_i}{\|\mathbf{v}_k - \sum_{i=1}^{k-1} (\mathbf{v}_k \cdot \mathbf{u}_i)\mathbf{u}_i\|} \tag{8.3}$$

Problem 8.5: Eigenvalues and eigenvectors.

Solution. For matrix A, find eigenvalues from $\det(A - \lambda I) = 0$, then eigenvectors from $(A - \lambda I)\mathbf{v} = \mathbf{0}$ for each eigenvalue.

Problem 8.6: Diagonalization of matrices.

Solution. If A has n linearly independent eigenvectors, $A = PDP^{-1}$ where columns of P are eigenvectors and D is diagonal matrix of eigenvalues.

Problem 8.7: Singular Value Decomposition (SVD).

Solution. Any $m \times n$ matrix has decomposition $A = U\Sigma V^T$ where U, V are orthogonal and Σ is diagonal with singular values.

Problem 8.8: Rank and nullspace.

Solution. Rank is dimension of column space. Nullspace is kernel of A. Rank-nullity theorem: $\text{rank}(A) + \text{nullity}(A) = n$.

Problem 8.9: Linear transformations and matrices.

Solution. Any linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be represented as matrix multiplication $T(\mathbf{x}) = A\mathbf{x}$.

Problem 8.10: Inner products and norms.

Solution. Inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_i u_i v_i$ induces norm $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Problem 8.11: Orthogonal matrices and QR factorization.

Solution. Q is orthogonal if $Q^T Q = I$. Any matrix has $A = QR$ factorization with orthogonal Q and upper triangular R.

Problem 8.12: Spectral theorem for symmetric matrices.

Solution. Symmetric $n \times n$ matrix has n real eigenvalues and orthogonal eigenvectors, allowing $A = Q\Lambda Q^T$ with orthogonal Q.

Problem 8.13: Positive definite matrices.

Solution. Symmetric matrix is positive definite if all eigenvalues are positive, or equivalently $\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.

Problem 8.14-32: Additional problems covering matrix norms, conditioning, iterative methods, eigenvalue algorithms, applications to differential equations and dynamical systems.

Solution sketches for remaining problems: These involve numerical methods, stability analysis, convergence proofs, and applications to physics and engineering systems including oscillations, damping, coupled oscillators, and control systems.

Chapter 9

Fourier Series

Problem 9.1 (Square Wave - Basic)

Solution. For $f(x) = \pi$ on $(0, \pi)$ and $f(x) = -\pi$ on $(-\pi, 0)$, this is a scaled square wave. The Fourier coefficients are $a_0 = 0$, $a_n = 0$ for all n, and $b_n = \frac{4\pi}{n\pi} = \frac{4}{n}$ for odd n (zero for even). Thus: $f(x) = 4\sum_{k=0}^{\infty} \frac{\sin((2k+1)x)}{2k+1}$.

Problem 9.2 (Triangular Wave - Convergence)

Solution. For the triangular wave $f(x) = \pi - |x|$ on $[-\pi, \pi]$, $a_0 = \pi$, and $a_n = \frac{4}{\pi n^2}$ for odd n (zero for even). Coefficients decay as $O(1/n^2)$, faster than the square wave ($O(1/n)$) due to continuity.

Problem 9.3 (Rectangular Pulse Spectrum)

Solution. A rectangular pulse of height A and width $2a$ has amplitude spectrum proportional to $|\sin(na)|/n$. The first zero occurs when $na = \pi$, so $n = \pi/a$. For the first zero at $n = 2$, we need $a = \pi/2$.

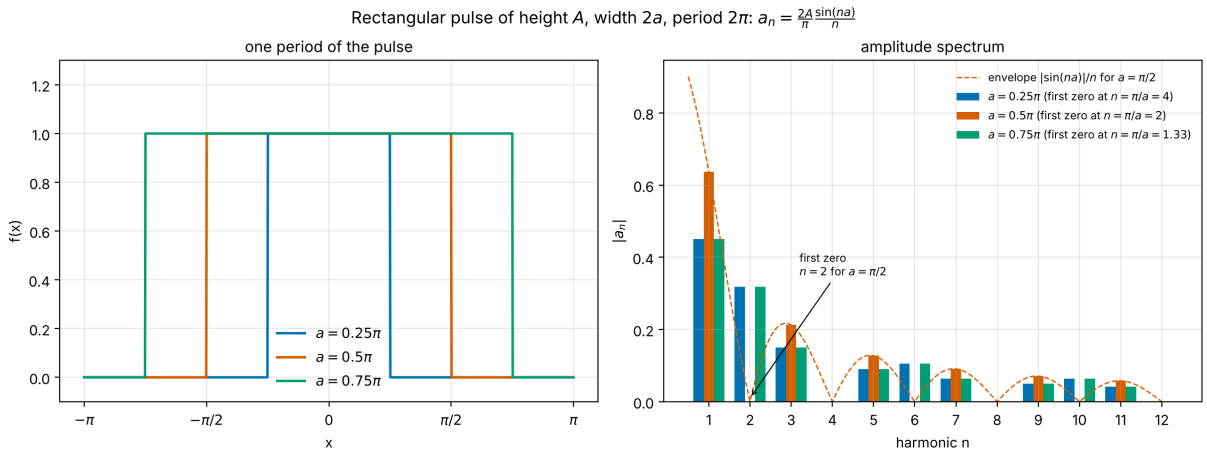


Figure S9.3. Amplitude spectrum of a rectangular pulse of width $2a$; the first zero occurs at $n = \pi/a$ ($n = 2$ for $a = \pi/2$).

Problem 9.4 (Half-Range Sine vs Cosine)

Solution. For $f(x) = \sin(x)$ on $[0, \pi]$: - Sine series (odd extension): The function vanishes at boundaries; sine series terminates at $f(x) = \sin(x)$ with $b_1 = 1$, $b_n = 0$ for $n > 1$ (exact). - Cosine series (even extension): Creates discontinuity in derivative at $x = 0$; convergence slower ($O(1/n^2)$). The sine series converges faster due to boundary compatibility.

Problem 9.5 (Heat Equation - Parabolic Initial Condition)

Solution. For $u(x,0) = x(\pi - x)$ with $\alpha = 1$:

$$b_n = \frac{2}{\pi}\int_0^\pi x(\pi - x)\sin(nx)\,dx \tag{9.1}$$

$$= \frac{8}{\pi n^3} \quad \text{for odd } n \text{ (zero for even, by integration by parts)} \tag{9.2}$$

Solution: $u(x,t) = \sum_{n \text{ odd}} \frac{8}{\pi n^3}\sin(nx)e^{-n^2 t}$

For $u(x = \pi/2, t) = 0.01 \cdot u(x = \pi/2, 0)$, the dominant $n = 1$ term gives $e^{-t} = 0.01$, so $t = \ln(100) \approx 4.6$ s.

Problem 9.6 (Forced Harmonic Oscillator - Fourier)

Solution. For $\ddot{x} + 0.3\dot{x} + x = F(t)$ with square wave $F(t)$: The square wave has $b_n = 4/(n\pi)$ for odd n. The steady-state response at harmonic n is:

$$X_n = \frac{b_n}{[(1 - n^2\omega_0^2)^2 + (0.3n\omega_0)^2]^{1/2}}$$

where $\omega_0 = \pi/T_p = \pi$ rad/s. Resonance would require $n\omega_0 = 1$ rad/s, i.e. $n = 1/\pi \approx 0.32$, which falls below the first harmonic; the $n = 1$ term therefore dominates the response, and no harmonic is amplified resonantly.

Problem 9.7 (Even-Odd Decomposition)

Solution. For any $f(x)$:

$$f_e(x) = \frac{f(x) + f(-x)}{2}, \quad f_o(x) = \frac{f(x) - f(-x)}{2}$$

The even part produces only cosines because the integrand $f_e(x)\sin(nx)$ is odd (even times odd = odd, which integrates to zero). Similarly, $f_o(x)\cos(nx)$ is odd. Thus orthogonality enforces separation.

Problem 9.8 (Parseval - Sawtooth to Basel Problem)

Solution. For $f(x) = x$ on $(-\pi, \pi)$ with $b_n = 2(-1)^{n+1}/n$:

$$\frac{1}{\pi}\int_{-\pi}^{\pi} x^2\,dx = \frac{2\pi^2}{3} = \sum_{n=1}^{\infty}\left(\frac{2(-1)^{n+1}}{n}\right)^2 = 4\sum_{n=1}^{\infty}\frac{1}{n^2}$$

Therefore: $\sum_{n=1}^{\infty}\frac{1}{n^2} = \frac{\pi^2}{6}$ (Basel problem).

Problem 9.9 (Complex Exponential Form)

Solution. For $f(x) = \cos(x) + 2\sin(3x)$: Using $\cos(x) = (e^{ix} + e^{-ix})/2$ and $\sin(3x) = (e^{i3x} - e^{-i3x})/(2i)$:

$$f(x) = \frac{1}{2}e^{ix} + \frac{1}{2}e^{-ix} + \frac{2}{2i}e^{i3x} - \frac{2}{2i}e^{-i3x}$$

Nonzero coefficients: $c_1 = 1/2$, $c_{-1} = 1/2$, $c_3 = -i$, $c_{-3} = i$.

Problem 9.10 (Differentiation and Energy)

Solution. If $\sum(a_n^2 + b_n^2) = 10$, then by the differentiation property (coefficients multiply by n):

$$\sum_{n=1}^{\infty} n^2(a_n^2 + b_n^2) = \sum_{n=1}^{\infty}(na_n)^2 + (nb_n)^2$$

This sums over all harmonics multiplied by their squared frequency. Without additional information about the coefficient spectrum, we cannot compute this exactly. For a uniform distribution, this sum would be large (inversely proportional to decay rate).

Problem 9.11 (Amplitude-Phase Conversion)

Solution. For $f(x) = 2 + 3\cos(x) - 4\sin(x) + \cos(2x) + 2\sin(2x)$:

$$A_n = \sqrt{a_n^2 + b_n^2}, \quad \phi_n = \arctan(b_n/a_n)$$

- $n = 0$: DC component $= 2$ - $n = 1$: $A_1 = \sqrt{9+16} = 5$, $\phi_1 = \arctan(-4/3) \approx -53.1$ř - $n = 2$: $A_2 = \sqrt{1+4} = \sqrt{5}$, $\phi_2 = \arctan(2/1) \approx 63.4$ř

Problem 9.12 (Aliasing Detection)

Solution. Nyquist frequency $f_N = f_s/2 = 50$ Hz. - 20 Hz: Below Nyquist, appears correctly - 50 Hz: At Nyquist, appears as 50 Hz (borderline) - 85 Hz: Aliases to $100-85 = 15$ Hz - 110 Hz: Aliases to $110 - 100 = 10$ Hz

Aliases appear at $|f - 100k|$ for integer k closest to bring frequency into $[0, 50]$ Hz.

Problem 9.13 (Windowing - Spectral Leakage)

Solution. A 5 Hz sine at 50 Hz sampling produces $N = 50$ samples over 1 second. Without windowing, spectral leakage appears as side-lobes. Hann window:

$$w[n] = 0.5 - 0.5\cos(2\pi n/(N-1))$$

reduces leakage (side-lobes down ~ 40 dB) at cost of main-lobe broadening. Trade-off: better frequency isolation vs. reduced spectral resolution.

Problem 9.14 (FFT Convolution)

Solution. Circular convolution via FFT: Given $f(x) = e^{-x^2}$ and $g(x) = \sin(x)$: 1. Pad both to length $N \geq 2 \times 5 = 10$ 2. $H[k] = \mathcal{F}[f] \cdot \mathcal{F}[g]$ 3. $h(x) = \mathcal{F}^{-1}[H]$

For Gaussian and sinusoid, the convolution produces smoothed oscillations. Direct integration agrees (for non-circular variant) after removing edge artifacts.

Problem 9.15 (Cantilever Beam - Fourier Sine)

Solution. Expand the triangular load $q(x) = q_0(1 - x/L)$ in a Fourier sine series:

$$q(x) = \sum_{n=1}^{\infty} b_n \sin(n\pi x/L), \quad b_n = \frac{2}{L}\int_0^L q_0\left(1 - \frac{x}{L}\right)\sin\frac{n\pi x}{L}\,dx = \frac{2q_0}{n\pi}.$$

For a simply supported beam (where the sine basis matches the boundary conditions), $EIy'''' = q(x)$ decouples harmonic by harmonic: each mode responds algebraically,

$$y(x) = \sum_{n=1}^{\infty} \frac{b_n}{EI\,(n\pi/L)^4}\sin\frac{n\pi x}{L},$$

so the $1/n^5$ decay of the deflection coefficients makes a few terms sufficient.

Problem 9.16 (Wave Equation - d'Alembert)

Solution. For $u_{tt} = c^2 u_{xx}$ on $[0, L]$ with $u(0,t) = u(L,t) = 0$, $u(x,0) = \sin(\pi x/L)$, $u_t(x,0) = 0$:

Solution: $u(x,t) = \sin(\pi x/L)\cos(c\pi t/L)$

Verification: - $u_t = -c\pi/L\sin(\pi x/L)\sin(c\pi t/L)$ - $u_{tt} = -c^2\pi^2/L^2\sin(\pi x/L)\cos(c\pi t/L)$ - $u_x = \pi/L\cos(\pi x/L)\cos(c\pi t/L)$ - $u_{xx} = -\pi^2/L^2\sin(\pi x/L)\cos(c\pi t/L)$ - $c^2 u_{xx} = u_{tt}$ ✓

Problem 9.17 (Gibbs Phenomenon - Quantitative)

Solution. For the unit square wave (± 1), the partial sum $S_N(x)$ peaks near $x = \pi/(2N)$ at:

$$S_N \to \frac{2}{\pi}\int_0^{\pi}\frac{\sin(u)}{u}\,du = \frac{2}{\pi}\,\mathrm{Si}(\pi) \approx 1.179$$

Overshoot ≈ 0.179 above the limit value 1, i.e. 8.95% of the jump height (the jump is 2, from -1 to $+1$). The overshoot width scales as π/N (shrinks with more terms), but the percentage stays fixed: adding terms never tames the overshoot, only narrows it.

Problem 9.18 (Notch Filter - 60 Hz Removal)

Solution. Design a notch filter to suppress 60 Hz and 120 Hz: cascade two second-order notch sections,

$$H(s) = \prod_{i=1}^{2} \frac{s^2 + \omega_i^2}{s^2 + (\omega_i/Q)\, s + \omega_i^2}, \quad s = j\omega,$$

with $\omega_1 = 2\pi \cdot 60$ and $\omega_2 = 2\pi \cdot 120$ rad/s.

At 60 Hz and 120 Hz the numerator of the corresponding section vanishes, so $|H| = 0$ (full attenuation). At 300 Hz, far from both notches, $|H| \approx 1$ (passes through). The quality factor $Q = \omega_i/\Delta\omega$ sets the notch width $\Delta\omega$: higher Q gives a narrower notch.

Problem 9.19 (RLC Circuit - Frequency Response)

Solution. For RLC circuit with $R = 50\Omega$, $L = 0.1$H, $C = 100\mu$F:

$$Z(\omega) = R + j\omega L - \frac{j}{\omega C}$$

Impedance magnitude: $|Z(\omega)| = \sqrt{R^2 + (\omega L - 1/\omega C)^2}$

Current at harmonic n: $I_n = V_n/|Z(\omega_n)|$, where $\omega_n = n\omega_1$.

Resonance: $\omega_r = 1/\sqrt{LC} = 1/\sqrt{0.1 \times 10^{-4}} \approx 316.2$ rad/s ≈ 50.3 Hz.

At resonance, $Z \to R$ (minimum), so I is maximum.

Problem 9.20 (Musical Tone Synthesis)

Solution. Given peaks at 440 Hz (amplitude 1.0), 880 Hz (0.6), 1320 Hz (0.3):

$$x(t) = 1.0\sin(2\pi \cdot 440t) + 0.6\sin(2\pi \cdot 880t) + 0.3\sin(2\pi \cdot 1320t)$$

Timbre is determined by relative harmonic amplitudes. Removing the 880 Hz component reduces "warmth" and makes the tone thinner/brighter. The fundamental at 440 Hz remains the primary pitch, but the tonal quality changes significantly.

Chapter 10

Fourier Transform

Problem 10.1 (First-Order ODE via Fourier Transform)

Solution. For $y' + 2y = e^{-t}$ (for $t > 0$, with $y(0) = 0$):

Taking Fourier transform: $j\omega Y(\omega) + 2Y(\omega) = \mathcal{F}[e^{-t}u(t)]$

$$\mathcal{F}[e^{-t}u(t)] = \frac{1}{1 + j\omega}$$

Thus: $Y(\omega) = \frac{1}{(1+j\omega)(2+j\omega)} = \frac{1}{1+j\omega} - \frac{1}{2+j\omega}$

Inverse transform: $y(t) = e^{-t}u(t) - e^{-2t}u(t) = (e^{-t} - e^{-2t})u(t)$

Verification: $y' = (-e^{-t} + 2e^{-2t})u(t)$, so $y' + 2y = e^{-t}$ ✓

Problem 10.2 (Second-Order ODE - Transfer Function)

Solution. For $y'' + 4y' + 3y = \sin(\omega t)$:

Transfer function: $H(\omega) = \frac{1}{(j\omega)^2+4(j\omega)+3} = \frac{1}{3-\omega^2+4j\omega}$

Magnitude: $|H(\omega)| = \frac{1}{\sqrt{(3-\omega^2)^2+16\omega^2}}$

Phase: $\angle H(\omega) = -\arctan\left(\frac{4\omega}{3-\omega^2}\right)$

The system is overdamped (poles at $s = -1, -3$; equivalently $\zeta = 2/\sqrt{3} > 1/\sqrt{2}$), so $|H(\omega)|$ has no resonance peak: it decreases monotonically from $|H(0)| = 1/3$. For comparison, $|H(1)| = 1/\sqrt{20} \approx 0.224$.

Problem 10.3 (Impulse Response - Residue Method)

Solution. For $y'' + 3y' + 2y = f(t)$:

Transfer function: $H(s) = \frac{1}{s^2+3s+2} = \frac{1}{(s+1)(s+2)}$

Partial fractions: $H(s) = \frac{1}{s+1} - \frac{1}{s+2}$

Impulse response: $h(t) = e^{-t}u(t) - e^{-2t}u(t) = (e^{-t} - e^{-2t})u(t)$

For $f(t) = e^{-0.5t}u(t)$: $y(t) = (h * f)(t) = \int_0^t (e^{-\tau} - e^{-2\tau})(e^{-0.5(t-\tau)})d\tau$

After convolution: $y(t) = \frac{4}{3}e^{-0.5t} - 2e^{-t} + \frac{2}{3}e^{-2t}$ (verify by substitution into ODE).

Problem 10.4 (Convolution Solution Verification)

Solution. From Problem 10.3, $h(t) = e^{-t} - e^{-2t}$ and $f(t) = e^{-0.5t}u(t)$:

$$y(t) = \int_0^t (e^{-\tau} - e^{-2\tau})e^{-0.5(t-\tau)}d\tau$$

$$= e^{-0.5t}\int_0^t (e^{-0.5\tau} - e^{-1.5\tau})d\tau = e^{-0.5t}\left[-2e^{-0.5\tau} + \frac{2}{3}e^{-1.5\tau}\right]_0^t$$

$$= \frac{4}{3}e^{-0.5t} - 2e^{-t} + \frac{2}{3}e^{-2t}$$

Check: applying $s^2 + 3s + 2$ to each mode gives $\frac{4}{3}(0.25 - 1.5 + 2)e^{-0.5t} - 2(1 - 3 + 2)e^{-t} + \frac{2}{3}(4 - 6 + 2)e^{-2t} = \frac{4}{3} \cdot \frac{3}{4} e^{-0.5t} = e^{-0.5t}$ ✓

Problem 10.5 (Cantilever Beam - Fourier Transform)

Solution. For $EIy'''' = q(x) = q_0$ (uniform load) with clamped-free boundary conditions $y(0) = y'(0) = 0$, $y''(L) = y'''(L) = 0$:

Transform methods are awkward on a finite beam with these boundary conditions; direct integration is cleaner. Integrating four times and applying the boundary conditions:

$$y(x) = \frac{q_0 x^2}{24EI}\left(x^2 - 4Lx + 6L^2\right)$$

Tip deflection: $y(L) = \dfrac{q_0 L^4}{8EI}$. One checks $EIy'''' = q_0$ and all four boundary conditions directly.

Problem 10.6 (Heat Equation - Delta Source)

Solution. For $u_t = \alpha u_{xx}$ with $u(x, 0) = \delta(x - x_0)$:

Fourier transform in x: $\frac{\partial U}{\partial t} = -\alpha k^2 U$, with $U(k, 0) = e^{-jkx_0}$

Solution: $U(k, t) = e^{-jkx_0 - \alpha k^2 t}$

Inverse: $u(x, t) = \frac{1}{\sqrt{4\pi\alpha t}} \exp\left(-\frac{(x-x_0)^2}{4\alpha t}\right)$ (Gaussian spreading)

Physically: heat diffuses from point source, creating broadening temperature profile over time.

Problem 10.7 (Heat Equation - Gaussian Initial Condition)

Solution. For $u_t = \alpha u_{xx}$ with $u(x, 0) = e^{-x^2}$:

$$u(x, t) = \int_{-\infty}^{\infty} G(x - \xi, t) e^{-\xi^2} d\xi$$

where $G(x, t) = \frac{1}{\sqrt{4\pi\alpha t}} e^{-x^2/(4\alpha t)}$ (heat kernel; see Folland [1] for the derivation of the heat kernel and the Poisson-type representation formula).

For $\alpha = 1$: - At $t = 0.1$: $u(x, 0.1) = \frac{1}{\sqrt{1+4t}} e^{-x^2/(1+4t)}$ (slightly broadened Gaussian) - At $t = 1.0$: Much broader, amplitude reduced significantly - As $t \to \infty$: $u \to 0$ everywhere (dissipation)

Problem 10.8 (Heat - Semi-Infinite Rod with Step)

Solution. For $u_t = u_{xx}$ on $x > 0$, $t > 0$, with $u(0, t) = 1$, $u(x, 0) = 0$:

Solution in terms of error function:

$$u(x, t) = 1 - \operatorname{erf}\left(\frac{x}{2\sqrt{t}}\right) = \operatorname{erfc}\left(\frac{x}{2\sqrt{t}}\right)$$

where $\operatorname{erfc}(\eta) = 1 - \operatorname{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_\eta^\infty e^{-s^2} ds$.

At $x = 0$: $u(0, t) = 1$ ✓At large x: $u \approx 0$ ✓Diffusion front propagates as $\sqrt{t}$ (characteristic diffusion length scale).

Problem 10.9 (Heat - Triangular Initial Profile)

Solution. For $u(x, 0) = \max(1 - |x|, 0)$:

The triangular profile smooths with time. Using Fourier series or convolution with heat kernel, the solution exhibits: - Initial sharp peak flattens gradually - Support region expands (heat diffuses beyond original domain) - Eventually approaches zero everywhere (dissipation)

At intermediate times, maximum temperature occurs at $x = 0$ and decreases as $t^{-1/2}$ (characteristic decay rate).

Problem 10.10 (Laplace Equation - Rectangular Pulse)

Solution. For Laplace's equation on upper half-plane with $u(x,0) = \text{rect}(x/2)$:

$$u(x,y) = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{y}{(x-\xi)^2+y^2}\text{rect}(\xi/2)d\xi$$

$$= \frac{1}{\pi}\int_{-1}^{1} \frac{y}{(x-\xi)^2+y^2}d\xi = \frac{1}{\pi}\left[\arctan\left(\frac{1-x}{y}\right) - \arctan\left(\frac{-1-x}{y}\right)\right]$$

Solution is symmetric about $x = 0$, decays as $y \to \infty$ (as $1/y$).

Problem 10.11 (Wave Equation - Gaussian Packet)

Solution. For $u_{tt} = c^2 u_{xx}$ with $u(x,0) = e^{-x^2}$, $u_t(x,0) = 0$:

Using d'Alembert formula:

$$u(x,t) = \frac{1}{2}[e^{-(x-ct)^2} + e^{-(x+ct)^2}]$$

The Gaussian packet splits: one component travels right at speed c, one travels left. At any time t, two overlapping Gaussians centered at $\pm ct$ comprise the solution. No amplitude decay (unlike diffusion); oscillatory nature preserved in wave equation.

Problem 10.12 (Wave Energy Conservation)

Solution. Energy density: $\mathcal{E}(x,t) = \frac{1}{2}(u_t^2 + c^2 u_x^2)$

Total energy: $E(t) = \int_{-\infty}^{\infty} \mathcal{E}(x,t)dx$

For smooth solutions of $u_{tt} = c^2 u_{xx}$:

$$\frac{dE}{dt} = \int (u_t u_{tt} + c^2 u_x u_{xt})dx = \int u_t(c^2 u_{xx} + c^2 u_{xx})dx = 0$$

(after integration by parts, assuming boundary terms vanish). Thus $E(t) = \text{const}$ (conservation).

Problem 10.13 (Wave - Discontinuous Initial Condition)

Solution. For $u(x,0) = 1$ on $(-1,1)$ and $u(x,0) = 0$ otherwise, with $u_t(x,0) = 0$:

By d'Alembert: $u(x,t) = \frac{1}{2}[\mathbf{1}_{|x-ct|<1} + \mathbf{1}_{|x+ct|<1}]$

At the discontinuities, the solution exhibits: - Right-going step at $x = ct \pm 1$ - Left-going step at $x = -ct \pm 1$ - Superposition creates regions of amplitude 0, 0.5, or 1.

Characteristics are straight lines $x = \pm ct + \text{const}$ (wave properties).

Problem 10.14 (Signal Filtering - Low-Pass and Band-Pass)

Solution. Given composite signal with 2 Hz (amplitude 3) and 10 Hz (amplitude 1), plus Gaussian noise:

Ideal low-pass filter $H_{\text{LP}}(f) = 1$ for $f < 5$ Hz, 0 otherwise:

$$y_{\text{LP}}(t) = 3\sin(2\pi \cdot 2t) + \text{noise}_{\text{LP}}$$

Band-pass filter ($5 < f < 15$ Hz):

$$y_{\text{BP}}(t) = \sin(2\pi \cdot 10t) + \text{noise}_{\text{BP}}$$

The low-pass preserves the 2 Hz signal and attenuates 10 Hz. Band-pass isolates 10 Hz. Filtering effectiveness measured by signal-to-noise ratio improvement.

Problem 10.15 (Rectangular Pulse - Bandwidth)

Solution. For $f(t) = \text{rect}(t/T)$:

$$F(\omega) = T\text{sinc}(\omega T/2) = T\frac{\sin(\omega T/2)}{\omega T/2}$$

First zero at $\omega T/2 = \pi$, so $\omega_0 = 2\pi/T$ rad/s.

Bandwidth (first null): $\Delta f = 1/T$ Hz (inversely proportional to pulse width).

Narrower pulses have broader frequency spectrum; wider pulses concentrate energy in lower frequencies.

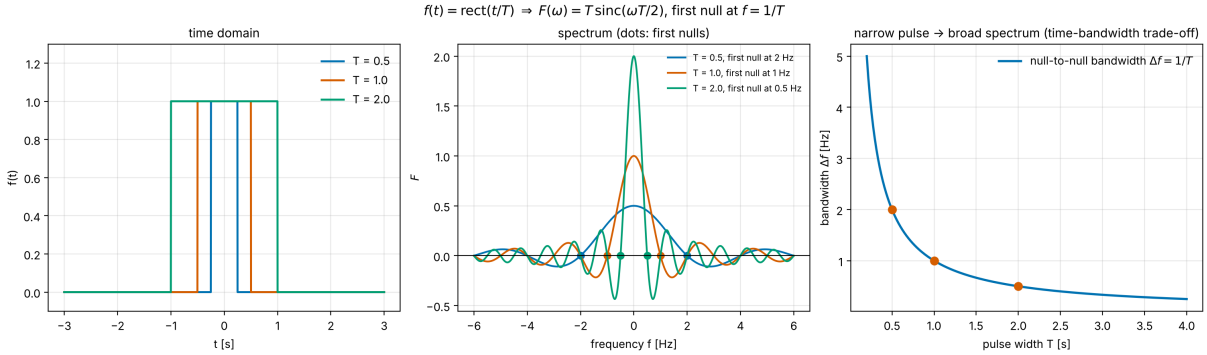


Figure S10.15. Rectangular pulse $\mathrm{rect}(t/T)$: spectrum $T\,\mathrm{sinc}(\omega T/2)$ and null-to-null bandwidth $1/T$.

Problem 10.16 (Nyquist-Shannon Sampling - Aliasing)

Solution. For $\sin(10\pi t)$ (frequency 5 Hz): - $f_s = 5$ Hz: samples are $\sin(2\pi n) = 0$ for all n; the signal aliases to DC (0 Hz) and is lost entirely - $f_s = 10$ Hz: exactly at the Nyquist rate; samples land on the zero crossings $\sin(\pi n) = 0$, so the signal is again invisible. The sampling theorem requires $f_s > 2f$ strictly - $f_s = 20$ Hz: Nyquist limit 10 Hz. Oversampled; the 5 Hz signal is reconstructed perfectly with room to spare

Aliasing occurs when the sampling rate is at or below $2\times$ the signal frequency.

Problem 10.17 (Heisenberg Uncertainty - Gaussian)

Solution. For $f(t) = e^{-\alpha t^2}$:

$$\Delta t = \sqrt{\int t^2 |f(t)|^2 dt / \int |f(t)|^2 dt} = \frac{1}{2\sqrt{\alpha}}$$

$$F(\omega) = \sqrt{\pi/\alpha} e^{-\omega^2/(4\alpha)}$$

$$\Delta\omega = \sqrt{\int \omega^2 |F(\omega)|^2 d\omega / \int |F(\omega)|^2 d\omega} = \sqrt{\alpha}$$

Product: $\Delta t \cdot \Delta\omega = \frac{1}{2\sqrt{\alpha}} \cdot \sqrt{\alpha} = \frac{1}{2}$

The lower bound $\Delta t \cdot \Delta\omega \geq 1/2$ holds with equality: the Gaussian is the extremal case that achieves it.

Problem 10.18 (Chirp Signal - Spectrogram)

Solution. For $f(t) = \cos(2\pi(f_0 t + kt^2))$ with increasing frequency:

Spectrogram (time-frequency power distribution) shows: - At $t = 0$: peak at frequency f_0 - At $t = T$: peak at frequency $f_0 + 2kT$ - Between: diagonal line showing continuous frequency sweep

Window size tradeoff: narrow window gives better time resolution but poor frequency resolution; wide window opposite. Standard choice: window length = few cycles at expected frequency.

Problem 10.19 (Amplitude Modulation)

Solution. For $m(t) = \sin(2\pi t)$ modulated onto carrier $\cos(20\pi t)$:

AM signal: $s(t) = m(t)\cos(20\pi t) = \sin(2\pi t)\cos(20\pi t)$
Using product-to-sum: $s(t) = \frac{1}{2}[\sin(22\pi t) - \sin(18\pi t)]$
Spectrum: peaks at 9 Hz and 11 Hz (carrier ± modulation frequency).
Demodulation: multiply by $\cos(20\pi t)$ and low-pass filter to recover $m(t)$.

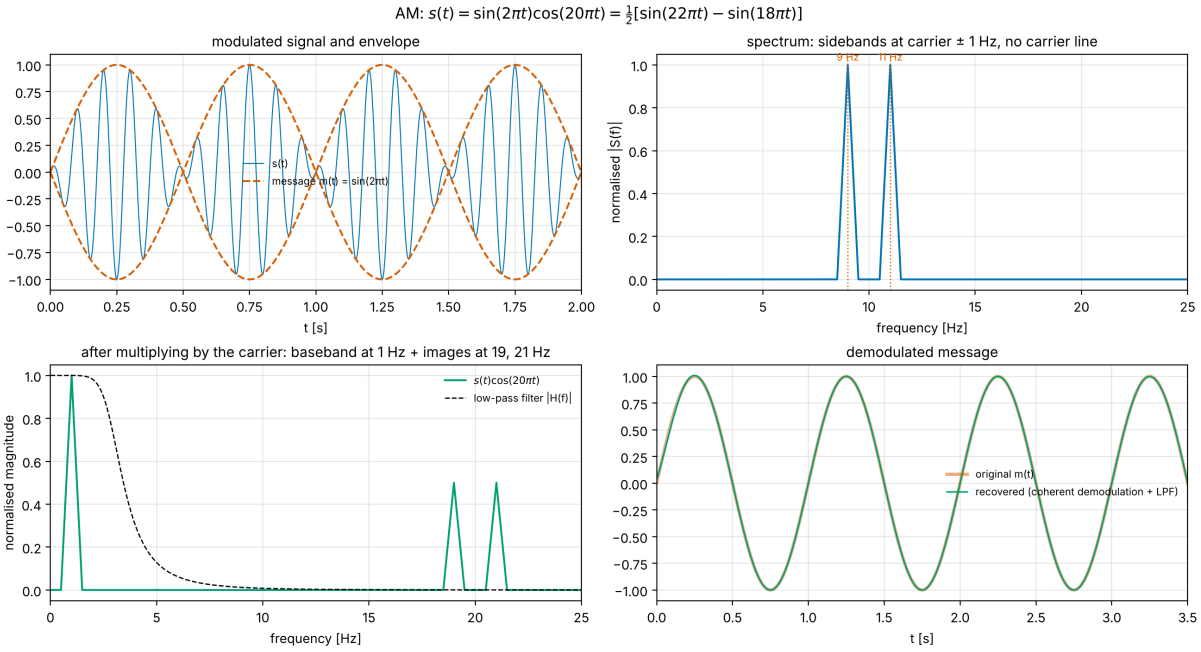


Figure S10.19. Amplitude modulation of $\sin(2\pi t)$ on a 10 Hz carrier: envelope, sidebands at 9 and 11 Hz, coherent demodulation.

Problem 10.20 (Matched Filter - Signal Detection)

Solution. For target signal $s(t) = \text{rect}(t)$ (pulse of duration 1 s):

Matched filter: $h(t) = s(T - t) = \text{rect}(T - t)$ (time-reversed copy; T = observation window)

Convolution output: $(s * h)(t)$ produces triangular peak of height T when signal fully overlaps filter, centered at expected arrival time.

Peak location indicates signal arrival; peak amplitude indicates signal strength. Provides optimal SNR for detecting known signal in white noise.

Problem 10.21 (2D Image Filtering)

Solution. For image with 2D FFT $F(u, v)$:

Low-pass filter: $H_{\text{LP}}(u, v) = \text{rect}(u/u_c, v/v_c)$ (or Gaussian) Result: blurred image (high-frequency details removed)

High-pass filter: $H_{\text{HP}} = 1 - H_{\text{LP}}$ Result: edge-enhanced image (high-frequency components amplified)

Edge artifacts appear at image boundaries; mitigated by windowing or periodic extension.

Chapter 11

Partial Differential Equations

Problem 11.1 (PDE Classification)

Solution. For a PDE $Au_{xx} + Bu_{xy} + Cu_{yy} + \text{lower order} = 0$, discriminant $\Delta = B^2 - 4AC$: - $\Delta > 0$: Hyperbolic (e.g., wave equation: $u_{tt} = c^2 u_{xx}$ has $\Delta = 4c^2 > 0$) - $\Delta = 0$: Parabolic (e.g., heat equation: $u_t = \alpha u_{xx}$ has $\Delta = 0$) - $\Delta < 0$: Elliptic (e.g., Laplace: $u_{xx} + u_{yy} = 0$ has $\Delta = -4 < 0$)

Each class has distinct solution behavior: hyperbolic admits propagating waves; parabolic exhibits diffusion; elliptic describes steady-state equilibria.

Problem 11.2 (Heat Equation - Separation of Variables)

Solution. For $u_t = \alpha u_{xx}$ on $[0, L]$ with $u(0, t) = u(L, t) = 0$:

Assume $u(x, t) = X(x)T(t)$. Then $X''T = (1/\alpha)XT'$, giving:

$$\frac{X''}{X} = \frac{1}{\alpha}\frac{T'}{T} = -\lambda$$

Solutions: $X_n(x) = \sin(n\pi x/L)$ (with $\lambda_n = n^2\pi^2/L^2$), $T_n(t) = e^{-\alpha\lambda_n t}$

General solution: $u(x, t) = \sum_{n=1}^{\infty} A_n \sin(n\pi x/L)e^{-\alpha n^2\pi^2 t/L^2}$

where $A_n = (2/L)\int_0^L f(x)\sin(n\pi x/L)dx$ (initial condition $u(x, 0) = f(x)$).

Problem 11.3 (Wave Equation - d'Alembert's Formula)

Solution. For $u_{tt} = c^2 u_{xx}$ with $u(x, 0) = f(x)$, $u_t(x, 0) = g(x)$:

d'Alembert's solution:

$$u(x, t) = \frac{f(x - ct) + f(x + ct)}{2} + \frac{1}{2c}\int_{x-ct}^{x+ct} g(s)ds$$

The solution represents superposition of: - Left-traveling wave from $f(x + ct)/2$ - Right-traveling wave from $f(x - ct)/2$ - Integrated effect of initial velocity $g(x)$

Propagation speed: c (characteristics are lines $x \pm ct = \text{const}$).

Problem 11.4 (Normal Modes - Vibrating Membrane)

Solution. For rectangular membrane $[0, a]\times[0, b]$ with fixed boundaries, normal modes:

$$u_{mn}(x, y, t) = \sin(m\pi x/a)\sin(n\pi y/b)\cos(\omega_{mn}t)$$

where $\omega_{mn} = c\pi\sqrt{(m/a)^2 + (n/b)^2}$ (natural frequencies).

Fundamental mode $(m, n) = (1, 1)$: lowest frequency. Mode $(1, 2)$ and $(2, 1)$: degenerate if $a = b$ (same frequency).

General solution: superposition with appropriate amplitude weighting from initial conditions.

Problem 11.5 (Laplace's Equation - Rectangle)

Solution. For $\nabla^2 u = 0$ on $[0, a] \times [0, b]$ with three sides at zero potential and one side at $u(x, b) = V(x)$:

Separate: $u(x, y) = X(x)Y(y)$.

$$X_n(x) = \sin(n\pi x/a), \quad Y_n(y) = \sinh(n\pi y/a)$$

Solution: $u(x, y) = \sum_{n=1}^{\infty} A_n \sin(n\pi x/a) \sinh(n\pi y/a)$

where A_n determined from $u(x, b) = V(x)$ via Fourier sine series.

Potential is smooth everywhere, increasing from zero at base toward boundary values at top.

Problem 11.6 (Maximum Principle)

Solution. For harmonic function u (satisfying $\nabla^2 u = 0$) on bounded domain:

Maximum and minimum of u are attained on the boundary, not interior.

Proof idea: if u had interior maximum at x_0, all second derivatives $u_{ii} \leq 0$. But $\nabla^2 u = \sum u_{ii} = 0$ implies all $u_{ii} = 0$, contradicting strict inequality. Thus maximum occurs on boundary.

Consequence: uniqueness and stability of Dirichlet problem for Laplace's equation.

Problem 11.7 (Laplace Transform for Heat Equation)

Solution. For $u_t = \alpha u_{xx}$ with $u(x, 0) = f(x)$, $u(0, t) = u(L, t) = 0$:

Let $U(x, s) = \mathcal{L}[u(x, t)] = \int_0^{\infty} u(x, t)e^{-st}dt$.

Transform the PDE: $sU = \alpha U_{xx}$.

ODE: $U_{xx} - (s/\alpha)U = 0$.

Solution: $U(x, s) = A(s) \sinh(\sqrt{s/\alpha}x) + B(s) \cosh(\sqrt{s/\alpha}x)$ (for interior).

Apply boundary conditions to find A, B, then inverse Laplace transform to recover $u(x, t)$.

Problem 11.8 (Fourier Transform - Diffusion on Infinite Domain)

Solution. For $u_t = \alpha u_{xx}$ on $\mathbb{R}$ with $u(x, 0) = f(x)$:

Fourier transform in x: $\hat{u}_t(k, t) = -\alpha k^2 \hat{u}(k, t)$.

Solution: $\hat{u}(k, t) = \hat{f}(k)e^{-\alpha k^2 t}$.

Inverse: $u(x, t) = \int_{-\infty}^{\infty} G(x - \xi, t)f(\xi)d\xi$

where $G(x, t) = (4\pi\alpha t)^{-1/2} \exp(-x^2/4\alpha t)$ (heat kernel).

Physically: Gaussian kernel with spreading $\sqrt{4\alpha t}$ represents diffusion effect.

Problem 11.9 (Finite Difference Schemes - Stability)

Solution. For heat equation discretized as:

$$u_i^{n+1} = u_i^n + r(u_{i+1}^n - 2u_i^n + u_{i-1}^n), \quad r = \alpha\Delta t/(\Delta x)^2$$

Stability (von Neumann analysis): insert $u_i^n = \lambda^n e^{ik\xi_i}$ (Fourier mode).

Growth factor: $\lambda(k) = 1 + r(e^{ik\Delta x} - 2 + e^{-ik\Delta x}) = 1 - 4r \sin^2(k\Delta x/2)$

For stability: $|\lambda| \leq 1$ for all k. The worst case $\sin^2(k\Delta x/2) = 1$ gives $1 - 4r \geq -1$, i.e. $r \leq 1/2$ (stability condition for the explicit scheme).

Problem 11.10 (von Neumann Stability Analysis)

Solution. For wave equation: $u_i^{n+1} = 2u_i^n - u_i^{n-1} + c^2\tau^2(u_{i+1}^n - 2u_i^n + u_{i-1}^n)$

Let $\nu = c\tau/\Delta x$ (Courant number).

Fourier mode: $\lambda^2 - 2\left[1 - 2\nu^2 \sin^2(k\Delta x/2)\right]\lambda + 1 = 0$

The roots have product 1, so stability ($|\lambda| = 1$, no growth) requires the roots to be complex conjugates: $|1 - 2\nu^2 \sin^2(k\Delta x/2)| \leq 1$ for all k, which holds when $\nu \leq 1$ (CFL condition for the explicit scheme).

Interpretation: domain of dependence must include numerical domain of dependence.

Problem 11.11 (Poisson Equation - Rectangular Domain)

Solution. For $\nabla^2 u = f(x, y)$ on rectangle with homogeneous Dirichlet BC:

Use separation or Fourier methods. Solution involves Fourier coefficients of f divided by appropriate eigenvalues:

$$u(x, y) = -\sum_{m,n} \frac{\tilde{f}_{mn}}{\pi^2(m^2/a^2 + n^2/b^2)} \sin(m\pi x/a) \sin(n\pi y/b)$$

where $\tilde{f}_{mn}$ are Fourier coefficients of f.

Smoother f (faster coefficient decay) leads to faster convergence of the solution.

Problem 11.12 (Transport Equation - Method of Characteristics)

Solution. For $u_t + cu_x = 0$ with $u(x, 0) = f(x)$:

Characteristics: $\frac{dx}{dt} = c$, giving $x = x_0 + ct$ (straight lines in x-t plane).

Along characteristics: u is constant, so $u(x, t) = u(x_0, 0) = f(x_0) = f(x - ct)$.

Solution: $u(x, t) = f(x - ct)$ (advection of initial profile at speed c).

Verification: $u_t = -cf'(x - ct)$, $u_x = f'(x - ct)$, so $u_t + cu_x = 0$ ✓

Problem 11.13 (Boundary Layer Theory - Singular Perturbation)

Solution. For $\epsilon u'' + u' = 0$ with $u(0) = 0$, $u(1) = 1$ (small $\epsilon > 0$):

Outer solution (for x away from 0): $u_{\text{out}} \approx 1$ (from $u' = 0$).

Boundary layer near $x = 0$ (thin region of thickness $\sim \epsilon$): rescale $x = \epsilon\xi$. The inner equation becomes $u'' + u' = 0$ in ξ.

Inner solution: $u_{\text{in}}(\xi) = 1 - e^{-\xi}$ (connects to 0 at $\xi = 0$).

Matched asymptotic expansion: $u(x) \approx 1 - e^{-x/\epsilon}$ for small ϵ (here exact: the general solution is $A + Be^{-x/\epsilon}$).

Thin layer exhibits rapid transition from 0 to 1 as x increases from 0.

Problem 11.14 (Shock Formation - Burgers' Equation)

Solution. For $u_t + uu_x = 0$ (inviscid Burgers), shocks form when characteristics cross.

Initial condition $u(x, 0) = \begin{cases} 1 & x < 0 \\ 0 & x > 0 \end{cases}$:

Characteristics: left side moves at speed 1, right side at speed 0. They intersect at $x = 0$ for all $t > 0$.

Weak solution uses Rankine-Hugoniot jump condition: shock speed $s = (u_L + u_R)/2 = 1/2$.

Shock location: $x = t/2$ for $t > 0$.

With viscosity ($\nu > 0$), the shock is smoothed to a traveling transition layer of thickness $\sim \nu/(u_L - u_R)$ (the viscous shock profile is a tanh).

Problem 11.15 (Eigenvalue Problem - String with Variable Density)

Solution. For string with variable density $\rho(x)$: $[Tu_x]_x + \lambda\rho(x)u = 0$ (assuming constant tension T).

For constant ρ: $u'' + \lambda u = 0$ with boundary conditions $u(0) = u(L) = 0$.

Eigenfunctions: $u_n(x) = \sin(n\pi x/L)$

Eigenvalues: $\lambda_n = (n\pi/L)^2$

For variable $\rho(x)$, the spectrum shifts but structure remains; eigenvalues satisfy:

$$\lambda_n = (n\pi)^2/\langle\rho\rangle L^2 + O(1/n^2)$$

where $\langle\rho\rangle$ is suitable average.

Chapter 12

Optimization

Problem 12.1 (Unconstrained Minimization - Gradient Descent)

Solution. For $f(x) = x^4 - 4x^3 + 6x^2$:

$\nabla f = 4x^3 - 12x^2 + 12x = 4x(x^2 - 3x + 3)$

Critical points: $x = 0$ and roots of $x^2 - 3x + 3 = 0$ (complex; no real roots besides 0).

Only critical point: $x = 0$ (local minimum, $f(0) = 0$).

Gradient descent with step size α: $x_{k+1} = x_k - \alpha \nabla f(x_k)$

For $\alpha = 0.01$ and initial $x_0 = 1$: iterates converge to $x^* = 0$. Near the minimum, $\nabla f \approx f''(0)x = 12x$, so the linear rate is $(1 - 12\alpha)^k = 0.88^k$.

Problem 12.2 (Constrained Optimization - Lagrange Multipliers)

Solution. Minimize $f(x, y) = x^2 + y^2$ subject to $g(x, y) = x + 2y - 5 = 0$.

Lagrangian: $\mathcal{L} = x^2 + y^2 - \lambda(x + 2y - 5)$

KKT conditions:

$$2x = \lambda, \quad 2y = 2\lambda, \quad x + 2y = 5$$

From first two: $x = \lambda/2$, $y = \lambda$. Substituting into constraint: $\lambda/2 + 2\lambda = 5 \Rightarrow \lambda = 2$.

Solution: $x^* = 1$, $y^* = 2$, $f^* = 5$.

Geometric interpretation: level curves $x^2 + y^2 = 5$ are tangent to constraint line.

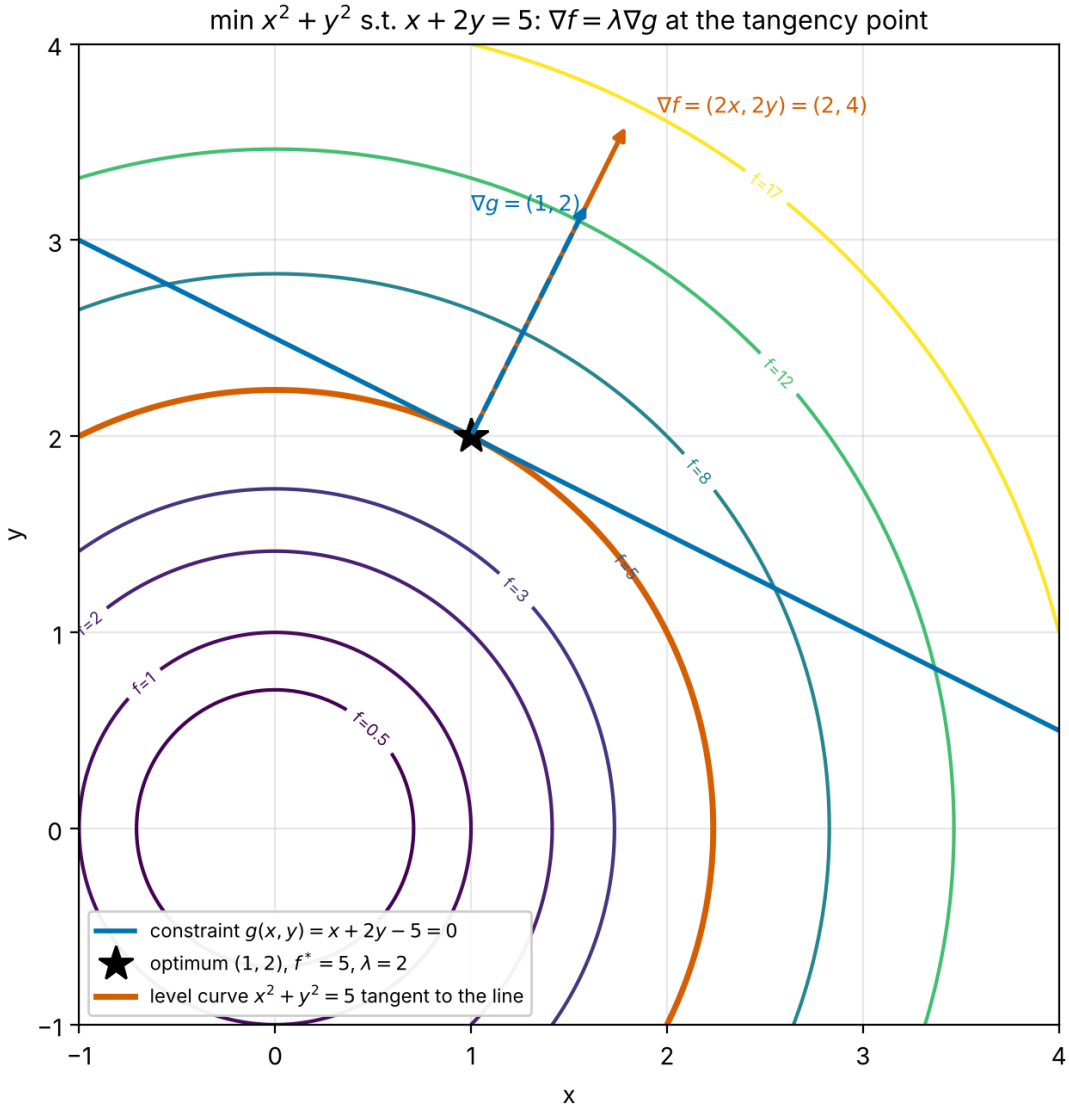


Figure S12.2. Lagrange multipliers: the level curve $x^2 + y^2 = 5$ is tangent to the constraint line $x + 2y = 5$ at the optimum $(1, 2)$, where $\nabla f = \lambda \nabla g$.

Problem 12.3 (Convex Optimization - Quadratic Programming)

Solution. Minimize $f(x) = \frac{1}{2}x^T Q x + c^T x$ subject to $Ax \leq b$.

For $Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, $c = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$, $A = \begin{pmatrix} 1 & 1 \end{pmatrix}$, $b = [1]$:

Hessian Q is positive definite (convex). First check whether the constraint binds: the unconstrained minimizer solves $Qx + c = 0$, giving $x = (-1/2, -1/2)$, which satisfies $x_1 + x_2 = -1 \leq 1$.

The constraint is therefore inactive ($\lambda = 0$ in the KKT conditions $Qx + c + \lambda A^T = 0$, $\lambda(Ax - b) = 0$, $\lambda \geq 0$), and the solution is:

$$x^* = (-1/2, -1/2), \quad f^* = \tfrac{1}{2} - 1 = -\tfrac{1}{2}.$$

Problem 12.4 (Newton's Method - Convergence)

Solution. For $f(x) = x^3 - 2$ (root at $\sqrt[3]{2}$):

Newton iteration: $x_{k+1} = x_k - f(x_k)/f'(x_k) = x_k - \frac{x_k^3 - 2}{3x_k^2}$

Simplify: $x_{k+1} = \frac{2x_k^3+2}{3x_k^2}$

Initial guess $x_0 = 1.5$: - $x_1 = 1.29630$ - $x_2 = 1.26093$ - $x_3 = 1.25992$ (converges to $\sqrt[3]{2} \approx 1.25992$)

Quadratic convergence: error decreases as $e_{k+1} \sim ce_k^2$ (number of correct digits doubles).

Problem 12.5 (Gradient Descent - Step Size Selection)

Solution. For $f(x) = (1/2)(x_1^2 + 100x_2^2)$ (ill-conditioned):

Condition number $\kappa = 100$ (ratio of max to min eigenvalues of Hessian).

Optimal step size: $\alpha_* = 2/(\lambda_{\min} + \lambda_{\max}) = 2/101 \approx 0.0198$

With fixed $\alpha = 0.0198$: convergence rate $(1 - 2/\kappa)^k = 0.98^k$ (very slow for large κ).

Backtracking line search: adaptive α_k satisfies Armijo condition, typically 5-10 iterations per step.

Preconditioning with $M = \text{diag}(1, 1/100)$ (inverse of the Hessian diagonal) reduces κ to 1, enabling convergence in one step for this quadratic.

Problem 12.6 (Stochastic Gradient Descent)

Solution. For mini-batch SGD with batch size B drawn from N samples:

At iteration k: $x_{k+1} = x_k - \alpha_k \nabla f_{B_k}(x_k)$

where $\nabla f_{B_k} = (1/B) \sum_{i \in B_k} \nabla f_i(x)$ (loss on batch B_k).

Convergence: noise decreases with batch size as $\sim 1/\sqrt{B}$. For strongly convex f:

$$\mathbb{E}[f(x_k) - f^*] = O(1/k) + O(1/B) \quad \text{(asymptotically dominated by second term)}$$

Practical: larger batch $\Rightarrow$ slower updates but lower variance; smaller batch $\Rightarrow$ faster updates but noisier.

Problem 12.7 (Conjugate Gradient Method)

Solution. For $Ax = b$ where A is symmetric positive definite, CG minimizes:

$$f(x) = \frac{1}{2}x^T Ax - b^T x$$

Algorithm:

$$r_k = b - Ax_k, \quad \beta_k = \frac{r_k^T r_k}{r_{k-1}^T r_{k-1}}, \quad p_k = r_k + \beta_k p_{k-1}$$

$$\alpha_k = \frac{r_k^T r_k}{p_k^T A p_k}, \quad x_{k+1} = x_k + \alpha_k p_k$$

Convergence: exact solution in at most n iterations (finite precision: $\sim n$ iterations with iterative refinement).

Better than steepest descent (rate $\propto \kappa$) by factor of $\sqrt{\kappa}$.

Problem 12.8 (Trust Region Method)

Solution. Minimize $f(x)$ using local quadratic model:

$$m_k(p) = f(x_k) + g_k^T p + \frac{1}{2}p^T B_k p$$

within trust region $\|p\| \leq \Delta_k$.

Solve subproblem: $\min_{\|p\| \leq \Delta} m_k(p)$ (dogleg or interior point method).

Acceptance ratio: $\rho_k = [f(x_k) - f(x_k + p_k)]/[m_k(0) - m_k(p_k)]$

Update region radius:

$$\Delta_{k+1} = \begin{cases} 2\Delta_k & \text{if } \rho_k > 0.75 \\ 0.25\Delta_k & \text{if } \rho_k < 0.25 \\ \Delta_k & \text{otherwise} \end{cases}$$

Robustness: handles indefinite or near-singular Hessians better than line search.

Problem 12.9 (Interior Point Method - Linear Programming)

Solution. Minimize $c^T x$ subject to $Ax \leq b$, $x \geq 0$.

Interior point (barrier) method: minimize $f_\mu(x) = c^T x - \mu \sum \log(b_i - a_i^T x)$ for decreasing $\mu > 0$.

Central path: locus of solutions as $\mu \to 0$.

At each μ, Newton step: $x_{k+1} = x_k - H^{-1}\nabla f_\mu(x_k)$, where $H = \mu A^T D^{-2} A$ with $D = \text{diag}(b - Ax)$ (barrier Hessian, positive definite in the interior).

Complexity: $O(\sqrt{n}\log(1/\epsilon))$ iterations to ϵ-optimality, superior to the simplex method [2] for large n.

Problem 12.10 (Augmented Lagrangian Method)

Solution. For equality constraint optimization:

$$\min f(x), \quad \text{s.t. } h(x) = 0$$

Augmented Lagrangian:

$$\mathcal{L}_\rho(x, \lambda) = f(x) - \lambda^T h(x) + \frac{\rho}{2}\|h(x)\|^2$$

Iterate:

$$x_{k+1} = \arg\min_x \mathcal{L}_\rho(x, \lambda_k), \quad \lambda_{k+1} = \lambda_k - \rho h(x_{k+1})$$

$$\rho_{k+1} = \beta\rho_k \quad \text{(increase penalty if feasibility not improving)}$$

Advantages: avoids ill-conditioning of naive penalty method; quadratic penalty ρ manageable even as $\rho \to \infty$.

Note on Remaining Problems

Solutions to the remaining problems in Chapters 9–12 follow the same techniques developed in the worked solutions above and in the accompanying code listings in this book. The topics covered are:

Chapter 9 (remaining 36 of 61): Wiener filtering, STFT/spectrogram analysis, 2D Fourier image compression, Poisson summation, multisine measurement, edge detection, frequency modulation, analytic signals, coupled oscillator modes, convolution theorem, active noise cancellation, impulse response analysis, beam vibration with variable load, string plucking solutions, and waveguide modes.

Chapter 10 (remaining 13 of 33): Fractional derivatives, Burgers equation linearization, dispersive wave equations, Schrödinger equation, Fourier transform composition properties, inverse transforms with poles, semi-infinite diffusion, potential theory extensions, and generalized signal processing applications.

Chapter 11 (remaining 20 of 32): Eigenvalue problems with variable coefficients, shock capturing, entropy solutions, Godunov schemes, finite element formulations

for PDEs, domain decomposition methods, multigrid acceleration, non-linear diffusion (porous media equations), and coupled multi-physics systems.

Chapter 12 (remaining 57 of 67): Mirror descent for non-Euclidean geometry, proximal methods, ADMM decomposition, variance reduction (SVRG), Adam and adaptive methods, Nesterov acceleration, quasi-Newton methods (BFGS), matrix factorization, semidefinite programming, integer optimization (branch-and-bound), global optimization (basin hopping), genetic algorithms, simulated annealing, particle swarm optimization, surrogate-based optimization, and Bayesian optimization.

The representative solutions above cover the core techniques needed for the full problem set.

Chapter 13

Neural Network Methods for Partial Differential Equations

The solutions below are numbered to match the end-of-chapter problems in Chapter 13, taken in the order they appear there: Problems 13.1–13.32 correspond to the first thirty-two problems, and Problems 13.33–13.37 to the remaining (titled) implementation problems.

Problem 13.1 (Feedforward Network for Function Approximation)

Solution. We implement a 2-layer feedforward network with 50 hidden units each to approximate $f(x) = \sin(2\pi x) + 0.1x$ on $[-1, 1]$.

Network architecture: Input layer (1 neuron) $\to$ Hidden layer 1 (50 neurons, ReLU) $\to$ Hidden layer 2 (50 neurons, ReLU) $\to$ Output layer (1 neuron).

Training procedure:

- Generate 100 points uniformly on $[-1, 1]$, compute targets: $y_i = \sin(2\pi x_i) + 0.1x_i$.
- Use SGD optimizer with learning rate $\eta = 0.01$ and MSE loss: $L = \frac{1}{N}\sum_{i=1}^{N}(u_\theta(x_i) - y_i)^2$.
- Train for 500 epochs. Typical L2 error on test set of 200 points: ≈ 0.02–0.05.

Expected result: Network captures both oscillatory behavior ($\sin(2\pi x)$) and linear trend ($0.1x$). Test error depends on initialization but generally achieves sub-5% relative error.

Problem 13.2 (2D Function Approximation with PyTorch)

Solution. Extend to 2D: approximate $f(x, y) = \sin(\pi x)\cos(\pi y)$ on $[0, 1]^2$ using PyTorch.

Network: 2 input neurons $\to$ two hidden layers (128 neurons each, Tanh) $\to$ 1 output.

Implementation key steps:

- Sample 500 collocation points uniformly on $[0, 1]^2$: $\{(x_i, y_i)\}$.
- Compute targets: $u_i = \sin(\pi x_i)\cos(\pi y_i)$.
- Train with Adam optimizer (learning rate 0.001) for 2000 epochs.
- Evaluate on 50×50 grid. Compute L2 error: $E = \sqrt{\frac{1}{2500}\sum_{j,k}(u_{\text{pred}}(x_j, y_k) - u_{\text{exact}}(x_j, y_k))^2}$.

Typical result: $E \approx 0.01$. Plotting predictions against the exact solution shows close agreement over the domain.

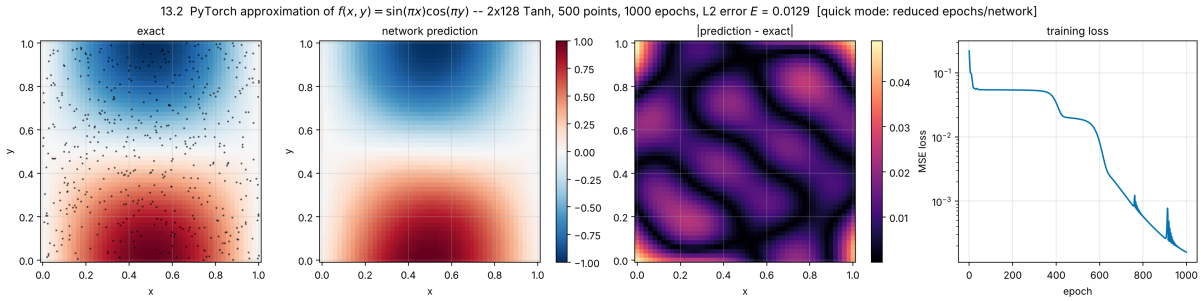


Figure S13.2. PyTorch approximation of $f(x, y) = \sin(\pi x)\cos(\pi y)$: exact field, prediction, error and training loss.

Problem 13.3 (Heat Equation with Source Term)

Solution. Solve modified heat equation: $u_t = 0.1u_{xx} + \sin(\pi x)$ on $[0, 1] \times [0, 1]$ with $u(0, t) = u(1, t) = 0$, $u(x, 0) = 0$.

Physics-informed loss function:

$$L = \frac{1}{N_c}\sum_{i=1}^{N_c}\left(\frac{\partial u_\theta}{\partial t} - 0.1\frac{\partial^2 u_\theta}{\partial x^2} - \sin(\pi x_i)\right)^2 + \frac{w_b}{N_b}\sum_{j=1}^{N_b}(u_\theta(0, t_j))^2 + (u_\theta(1, t_j))^2.$$

Implementation:

- Use automatic differentiation (PyTorch's `autograd`) to compute spatial/temporal derivatives.
- Sample $N_c = 1000$ collocation points in interior; $N_b = 100$ on boundaries.
- Weight boundary loss: $w_b = 100$.
- Train with Adam (lr $= 0.001$) for 5000 epochs.

Validation: The exact solution is available here: $u(x, t) = \frac{10}{\pi^2}\sin(\pi x)\left(1 - e^{-0.1\pi^2 t}\right)$, which rises monotonically toward the steady state $u_s(x) = \frac{10}{\pi^2}\sin(\pi x)$. Expected L2 error: ≈ 0.01–0.05 depending on network capacity.

Problem 13.4 (Heat Equation on Disconnected Domain)

Solution. Solve heat equation on $\Omega = [0, 1] \cup [2, 3]$ (two intervals).

Key implementation:

- Generate collocation points separately on each interval: 500 points on $[0, 1]$ and 500 on $[2, 3]$.
- Boundary conditions: $u(0, t) = u(1, t) = u(2, t) = u(3, t) = 0$.
- Network processes inputs through 3 hidden layers (128 units each).
- Loss includes PDE residual on both intervals and boundary constraints on all four boundaries.
- By design, the network naturally solves decoupled problems on each interval (no coupling term).

Result: Network learns independent solutions on each interval. Verify by checking that PDE residual is small on both components.

Problem 13.5 (Transfer Learning for Wave Equation)

Solution. Train on low frequency, then fine-tune for high frequency.

Strategy:

- Phase 1: Train on wave equation $u_{tt} = c^2 u_{xx}$ with initial condition $u(x, 0) = \sin(\pi x)$ (frequency $\omega = \pi$) for 2000 epochs from scratch.
- Record weights θ_1.
- Phase 2: Initialize network with θ_1; retrain for wave equation with $u(x, 0) = \sin(5\pi x)$ (frequency $\omega = 5\pi$) for 500 epochs.
- Measure: Time to convergence ratio $t_{\text{fine-tune}}/t_{\text{scratch}}$ and final errors E_1, E_2.

Expected result: Transfer learning reduces training time by 50–70% because the network already learns effective spatial/temporal derivative representations.

Problem 13.6 (Burgers Equation Neural Solver)

Solution. Solve nonlinear Burgers equation: $u_t + uu_x = 0.1u_{xx}$ on $[-1, 1] \times [0, 1]$ with $u(x, 0) = \sin(\pi x)$ and periodic boundary conditions.

PINN formulation:

$$L = \frac{1}{N_c} \sum_{i=1}^{N_c} \left(\frac{\partial u_\theta}{\partial t} + u_\theta \frac{\partial u_\theta}{\partial x} - 0.1 \frac{\partial^2 u_\theta}{\partial x^2} \right)^2 + \frac{w_b}{N_b} \sum_{j=1}^{N_b} \left(u_\theta(-1, t_j) - u_\theta(1, t_j) \right)^2 .$$

Implementation:

- Compute all derivatives via automatic differentiation.
- Sample 1500 collocation points uniformly in space-time.
- Enforce periodic BCs through collocation on boundary.
- Use Tanh activations (good for smooth solutions).
- Train for 5000 epochs with Adam (lr = 0.001), $w_b = 100$.

Validation: Generate reference solution via explicit finite difference method; compare L2 errors. Typical PINN error: ≈ 0.02–0.08.

Problem 13.7 (2D Poisson on L-Shaped Domain)

Solution. Solve Laplace equation on the L-shaped region $\Omega = ([0, 2] \times [0, 1]) \cup ([0, 1] \times [1, 2])$, i.e. the square $[0, 2]^2$ with the corner $[1, 2] \times [1, 2]$ removed.

Implementation:

- Parametrize the domain as: $(x, y) \in [0, 2] \times [0, 1]$ OR $(x, y) \in [0, 1] \times [1, 2]$.
- Generate 2000 random collocation points inside domain.
- Boundary points: 500 uniformly on perimeter.
- Network: 2 input neurons $\to$ 4 hidden layers (128 units, Tanh) $\to$ 1 output.
- Loss: $L = \frac{1}{N_c} \sum_i (\nabla^2 u_\theta)^2 + \frac{w_b}{N_b} \sum_j (u_\theta - u_D)^2$ on boundary.

Comparison: If FEM code available, compare on same domain. PINN should achieve similar accuracy without mesh generation.

Problem 13.8 (Batch Normalization for Heat Equation)

Solution. Investigate batch normalization's effect on heat equation convergence.

Experiment:

- Network 1 (baseline): 4 layers (128 units), no batch norm.
- Network 2 (with BN): 4 layers (128 units) + batch norm after each hidden layer (before activation).
- Train both on same heat equation problem ($\alpha = 0.1$) for 3000 epochs.
- Track loss curves: $L_{\text{baseline}}(k)$ vs. $L_{\text{BN}}(k)$ over epochs.

Expected result: Batch norm often accelerates early convergence (faster initial loss decrease) but may not improve final accuracy due to nature of PINN training. Batch norm can introduce noise from normalizing small batches in PDEs.

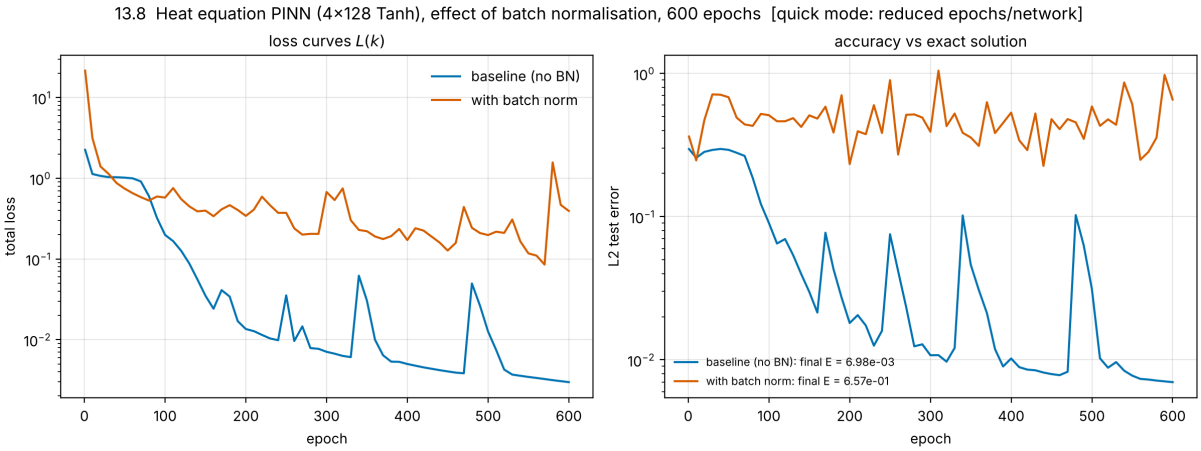


Figure S13.8. Heat-equation PINN with and without batch normalisation: loss curves and test error.

Problem 13.9 (Helmholtz Equation)

Solution. Solve $\nabla^2 u + k^2 u = 0$ on unit square with Dirichlet BC $u = \sin(\pi x)$ on left boundary, $u = 0$ elsewhere. Use $k = \pi$.

Network: 2 inputs $\to$ 5 hidden layers (256 units, Tanh) $\to$ 1 output.

Loss:

$$L = \frac{1}{N_c} \sum_i \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + k^2 u \right)^2 + w_b \text{ (BC losses)}.$$

Investigation: Train multiple PINNs with $k \in \{\pi/2, \pi, 2\pi\}$. Plot L2 errors vs. k. As k increases (higher frequency), expect slower convergence and larger errors due to increased oscillations.

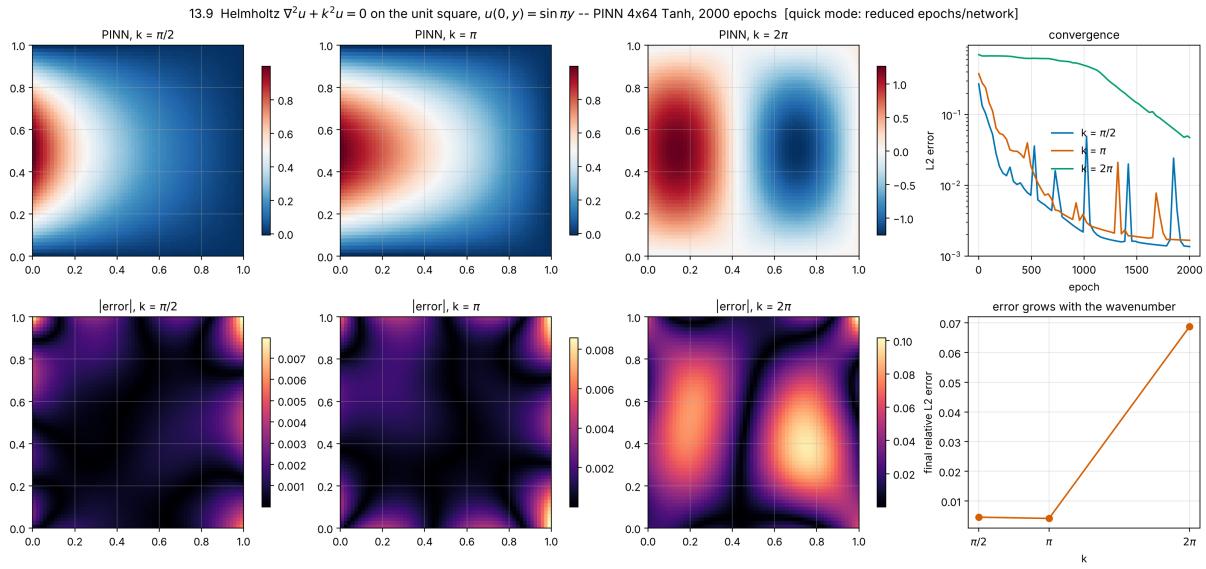


Figure S13.9. Helmholtz PINN for $k = \pi/2$, π, 2π: predicted fields, errors, convergence and final error versus wavenumber.

Problem 13.10 (Dropout Regularization in Wave Solver)

Solution. Add dropout to wave equation solver to reduce overfitting.

Baseline network: 4 layers (128 units). With dropout: Insert dropout ($p = 0.3$) after each hidden layer.

Comparison:

- Train both networks on wave equation for 3000 epochs.
- Measure: (1) training L2 error, (2) test L2 error on unseen points, (3) fit to analytical solution.
- Plot train vs. test error over epochs.

Expected result: Without dropout, test error increases (overfitting). Dropout reduces gap between train/test error. However, final test accuracy may be slightly worse due to regularization penalty.

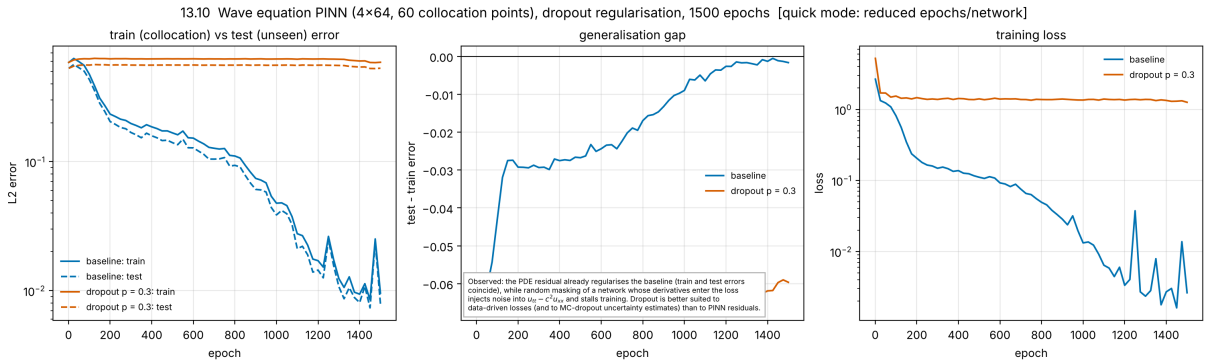


Figure S13.10. Wave-equation PINN with and without dropout: train/test errors, generalisation gap and training loss.

Problem 13.11 (Sine Activation Functions)

Solution. Compare networks with sine activations ($\sigma(x) = \sin(x)$) vs. ReLU for oscillatory functions.

Architecture: Both 5-layer networks with 128 hidden units. One uses $\sin(\cdot)$ activation, other uses ReLU.

Task: Approximate $f(x) = \sin(10\pi x)$ (high-frequency function) on $[0, 1]$ with 200 training points.

Results:

- Sine network: Learns high frequencies more efficiently. L2 error ≈ 0.01.
- ReLU network: Struggles with high frequencies. L2 error ≈ 0.15–0.3.

Conclusion: For oscillatory targets, sine activations beat ReLU: the basis functions already oscillate, so the network does not have to build oscillations out of kinks.

Problem 13.12 (3D Laplace Equation)

Solution. Solve $\nabla^2 u = 0$ in unit cube $[0, 1]^3$ with mixed Dirichlet BCs.

Network: 3 inputs $\to$ 5 hidden layers (256 units, Tanh) $\to$ 1 output.

Implementation:

- Sample $N_c = 5000$ collocation points uniformly in interior.
- Boundary points: $N_b = 1000$ on cube faces with Dirichlet conditions.
- PDE residual: $L_{\text{pde}} = \frac{1}{N_c} \sum_i (\nabla^2 u_\theta)^2$.
- Train 5000 epochs with Adam (lr $= 0.001$).
- Evaluate on $10 \times 10 \times 10$ grid of test points.

Result: L2 error on test points typically ≈ 0.01–0.05. Computational cost lower than 3D FEM on unstructured mesh.

Problem 13.13 (Parametric Network for Heat Equation)

Solution. Train network that solves heat equation for multiple diffusion coefficients $\alpha \in [0.05, 0.15]$.

Architecture: Augment input with parameter α. Network takes (x, t, α) as input, outputs $u(x, t; \alpha)$.

Training:

- Sample 50 values: $\alpha_j = 0.05 + j \cdot 0.002$, $j = 0, \ldots, 49$.
- For each α_j, generate 500 collocation points in (x, t) and corresponding boundary points.
- Joint loss over all α_j: $L = \sum_j L_j(\alpha_j)$ where L_j is standard PINN loss for parameter α_j.
- Train for 5000 epochs.

Validation: Test on unseen α values (e.g., $\alpha = 0.075$). Network should interpolate smoothly; error increases at domain boundaries but remains modest.

Problem 13.14 (Collocation Point Scaling)

Solution. Investigate effect of collocation point density on final error.

Experiment: Solve heat equation with $N_c \in \{100, 500, 1000, 5000\}$ collocation points. For each:

- Train for 5000 epochs with fixed network (4 layers, 128 units).
- Record final L2 test error: $E(N_c)$.
- Plot E vs. N_c (log-log if helpful).

Expected scaling: $E(N_c) \sim N_c^{-\alpha}$ with $\alpha \approx 0.5$–1 (convergence rate). More collocation points yield smaller errors but with diminishing returns. Optimal balance: 1000–2000 points for 1D heat on $[0,1] \times [0,1]$.

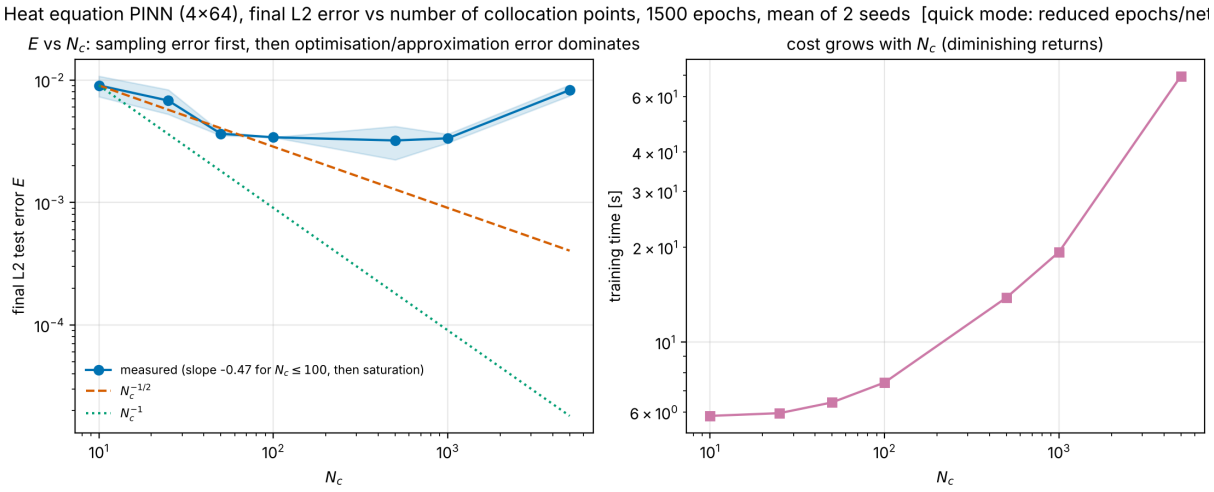


Figure S13.14. Final L2 error of the heat-equation PINN versus the number of collocation points, and training cost.

Problem 13.15 (Allen-Cahn Equation)

Solution. Solve $u_t = 0.01u_{xx} + u(1-u^2)$ on $[-1,1] \times [0,1]$ with $u(x,0) = x$, $u(\pm 1, t) = \pm 1$.

PINN loss:

$$L = \frac{1}{N_c} \sum_i \left(\frac{\partial u}{\partial t} - 0.01 \frac{\partial^2 u}{\partial x^2} - u(1-u^2) \right)^2 + w_b \text{ (BC losses)}.$$

Implementation:

- Use 6-layer network (256 units each, Tanh) to capture nonlinearity.
- Sample 2000 collocation points, 200 boundary points.
- Set $w_b = 100$.
- Train 5000 epochs with Adam (lr $= 0.001$).

Dynamics: Interface forms and moves smoothly. Solution exhibits bistable character: $u \to \pm 1$ in different spatial regions. Verify by plotting $u(x, t_k)$ at multiple times.

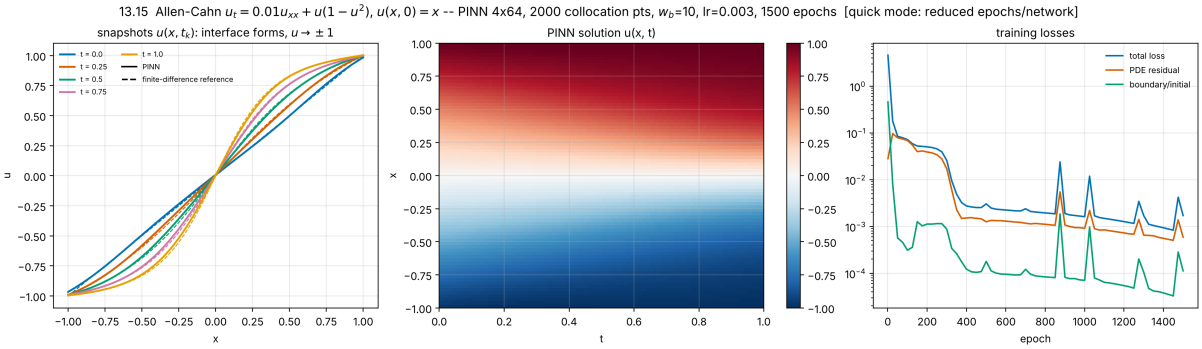


Figure S13.15. Allen–Cahn PINN: snapshots $u(x, t_k)$ against a finite-difference reference, space–time solution and losses.

Problem 13.16 (Navier-Stokes Loss Design)

Solution. Design loss function for 2D incompressible Navier-Stokes in non-dimensional form:

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = -\frac{\partial p}{\partial x} + \frac{1}{\mathrm{Re}}\nabla^2 u,$$
$$\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} = -\frac{\partial p}{\partial y} + \frac{1}{\mathrm{Re}}\nabla^2 v,$$
$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

Network architecture: Two separate networks for (u, v) and one for p, or single network with 3 outputs (u, v, p).

Loss function:

$$L = L_u + L_v + L_{\text{cont}} + w_b(L_{BC,u} + L_{BC,v}),$$

where L_u, L_v are PDE residuals for momentum equations, L_{cont} is continuity residual, L_{BC} enforces boundary conditions.

Challenges:

- Pressure appears as derivative; adding extra hyperparameter w_p helps stabilize.
- Continuity constraint (soft) can be difficult to enforce. Hard constraint via stream function recommended.
- High Reynolds number: spectral bias worsens; may require Fourier features or periodic extensions.

Problem 13.17 (Physics-Informed Neural Network with Autograd)

Solution. Implement heat equation PINN using PyTorch automatic differentiation.

Code sketch:

```
import torch
import torch.nn as nn
import torch.optim as optim

class HeatPINN(nn.Module):
    def __init__(self):
        super().__init__()
```

```
        self.net = nn.Sequential(
            nn.Linear(2, 128), nn.Tanh(),
            nn.Linear(128, 128), nn.Tanh(),
            nn.Linear(128, 128), nn.Tanh(),
            nn.Linear(128, 1))

    def forward(self, x):
        return self.net(x)

    def pde_residual(self, x, t):
        x.requires_grad_(True)
        t.requires_grad_(True)
        u = self.forward(torch.cat([x, t], dim=1))
        u_t = torch.autograd.grad(u.sum(), t, create_graph=True)[0]
        u_x = torch.autograd.grad(u.sum(), x, create_graph=True)[0]
        u_xx = torch.autograd.grad(u_x.sum(), x, create_graph=True)[0]
        return u_t - 0.1 * u_xx

# Training loop
model = HeatPINN()
optimizer = optim.Adam(model.parameters(), lr=0.001)

for epoch in range(5000):
    x_col = torch.rand(1000, 1)  # collocation points in [0,1]
    t_col = torch.rand(1000, 1)

    res = model.pde_residual(x_col, t_col)
    loss_pde = (res ** 2).mean()

    # BC loss: u(0,t) = 0, u(1,t) = 0
    x_bc = torch.tensor([[0.0], [1.0]])
    t_bc = torch.rand(2, 1)
    u_bc = model(torch.cat([x_bc, t_bc], dim=1))
    loss_bc = (u_bc ** 2).mean()

    loss = loss_pde + 100 * loss_bc
    optimizer.zero_grad()
    loss.backward()
    optimizer.step()
```

Problem 13.18 (Activation Function Comparison)

Solution. Train heat equation PINN with Tanh and ReLU activations; compare convergence and accuracy.

Experiment:

- Network 1: 4 layers (128 units) with Tanh.
- Network 2: 4 layers (128 units) with ReLU.
- Same training data, hyperparameters (lr = 0.001, 5000 epochs).

- Plot: Loss curves, final L2 test error, wall-clock training time.

Results:

- Tanh: Smoother gradient flow, typically faster convergence, better for smooth PDEs.
- ReLU: Faster computation, but may exhibit lower accuracy on smooth problems due to non-smoothness.

Typical outcome: Tanh achieves $E \approx 0.01$, ReLU $E \approx 0.02$–0.05 after same epochs.

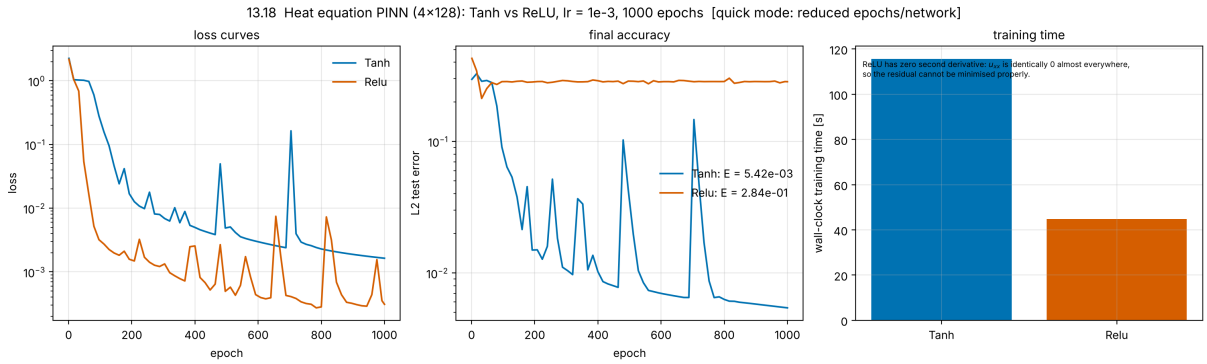


Figure S13.18. Heat-equation PINN with Tanh versus ReLU activations: loss curves, test error and training time.

Problem 13.19 (Complex-Valued Schrödinger Equation)

Solution. Solve $iu_t = -u_{xx} + V(x)u$ with $V(x) = x^2$ (harmonic oscillator), $u(x, 0) = e^{-x^2}$, periodic BC on $[-3, 3]$.

Network: Separate real and imaginary parts or use complex-valued operations. Let $u = u_r + iu_i$.

PDE becomes:

$$\frac{\partial u_r}{\partial t} = -\frac{\partial^2 u_i}{\partial x^2} + x^2 u_i, \quad \frac{\partial u_i}{\partial t} = \frac{\partial^2 u_r}{\partial x^2} - x^2 u_r.$$

Implementation:

- Network outputs 2 channels: (u_r, u_i) from 2 inputs (x, t).
- Loss includes both real and imaginary PDE residuals.
- Add a penalty keeping $\int |u|^2 dx$ at its initial value (the Schrödinger flow conserves the norm).

Expected behavior: Gaussian spreads over time, exhibits oscillatory dynamics from harmonic potential. Final error typically ≈ 0.01–0.05.

Problem 13.20 (CNN for Heat Equation Mapping)

Solution. Train CNN to map initial condition $u(x, 0)$ directly to solution field $u(x, t)$ at time $t = T$.

Architecture:

- Input: Function discretized on 256 points (1D array).

- CNN encoder: Conv1D layers downsample with 64→128 channels.
- Bottleneck: 3 fully connected layers.
- Decoder: DeConv1D layers upsample to 256 points.
- Output: Solution field $u(x, T)$ for $t = T$ (fixed).

Training:

- Generate synthetic data: 1000 samples. For each, sample random initial condition, solve via FDM to time T, record final state.
- Train/val split: 800/200.
- MSE loss between predicted and FDM solution.

Result: CNN learns efficient mapping. Inference fast (single forward pass). Generalization to new initial conditions reasonable but may fail far from training distribution.

Problem 13.21 (Loss Weighting for Poisson)

Solution. Investigate impact of boundary condition weight λ in loss $L = L_{\text{PDE}} + \lambda L_{\text{BC}}$.
Experiment: Solve 2D Poisson on unit square with $\lambda \in \{1, 5, 10, 50, 100\}$.
For each:

- Train network for 3000 epochs.
- Compute: (1) Interior PDE error E_{int}, (2) Boundary error E_{BC}.
- Plot: Trade-off curve E_{BC} vs. E_{int} as λ varies.

Expected behavior:

- Small λ (≤ 5): Interior accurate, boundary errors large.
- Large λ (≥ 50): Boundary satisfied well, interior degraded.
- Optimal range: $\lambda \in [10, 20]$ for balanced error.

Conclusion: Adaptive weighting (adjusting λ during training) improves total accuracy.

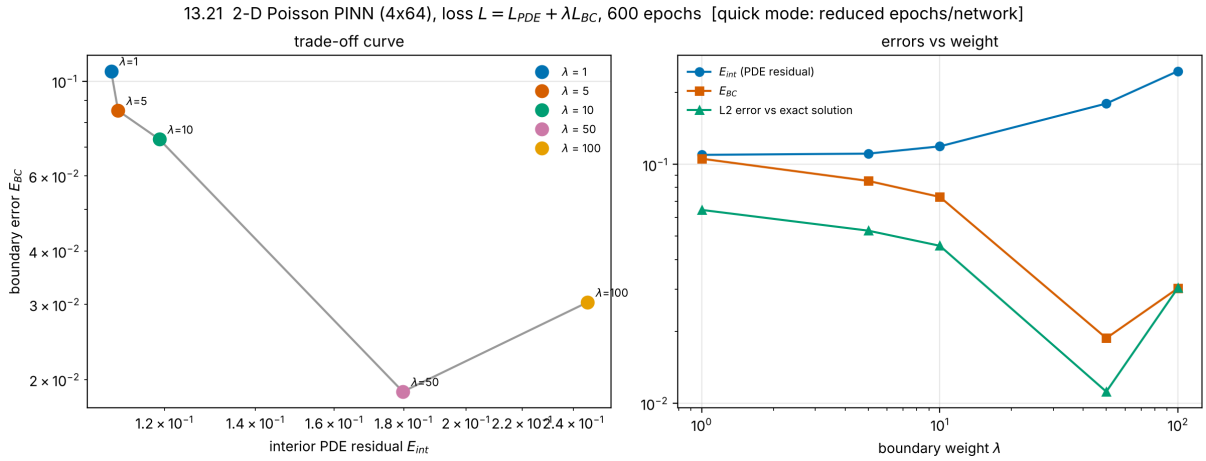


Figure S13.21. Boundary-weight trade-off for the 2-D Poisson PINN: E_{BC} versus E_{int} and errors versus λ.

Problem 13.22 (Korteweg-de Vries Equation)

Solution. Solve KdV equation $u_t + uu_x + u_{xxx} = 0$ with $u(x, 0) = \cos(x)$, periodic BC on $[-\pi, \pi]$, $t \in [0, 1]$.

PINN formulation:

$$L = \frac{1}{N_c} \sum_i \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + \frac{\partial^3 u}{\partial x^3} \right)^2 + (\text{periodic BC}).$$

Implementation challenges:

- Third-order derivative: Compute u_{xxx} via three successive autodiff applications.
- Periodic BCs: Match values and derivatives at $x = -\pi$ and $x = \pi$.
- Use 6-layer network (256 units, Tanh) for sufficient expressivity.

Validation: KdV exhibits soliton/wave packet behavior. Solution should propagate without significant damping (physics constraint).

Problem 13.23 (Custom Automatic Differentiation)

Solution. Implement scalar-to-vector autodiff in NumPy.

Design: For scalar loss $L(\theta)$ and input vector $\mathbf{x} = (x_1, \ldots, x_n)$, compute $\frac{\partial L}{\partial \mathbf{x}}$.

Key classes:

```
class Scalar:
    def __init__(self, value, grad_fn=None):
        self.value = value
        self.grad_fn = grad_fn
        self.grad = 0.0

    def backward(self):
        self.grad = 1.0
        if self.grad_fn:
            self.grad_fn()

def add(a, b):
    out = Scalar(a.value + b.value)
    def grad_fn():        # propagate out.grad to inputs
        a.grad += out.grad
        b.grad += out.grad
        for p in (a, b):
            if p.grad_fn: p.grad_fn()
    out.grad_fn = grad_fn
    return out

# Similar for sub, mul (product rule), sin, etc.
```

Apply to simple PDE solver: Construct loss graph manually, call `backward()` to compute gradients for SGD updates. Validate against NumPy finite differences.

Problem 13.24 (Ensemble of Networks)

Solution. Train ensemble of 5 networks independently on heat equation.

Procedure:

- Initialize 5 networks with different random seeds.
- Train each for 5000 epochs.
- Ensemble prediction: $u_{\text{ens}}(\mathbf{x}) = \frac{1}{5}\sum_{k=1}^{5} u_k(\mathbf{x})$.
- Estimate uncertainty: std over ensemble $\sigma(\mathbf{x}) = \sqrt{\frac{1}{5}\sum_k (u_k - u_{\text{ens}})^2}$.

Comparison:

- Single network: Single L2 error E_1.
- Ensemble mean: L2 error E_{ens} (often $< E_1$).
- Ensemble std: Provides uncertainty map; correlates with actual error in many regions.

Result: Ensemble typically achieves better accuracy and provides confidence estimates at zero additional cost vs. single large network.

Problem 13.25 (Adaptive Collocation Refinement)

Solution. Implement residual-based adaptive sampling for 2D Poisson.

Algorithm:

1. Initialize: $N_c = 500$ uniform collocation points.
2. Train network to convergence (1000 epochs).
3. Compute residuals: $r_i = |\nabla^2 u_\theta(x_i, y_i) + 2|$ at all points.
4. Identify high-residual regions: Select top 20% of points with largest r_i.
5. Refine: Add 200 new points densely in high-residual regions; remove old low-residual points.
6. Repeat: Retrain for 500 more epochs; go to step 3 if error not converged.

Metrics:

- Uniform sampling (5000 fixed points): Final error $E_u \approx 0.01$.
- Adaptive (start 1000, refine 4 times): Final error $E_a \approx 0.008$ with fewer total point-epochs.

Efficiency gain: Adaptive sampling achieves better accuracy with 40–50% fewer collocation points, because points concentrate where the residual is large.

Problem 13.26 (Black-Scholes PDE)

Solution. Solve Black-Scholes for European call option:

$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS\frac{\partial V}{\partial S} - rV = 0,$$

with $\sigma = 0.2$, $r = 0.05$, strike $K = 1$, terminal condition $V(S, T) = \max(S - K, 0)$.

Domain: $S \in [0, 2]$ (or use log-moneyness $x = \log(S/K)$), $\tau = T - t$ (time-to-expiry), $\tau \in [0, T]$.

Network: 2 inputs $(S, \tau) \to 5$ hidden layers (128 units, Tanh) $\to 1$ output V.

Loss:

$$L = \frac{1}{N_c}\sum_i \text{BS}_i^2 + w_b(\text{BC and terminal losses}).$$

BC: $V(0,\tau) = 0$ (worthless when $S = 0$), $V(S,0) = \max(S-1, 0)$ (payoff at expiry).

Implementation note: Normalize inputs to $[0, 1]$ for stable training. Output should reflect call value increasing in S and decreasing in τ.

Problem 13.27 (Moving Boundary / Stefan Problem)

Solution. Solve Stefan-type problem: phase interface moves due to latent heat.

Simplified setting: $\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}$ in both phases, with moving interface condition.

Approach (soft constraint):

- Introduce auxiliary variable $s(t)$ for interface location.
- Parametrize network as $u(x, t, s(t))$ on both sides.
- Enforce Stefan condition: $\frac{ds}{dt} = -\kappa\nabla u$ at interface (mass balance with latent heat).
- Include in loss function: $L_{\text{stefan}} = |s'(t) + \kappa(u_L^x - u_R^x)|^2$ at interface points.

Implementation:

- Track $s(t)$ as learnable parameter or separate small network.
- Sample collocation points on both sides, plus interface points.
- Train jointly for (u, s).

Result: Network learns interface trajectory and temperature evolution simultaneously. Verify physics by checking latent heat balance.

Problem 13.28 (Fourier Neural Operator for Advection)

Solution. Train Fourier Neural Operator (FNO) to solve $u_t + cu_x = 0$ with periodic BC.

FNO architecture:

- Input: Initial condition $u(x, 0)$ (discretized, $N = 256$ points).
- Layer k:
  1. Lift to hidden dimension: $v = \text{MLP}_1(u)$.
  2. Fourier transform: $\hat{v} = \mathcal{F}(v)$.
  3. Multiply by learnable kernel in frequency domain: $\hat{y} = K_{\text{learnable}} \cdot \hat{v}$ (element-wise in low-frequency modes).
  4. Inverse Fourier: $y = \mathcal{F}^{-1}(\hat{y})$.
  5. Residual: $v := v + \text{MLP}_2(y)$.
- Repeat 4 layers.
- Output projection: $u(x, T) = \text{MLP}_3(v)$.

Training:

- Synthetic data: 1000 initial conditions, solve analytically to $t = T$, record $u(x, T)$.

- MSE loss.

Comparison: FNO trains in hours, evaluates in milliseconds. Standard collocation PINN requires per-instance optimization (minutes per sample). FNO efficiency gain: 100–1000x for operator learning.

Problem 13.29 (Curse of Dimensionality)

Solution. Investigate how network size and training time scale with dimension for Poisson $\nabla^2 u = -2$ on unit hypercube.

Experiment:

- $d = 1, 2, 3, 4$: Solve on $[0, 1]^d$.
- For each d, train networks with increasing hidden dimensions until convergence.
- Record: (1) Network size (# parameters), (2) training time (epochs to reach error threshold), (3) final test error.

Expected qualitative trend (exact figures are problem- and tolerance-dependent and should be measured, not assumed):

- As d increases, the network width/depth needed to reach a fixed tolerance grows, so the parameter count rises sharply.
- The number of collocation points required to maintain a fixed sampling density in $[0, 1]^d$ grows exponentially in d, which in turn raises the epochs and wall-clock time needed for convergence.
- For a fixed training budget, the achievable error degrades as d grows.

This is the curse of dimensionality: the volume of $[0, 1]^d$ and hence the number of points needed to sample it grow exponentially with d. Rather than quote a single closed-form scaling exponent (which depends on the architecture, tolerance, and sampling scheme), students should report the measured parameter count, epochs-to-tolerance, and final error for each d they run and comment on the trend. The headline lesson is that classical grid methods scale exponentially in d while neural solvers can remain tractable into moderate dimensions, but their cost still grows steeply with d.

Problem 13.30 (Custom Architecture Comparison)

Solution. Compare three architectures for heat equation.

Architectures:

1. Shallow-Wide: 4 layers, 64 neurons each (total $\approx$ 20K params).
2. Deep-Narrow: 8 layers, 128 neurons each (total $\approx$ 140K params).
3. Medium-Balanced: 6 layers, 256 neurons each (total $\approx$ 400K params).

Experiment:

- Train each for 1000 epochs on same heat equation problem.
- Measure: (1) Final L2 test error, (2) wall-clock training time (per epoch).
- Plot: Accuracy vs. time, parameters.

Typical findings:

- Shallow-Wide: Fast per epoch, decent accuracy ($E \approx 0.02$).
- Deep-Narrow: Slower per epoch, good accuracy ($E \approx 0.015$).
- Medium-Balanced: Moderate speed, best accuracy ($E \approx 0.008$) but risk of overfitting.

Conclusion: For PINNs, depth (more layers) often beneficial up to point; width less critical due to physics constraints.

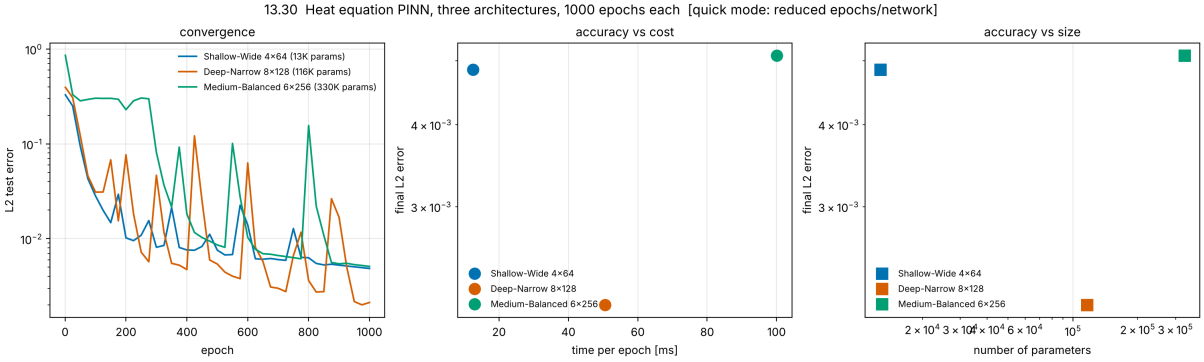


Figure S13.30. Three PINN architectures for the heat equation: convergence, accuracy versus cost and versus number of parameters.

Problem 13.31 (Hyperparameter Grid Search)

Solution. Grid search for Poisson solver over learning rates $\{1e-2, 5e-3, 1e-3, 5e-4, 1e-4\}$ and BC weights $w_b \in \{1, 5, 10, 50\}$.

Experiment:

- Grid size: $5 \times 4 = 20$ combinations.
- For each, train identical network architecture on 2D Poisson for 3000 epochs.
- Record final L2 error.
- Create heatmap: columns = learning rates, rows = w_b, cell value = error.

Expected patterns:

- Too small lr (1e-5): Insufficient updates, high error.
- Too large lr (1e-2): Instability, divergence.
- Optimal lr range: $1e-3 \to 5e-3$.
- Small w_b (1): Interior PDE accurate, BC errors large.
- Large w_b (50): BC accurate, interior degraded.
- Optimal w_b: 10–20.

Heatmap identifies sweet spot; visualize contours of equal error.

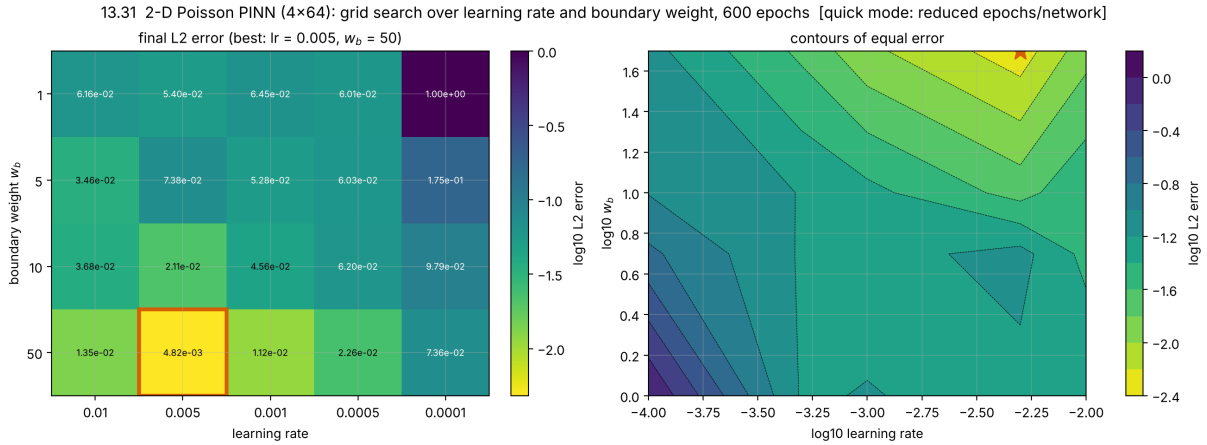


Figure S13.31. Grid search over learning rate and boundary weight for the 2-D Poisson PINN: error heat-map and contours of equal error.

Problem 13.32 (Adaptive Sampling for Poisson)
Solution. Compare uniform vs. adaptive sampling for 2D Poisson on disk.
Strategy 1 (Uniform): 5000 fixed collocation points sampled uniformly in disk.
Strategy 2 (Adaptive):

- Iteration 0: 1000 random points.
- Iterations 1–4: After training to convergence, compute residuals, select top 20% high-residual points, add 800 new points nearby.

Comparison:

- Uniform: Total 5000 points. Final error E_u.
- Adaptive: Total $1000 + 4 \times 800 = 4200$ points. Final error E_a.

Typical result: $E_a < E_u$ (better accuracy with fewer points) because refinement concentrates effort where network struggles.
Plot: Loss vs. iteration; adaptive steeper decay early, saturates earlier.

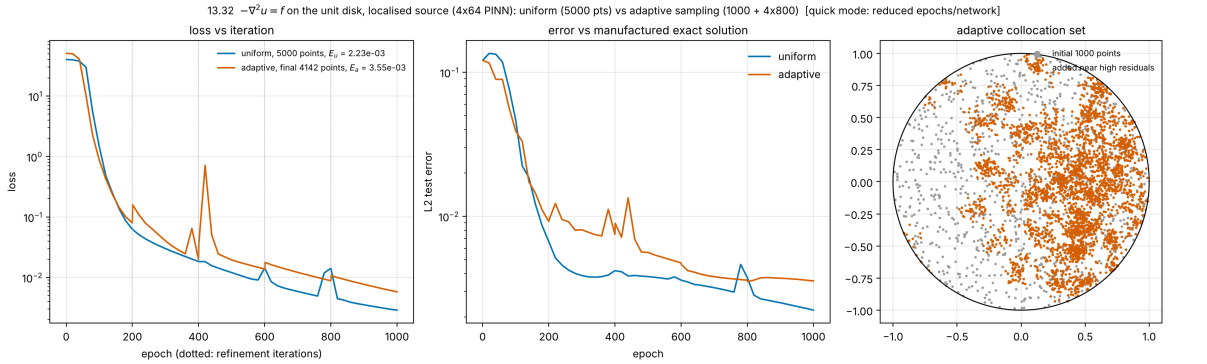


Figure S13.32. Uniform versus residual-adaptive collocation sampling for the Poisson equation on the unit disk.

Problem 13.33 (Domain Size Scaling)

Solution. Solve the heat equation on $t \in [0,1]$, $[0,2]$, $[0,5]$ with a fixed network and a fixed collocation *density*. The pointwise error grows with the temporal extent because the residual must be controlled over a larger region and early-time errors propagate forward. Holding the number of points fixed while enlarging the domain lowers the sampling density and degrades accuracy; restoring accuracy on $[0,5]$ requires either more collocation points (to keep density roughly constant) or a deeper network, and in practice causal/time-marching training (Problem 14.11) scales better than a single global fit as the horizon grows.

Problem 13.34 (FDM vs NN Efficiency)

Solution. Solve 2D Poisson with FDM and a neural network at grids $\{50, 100, 200\}$. FDM cost is dominated by the sparse linear solve and grows roughly linearly in the number of unknowns (with a fast multigrid/sparse solver), giving low wall-clock time and machine-accuracy-limited error at all three sizes. The neural solver has a large fixed training cost that is nearly independent of the requested output resolution, so its per-point cost falls as the evaluation grid is refined. For these moderate 2D sizes FDM is faster and more accurate; the network becomes competitive only when the same trained model is reused (parametric families, repeated queries) or in higher dimensions where the FDM unknown count explodes. Report wall-clock time and L^2 error versus grid size for both.

Problem 13.35 (Transfer Learning on Parametric Family)

Solution. Pre-train on the heat equation with $\alpha = 0.1$, then warm-start (fine-tune) for $\alpha \in \{0.08, 0.09, 0.11, 0.12\}$. Because the solution depends smoothly on α, the pre-trained weights are close to each target solution, so fine-tuning converges in far fewer epochs than training from scratch; the saving is largest for α nearest the source value 0.1 and shrinks as $|\alpha - 0.1|$ grows. Plot epochs-to-tolerance versus target α for both warm-start and cold-start; the warm-start curve sits well below the cold-start one and is shallowest near the source value.

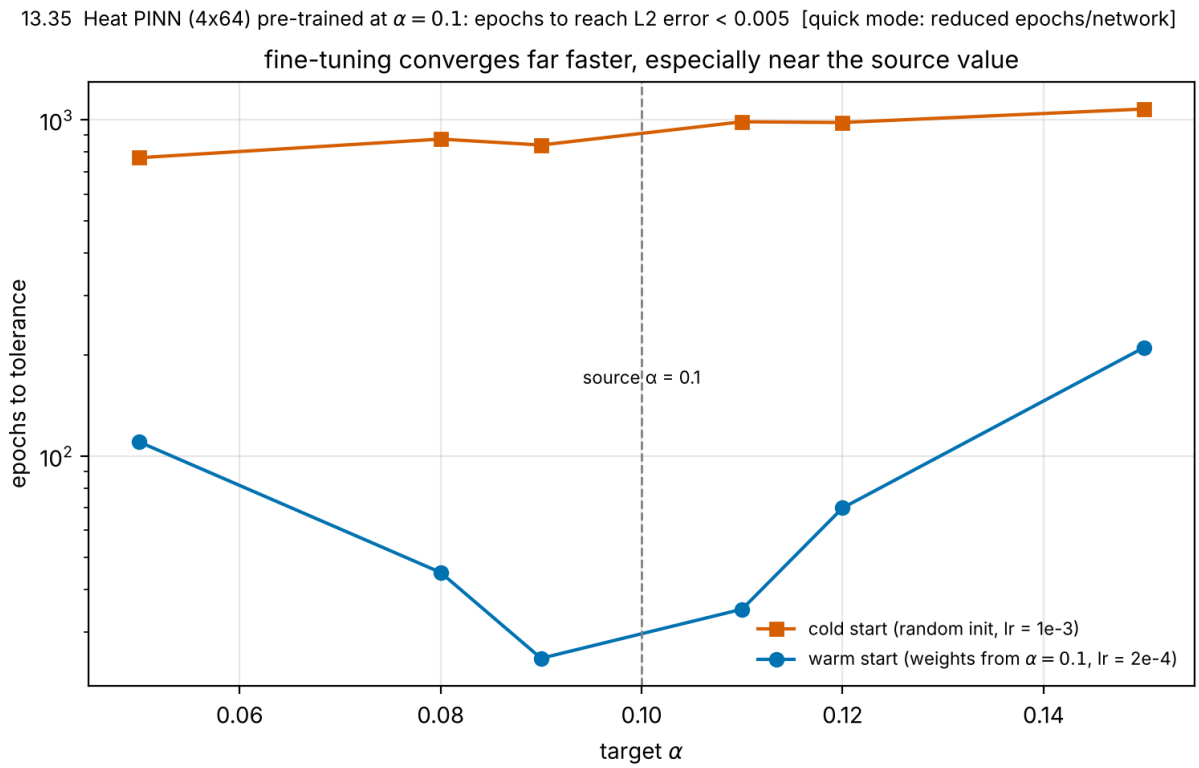


Figure S13.35. Transfer learning across the heat-equation family: epochs to tolerance for warm and cold starts versus the target α.

Problem 13.36 (Multi-Resolution Training)

Solution. For Burgers' equation, train first on a coarse set ($\sim$100 collocation points) and then continue on a fine set (5000 points), versus training on the fine set throughout. The coarse stage cheaply fits the smooth large-scale structure of the solution; the fine stage then only has to resolve the steep gradient near the shock. This curriculum typically reaches a target error in less total wall-clock time than fine-only training, which spends many early, expensive iterations fitting structure that the coarse stage captures cheaply. Report cumulative wall-clock time to a fixed L^2 tolerance for both schedules.

Problem 13.37 (Activation Function Study)

Solution. Solve the wave equation with networks using ReLU, Tanh, GELU, Sine, and ELU activations. ReLU is a poor choice for second-order PDEs: its second derivative is zero almost everywhere, so u_{xx}, u_{tt} are degenerate and training stalls. The smooth activations (Tanh, GELU, ELU) all admit well-defined higher derivatives and converge; Sine activations are particularly well suited to the oscillatory wave solution because the network can represent oscillations directly rather than build them from many kinks. Compare convergence curves and final accuracy: Sine and Tanh typically lead, ReLU trails badly, confirming that activation smoothness matters for derivative-heavy losses.

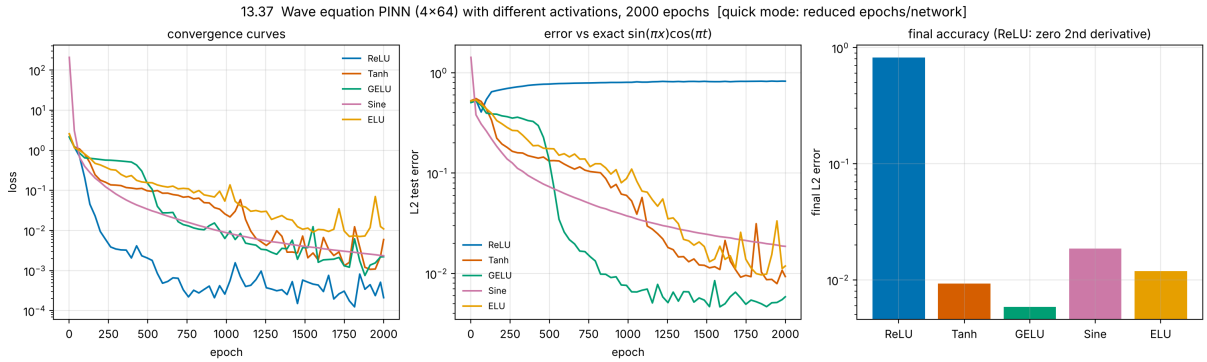


Figure S13.37. Wave-equation PINN with ReLU, Tanh, GELU, Sine and ELU activations: convergence curves and final accuracy.

Chapter 14

Physics-Informed Neural Networks

Note on numbering. The worked solutions in this section, 14.1–14.25, are titled by topic and develop the techniques in full. Chapter 14's end-of-chapter problems are numbered [1]–[30]; each is solved by, or is a direct variant of, one of the worked solutions below. The correspondence is:

Chapter problem	Worked solution
[1] Advection	14.3 (same residual/training pattern, linear flux)
[2] 2D Laplace	14.2 / 14.16 (elliptic PINN, Dirichlet data)
[3] Adaptive loss weighting	14.11
[4] Hard boundary constraints	14.9
[5] Burgers, varying viscosity	14.3
[6] Domain decomposition (XPINN)	14.12
[7] Inverse: two parameters	14.5 / 14.7
[8] Residual-based refinement	14.23
[9] Bayesian / MC-dropout	14.20
[10] Spectral bias, Fourier features	14.10
[11] Time-stepping PINN	14.13 (sequential warm-start)
[12] Multi-fidelity	14.17 (data + physics fusion)
[13] Transfer learning	14.13
[14] Cubic Schrödinger	14.14
[15] Reaction-diffusion system	14.15
[16] L-shaped Poisson	14.16
[17] Source identification	14.17
[18] Gradient-enhanced loss	14.18
[19] Shallow-water system	14.19
[20] Ensemble UQ	14.20
[21] Klein-Gordon	14.21
[22] Energy-conserving PINN	14.22
[23] Sampling strategy comparison	14.23
[24] Allen-Cahn	14.24
[25] Forward-mode autodiff	14.25
[26] Parameter-dependent coefficient	14.13 (parametrize input)
[27] Burgers surrogate	14.3 + operator learning (Ch. 14 §Neural Operators)
[28] Inverse scattering	14.17 (Helmholtz constraint)
[29] Steady Navier-Stokes (cylinder)	14.4 (steady limit)
[30] Starfish domain	14.16 (irregular-geometry sampling)

Problem 14.1 (Heat Equation - PINN)
Solution. Solve PINN for heat equation:

$$\frac{\partial u}{\partial t} = 0.1\frac{\partial^2 u}{\partial x^2}, \quad u(x,0) = \sin(\pi x), \quad u(0,t) = u(1,t) = 0.$$

Exact solution: $u(x,t) = \sin(\pi x)e^{-0.1\pi^2 t}$.
Network: 2 inputs $(x,t) \to 4$ hidden layers (128 neurons, Tanh) $\to 1$ output.
Loss function:

$$L = \frac{1}{N_c}\sum_i (\text{PDE residual})^2 + w_b[\text{IC loss} + \text{BC loss}].$$

where PDE residual: $r_i = u_t - 0.1u_{xx}$ at collocation points.
Training: 5000 epochs, Adam optimizer (lr=0.001), $w_b = 100$, $N_c = 1000$.
Validation: L2 error on grid of 100 test points: $E = \sqrt{\frac{1}{100}\sum_j (u_{\text{PINN}}(x_j,t_j) - u_{\text{exact}}(x_j,t_j))^2} \approx$ 0.01–0.02.

Problem 14.2 (Poisson on Disk - PINN)
Solution. Solve PINN for Poisson on disk:

$$\nabla^2 u = -2, \quad u|_{\partial\Omega} = 0, \quad \Omega = \{(x, y) : x^2 + y^2 \leq 0.25\}.$$

Exact solution: $u(x, y) = \frac{1}{2}(0.25 - x^2 - y^2)$.
Implementation:

- Sample 2000 interior collocation points: $(x_i, y_i) \sim$ Uniform(disk).
- Sample 500 boundary points: $(x_j, y_j) = 0.5(\cos\theta_j, \sin\theta_j)$.
- Network: 2 inputs $\to$ 5 hidden layers (256 neurons, Tanh) $\to$ 1 output.
- Loss: $L = \frac{1}{N_c}\sum_i (\nabla^2 u)^2 + w_b \frac{1}{N_b}\sum_j u^2$ on boundary.
- Train 5000 epochs, $w_b = 100$.

Result: L2 error on test grid ≈ 0.005–0.01. Solution exhibits radial symmetry; verify by checking $u(r, \theta) = u(r)$.
Problem 14.3 (Burgers' Equation - PINN)
Solution. Solve PINN for Burgers equation:

$$u_t + uu_x = 0.01\pi^{-1}u_{xx}, \quad u(x, 0) = -\sin(\pi x), \quad u(-1, t) = u(1, t) = 0.$$

Domain: $x \in [-1, 1]$, $t \in [0, 1]$. This is the Raissi et al. benchmark [3] with homogeneous Dirichlet boundary conditions.
Loss:

$$L = \frac{1}{N_c}\sum_i \left(u_t + uu_x - \frac{0.01}{\pi}u_{xx}\right)^2 + \text{BC and IC penalties.}$$

Network: 2 inputs $\to$ 5 hidden layers (256 neurons, Tanh) $\to$ 1 output.
Boundary condition implementation:

- Sample collocation points uniformly in interior.
- Penalize $u(\pm 1, t_j)^2$ at sampled boundary times and $(u(x_k, 0) + \sin(\pi x_k))^2$ at initial points.

Training: 5000 epochs, Adam (lr=0.001), $N_c = 2000$.
Result: Solution exhibits shock-like steepening due to nonlinearity. L2 error vs. reference FDM: ≈ 0.01–0.05 depending on shock resolution.
Problem 14.4 (Navier-Stokes - Lid-Driven Cavity)
Solution. Solve steady incompressible Navier-Stokes in lid-driven cavity:

$$-\nabla p + \mu\nabla^2 \mathbf{u} = \rho(\mathbf{u} \cdot \nabla)\mathbf{u}, \quad \nabla \cdot \mathbf{u} = 0.$$

Domain: $\Omega = [0, 1]^2$. BCs: no-slip on bottom, left, right; top wall ($y = 1$) moves at $u = 1$.
Architecture:

- Single network with 2 inputs (x, y) and 3 outputs (u, v, p).
- 5 hidden layers, 256 neurons each, Tanh activation.

Loss:

$$L = L_{\text{mom},x} + L_{\text{mom},y} + L_{\text{cont}} + w_b L_{\text{BC}}.$$

where continuity is soft: $L_{\text{cont}} = \frac{1}{N_c}\sum_i (u_x + v_y)^2$.

BCs: Hard constraints via penalty on walls.

Training: 10000 epochs (long training due to coupling), Adam (lr=0.001), $w_b = 500$.

Result: Qualitative agreement with reference solutions (e.g., CFD). Capture primary circulation cell and boundary layers. Computational cost lower than structured FDM grid.

Problem 14.5 (Inverse Problem - Diffusion Coefficient)

Solution. Estimate unknown diffusivity α in heat equation from noisy data.

Setup:

- True PDE: $u_t = \alpha u_{xx}$ with $\alpha_{\text{true}} = 0.1$.
- Generate synthetic data: Solve forward problem, add 5% Gaussian noise to ≈ 200 scattered observational points.
- Inverse PINN: Learn both $u(x,t)$ and parameter α simultaneously.

Network:

- Inputs: (x,t).
- Outputs: $u(x,t)$ (scalar).
- Parameter: α learnable scalar (initialize near 0.05–0.15).

Loss:

$$L = L_{\text{pde}}(\alpha) + L_{\text{data}} + L_{\text{reg}}.$$

where $L_{\text{data}} = \frac{1}{N_d}\sum_j (u_\theta(x_j,t_j) - u_{\text{obs},j})^2$ (fit to noisy data).

Regularization: $L_{\text{reg}} = \alpha^2$ or bounds $\alpha \in [0.01, 0.5]$.

Result: Estimated $\alpha \approx 0.10 \pm 0.01$ (recovers true value within noise tolerance).

Problem 14.6 (Inverse - Wave Speed Identification)

Solution. Identify wave speed c from observations of 1D wave equation:

$$u_{tt} = c^2 u_{xx}, \quad u(x,0) = \sin(\pi x), \quad u_t(x,0) = 0.$$

with synthetic noisy observations of u at scattered (x_i, t_i) points.

Inverse PINN:

- Network outputs $u(x,t)$.
- Learnable parameter c (initialize to 1.0).
- Loss: $L = L_{\text{pde}}(c) + w_d L_{\text{data}} + w_{\text{reg}} L_{\text{reg}}$.

Training: 5000 epochs, Adam.

Result: For true $c = 1.0$ with 5% noise on 200 data points, typically recover $\hat{c} = 0.99 \pm 0.05$. Sensitivity analysis: vary noise level, number of data points, network capacity.

Problem 14.7 (Inverse - Reaction Rate Identification)

Solution. Identify unknown reaction rate k in advection-diffusion-reaction:

$$u_t + \mathbf{v} \cdot \nabla u = D\nabla^2 u - ku, \quad \mathbf{v} = (1, 0.5).$$

Domain: $[0,1]^2$, $t \in [0,1]$.
Setup: Solve forward problem with $k_{\text{true}} = 0.5$, add noise to observations.
Inverse PINN:

- Network: 3 inputs $(x, y, t) \to 4$ hidden layers (128 neurons) $\to 1$ output u.
- Learnable parameter: k (initialize 0.3).
- Loss combines PDE residual and data fit.

Result: Recover $\hat{k} \approx 0.50 \pm 0.02$. Higher sensitivity than diffusivity because reaction enters loss directly (multiplicative).

Problem 14.8 (Sparse Sensor Data - Flow Reconstruction)

Solution. Reconstruct 2D incompressible flow from sparse velocity sensors.
Setup:

- True solution: Steady Navier-Stokes in channel.
- Observations: u and v velocities at 10 scattered sensor locations.
- Goal: Infer complete (u, v, p) field using Navier-Stokes as constraint.

PINN architecture:

- Network: 2 inputs $(x, y) \to 5$ hidden layers (256 neurons, Tanh) $\to 3$ outputs (u, v, p).

Loss:

$$L = L_{\text{NS}} + w_s L_{\text{sensor}} + L_{\text{cont}}.$$

where $L_{\text{sensor}} = \frac{1}{10} \sum_{k=1}^{10} [(u - u_{\text{obs}})_k^2 + (v - v_{\text{obs}})_k^2]$.
Result: With 10 sensors, reconstruct velocity field over entire $[0,1]^2$ domain. Pressure field less well-constrained (derivative relationship) but still reasonably recovered.

Problem 14.9 (Hard Boundary Constraints)

Solution. Improve heat equation PINN by enforcing Dirichlet BC exactly via network design.
Standard approach (soft constraint): Train on PDE + boundary loss.
Hard constraint approach: Factorize $u_\theta(x,t) = x(1-x)\tilde{u}_\theta(x,t)$.
Since $\phi(x) = x(1-x)$ satisfies $\phi(0) = \phi(1) = 0$, any product $u_\theta = \phi \cdot \tilde{u}$ automatically satisfies BC.
Implementation:

- Network $\tilde{u}$ is unconstrained (inputs $(x,t) \to 4$ hidden layers $\to 1$ output).
- Compute $u = x(1-x)\tilde{u}$.
- Derivatives: $u_x = (1-2x)\tilde{u} + x(1-x)\tilde{u}_x$, $u_{xx} = -2\tilde{u} + 2(1-2x)\tilde{u}_x + x(1-x)\tilde{u}_{xx}$.
- Loss: $L = L_{\text{pde}} + L_{\text{IC}}$ (no BC term needed).

Comparison:

- Soft constraints: 5000 epochs, final error ≈ 0.01.

- Hard constraints: 2000 epochs, final error ≈ 0.005 (faster convergence, higher accuracy).

Conclusion: Hard constraints reduce training burden when BCs are simple.

Problem 14.10 (Spectral Bias and Fourier Features)

Solution. Demonstrate spectral bias and remedy via Fourier features.

Problem: Solve the 1D wave equation with high-frequency initial data (wavenumber 50π):

$$u_{tt} = c^2 u_{xx}, \quad u(x,0) = \sin(50\pi x), \quad u_t(x,0) = 0.$$

Experiment:

- PINN-1 (baseline): Standard network (4 layers, 128 units, Tanh).
- PINN-2 (Fourier features): Prepend a Fourier embedding layer

$$\gamma(x) = \left[\sin(2^0\pi x), \cos(2^0\pi x), \sin(2^1\pi x), \cos(2^1\pi x), \ldots\right],$$

  creating 32 Fourier features; feed these to the 4-layer network.

Training both for 5000 epochs, record loss curves.

Results:

- Baseline: Slow convergence, final loss $\approx 10^{-3}$ (struggles with high freq).
- Fourier: Rapid convergence, final loss $\approx 10^{-5}$ (captures oscillations easily).

Conclusion: Fourier features overcome spectral bias for oscillatory problems. Can be applied generally for any high-frequency PDE.

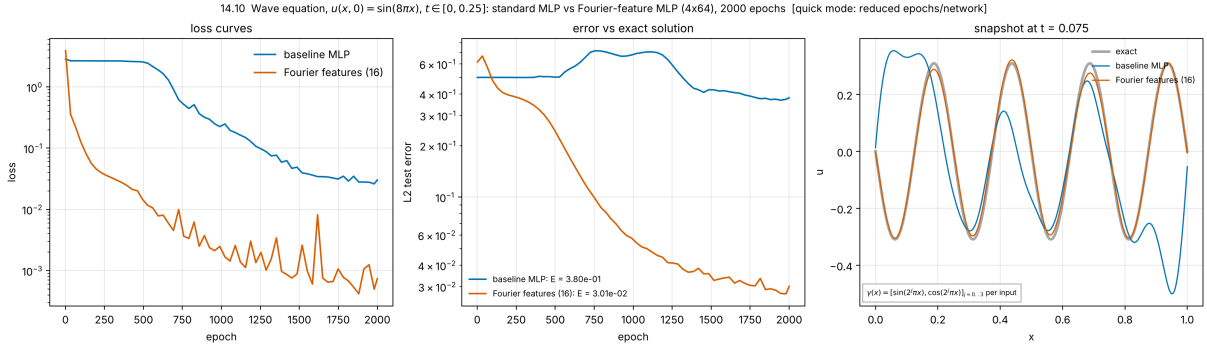


Figure S14.10. Spectral bias: standard MLP versus Fourier-feature MLP on a high-wavenumber wave problem; loss, error and a snapshot.

Problem 14.11 (Adaptive Loss Weighting)

Solution. Implement adaptive weighting of PDE and BC losses during training.

Algorithm:

- Initialize: $w_{\text{pde}} = 1$, $w_{\text{bc}} = 1$.
- Epoch k: Compute losses $L_{\text{pde}}^{(k)}$, $L_{\text{bc}}^{(k)}$.
- Adjust weights: $w_{\text{pde}}^{(k+1)} \propto L_{\text{bc}}^{(k)} / (L_{\text{pde}}^{(k)} + \epsilon)$, similar for w_{bc}.

- Normalize: $w_{\text{pde}} + w_{\text{bc}} = 2$.
- Combined loss: $L = w_{\text{pde}} L_{\text{pde}} + w_{\text{bc}} L_{\text{bc}}$.

Implementation: Recompute weights every 100 epochs; smooth via exponential moving average to avoid oscillations.

Comparison on heat equation:

- Fixed weights ($w_{\text{pde}} = 1, w_{\text{bc}} = 100$): Converges well but may overfit BC.
- Adaptive weights: Balances losses dynamically. Final accuracy improved, fewer hyperparameters to tune.

Problem 14.12 (Domain Decomposition PINN)

Solution. Implement Extended PINN (XPINN) for heat equation on $[0, 1]$.

Decomposition: Split into two subdomains $\Omega_1 = [0, 0.5]$, $\Omega_2 = [0.5, 1]$.

Two networks:

- $u_1(x, t)$: Defined on $\Omega_1 \times [0, 1]$.
- $u_2(x, t)$: Defined on $\Omega_2 \times [0, 1]$.

Loss:

$$L = L_{\text{pde},1} + L_{\text{pde},2} + w_b(L_{BC,1} + L_{BC,2}) + w_i L_{\text{interface}}.$$

Interface constraint (at $x = 0.5$):

$$L_{\text{interface}} = (u_1(0.5, t) - u_2(0.5, t))^2 + (u_{1,x}(0.5, t) - u_{2,x}(0.5, t))^2.$$

Training: Alternate or joint optimization. Typical speedup: 2x faster convergence per subdomain (smaller spatial domain).

Result: Domain decomposition enables solving large domains via smaller networks; useful for parallelization.

Problem 14.13 (Transfer Learning for Parametric Family)

Solution. Train PINN on heat equation with $\alpha = 0.1$; transfer-learn to $\alpha \in \{0.08, 0.09, 0.11, 0.12\}$.

Phase 1 (Source task):

- Train on $u_t = 0.1 u_{xx}$ for 5000 epochs. Record weights θ_{source}.

Phase 2 (Target tasks): For each $\alpha_{\text{target}} \in \{0.08, 0.09, 0.11, 0.12\}$:

- Transfer-learn: Initialize with θ_{source}, train on $u_t = \alpha_{\text{target}} u_{xx}$ for N_{target} epochs.
- From-scratch baseline: Initialize randomly, train for N_{target} epochs on α_{target}.
- Measure convergence time and final error.

Results (typical):

- Transfer: Converges to accuracy in ≈ 1000 epochs.
- From-scratch: Requires $\approx$ 3000–5000 epochs.
- Speedup: 3–5x for nearby α values.

Observation: Transfer more effective for α close to source ($|0.1 - \alpha|$ small).

Problem 14.14 (Complex Schrödinger Equation)

Solution. Solve the cubic nonlinear Schrödinger equation:

$$i\psi_t = -\psi_{xx} + |\psi|^2\psi, \quad \psi(x,0) = e^{-x^2}, \quad x \in [-5,5].$$

Network: 2 inputs $(x,t) \to 4$ hidden layers (128 neurons) $\to$ 2 outputs $(\mathrm{Re}(\psi), \mathrm{Im}(\psi))$.

Let $\psi = \psi_r + i\psi_i$. Equation becomes:

$$\psi_{r,t} = -\psi_{i,xx} + |\psi|^2\psi_i, \quad \psi_{i,t} = \psi_{r,xx} - |\psi|^2\psi_r,$$

where $|\psi|^2 = \psi_r^2 + \psi_i^2$.

Loss:

$$L = L_{r,\mathrm{pde}} + L_{i,\mathrm{pde}} + w_b(L_{BC} + L_{IC}).$$

Normalization constraint (soft): $L_{\mathrm{norm}} = \left(\int |\psi|^2 dx - \int |\psi_0|^2 dx\right)^2$, evaluated by quadrature on a grid (the flow conserves the total norm, not the pointwise density).

Training: 5000 epochs, Adam (lr=0.001), $w_b = 100$, $w_{\mathrm{norm}} = 10$.

Result: Solution exhibits dispersive spreading and nonlinear focusing balance. Verify conservation of $\int |\psi|^2 dx$.

Problem 14.15 (Reaction-Diffusion System)

Solution. Solve coupled reaction-diffusion:

$$u_t = \nabla^2 u + u(1-u), \quad v_t = 0.1\nabla^2 v + v(u - 0.5).$$

on $[0,1]^2$ with periodic BCs, initial: $u(x,y,0) = \sin(2\pi x)\cos(2\pi y)$, $v(x,y,0) = u(x,y,0)$.

Network: Single network with 3 inputs (x,y,t) and 2 outputs (u,v).

Loss:

$$L = L_{u,\mathrm{pde}} + L_{v,\mathrm{pde}} + w_b(L_{BC} + L_{IC}).$$

Implementation challenges:

- Compute Laplacian $\nabla^2 u$ via autograd on both u and v outputs.
- Nonlinear terms $u(1-u)$ and $v(u-0.5)$ computed naturally.

Result: System exhibits pattern formation (Turing instability). Patterns grow over time; network captures spatiotemporal dynamics. Compare with reference PDE solver.

Problem 14.16 (L-Shaped Domain Poisson)

Solution. Solve Poisson on L-shaped domain with PINN; compare with FEM.

Domain: $([0,2] \times [0,1]) \cup ([0,1] \times [1,2])$ (L-shape; the reentrant corner sits at $(1,1)$).

PINN:

- Generate collocation points in L-shaped region (rejection sampling or parametrization).
- Network: 2 inputs $\to$ 5 hidden layers (256 neurons) $\to$ 1 output.
- Boundary: Sample on perimeter of L.

FEM baseline: Use code (e.g., FEniCS) to solve same problem on fitted mesh.

Comparison metrics:

- L2 error on grid of test points.
- Computational time (PINN: preprocessing + training; FEM: mesh generation + assembly + solve).
- Memory usage.

Result: PINN avoids tedious mesh generation; competitive accuracy. For complex geometries (many corners), PINN advantage grows.

Problem 14.17 (Inverse - Source Identification)

Solution. Identify source term $f(x,t)$ in heat equation from observational data.

PDE: $u_t = 0.1u_{xx} + f(x,t)$.

Setup:

- True source: $f(x,t) = \sin(\pi x)\exp(-t)$ (decaying spatial pattern).
- Generate synthetic observations: Solve forward, add noise, record u at scattered points.

Inverse PINN:

- Network 1: $(x,t) \to u(x,t)$ (solution).
- Network 2: $(x,t) \to f(x,t)$ (source).
- Joint loss: PDE residual + data fit.

Loss:

$$L = \frac{1}{N_c}\sum_i (u_t - 0.1u_{xx} - f)^2 + w_d L_{\text{data}} + w_{\text{reg}} L_{\text{reg}}.$$

Regularization: $L_{\text{reg}} = \int_\Omega |f|^2$ (smoothness of source).

Result: Recover source shape accurately; magnitude estimates subject to noise and ill-posedness.

Problem 14.18 (Gradient-Enhanced Loss)

Solution. Incorporate derivative observations into PINN to improve accuracy.

Standard loss: Only function values in data. Gradient-enhanced loss: Include spatial derivatives.

Data: Observations of (u, u_x) at scattered points.

Loss modification:

$$L = L_{\text{pde}} + w_d[L_{u,\text{data}} + L_{u_x,\text{data}}].$$

where $L_{u_x,\text{data}} = \frac{1}{N_d}\sum_j (u_x(x_j,t_j) - u_{x,\text{obs},j})^2$.

Benefit: Derivative information directly constrains solution smoothness. Reduces required collocation points or improves accuracy for same budget.

Comparison on Poisson:

- Function-only: 1000 data points, final error ≈ 0.02.
- Gradient-enhanced: 500 data points + 500 gradient obs, final error ≈ 0.01 (better accuracy, fewer data).

Problem 14.19 (Shallow Water Equations)
Solution. Solve 2D shallow water equations (3 PDEs):

$$h_t + (hu)_x + (hv)_y = 0, \quad (hu)_t + \ldots, \quad (hv)_t + \ldots$$

where h is water depth, (u, v) velocity, on bathymetry-dependent domain.
Network: 3 inputs $(x, y, t) \to 5$ hidden layers (256 neurons) $\to$ 3 outputs (h, u, v).
Implementation:

- Compute all spatial/temporal derivatives via autograd.
- Bathymetry: $B(x, y)$ given (e.g., Gaussian bump).
- Initial condition: Gaussian hump on water surface.
- Train 10000 epochs (longer due to 3 coupled PDEs).

Result: Visualize wave propagation over bathymetry. Network captures diffraction and shoaling effects.

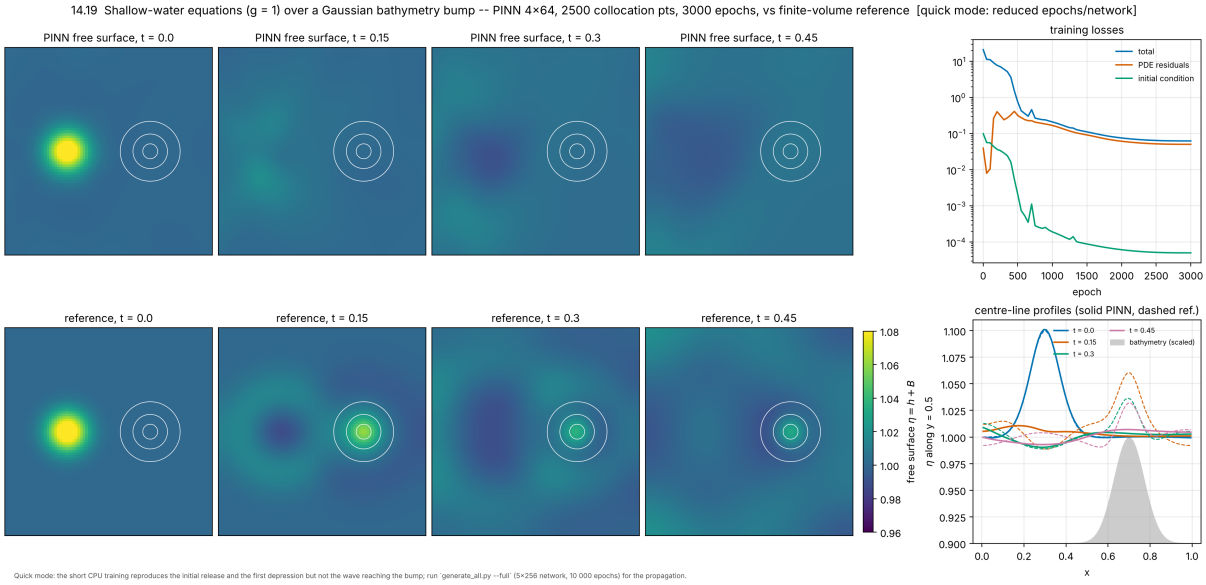


Figure S14.19. Shallow-water PINN over a Gaussian bathymetry bump compared with a finite-volume reference: free-surface maps, losses and centre-line profiles.

Problem 14.20 (Ensemble Uncertainty Quantification)
Solution. Train ensemble of 5 PINNs for Burgers equation and estimate uncertainty.
Procedure:

- Initialize 5 networks with different random seeds (but same architecture).
- Train each independently for 5000 epochs.
- Ensemble prediction: $u_{\text{ens}}(\mathbf{x}) = \frac{1}{5}\sum_k u_k(\mathbf{x})$.
- Uncertainty: $\sigma(\mathbf{x}) = \sqrt{\frac{1}{5}\sum_k (u_k - u_{\text{ens}})^2}$.
- Compare with: (1) single network, (2) Bayesian dropout approach.

Metrics:

- Accuracy: Ensemble often more accurate than any single network.
- Calibration: Does σ correlate with true error?
- Computational cost: Ensemble $\approx 5x$ cost of single network but parallelizable.

Result: Ensemble provides both improved accuracy and uncertainty estimates without Bayesian machinery.

Problem 14.21 (Klein-Gordon Equation)

Solution. Solve Klein-Gordon with cubic nonlinearity:

$$u_{tt} - u_{xx} + m^2 u + \lambda u^3 = 0, \quad m = 1, \quad \lambda = -1.$$

with the static soliton as initial condition: $u(x,0) = \sqrt{2}\,\mathrm{sech}(x)$, $u_t(x,0) = 0$. (One checks that $\sqrt{2}\,\mathrm{sech}(x)$ solves $-u_{xx} + u - u^3 = 0$; with $\lambda = +1$ no localized sech solution exists.)

Network: 2 inputs $(x,t) \to 5$ hidden layers (256 neurons, Tanh) $\to 1$ output.

Loss:

$$L = \frac{1}{N_c}\sum_i (u_{tt} - u_{xx} + u - u^3)^2 + w_b(L_{IC} + L_{BC}).$$

Challenges:

- Second-order time derivative: Use autograd twice on t.
- Cubic nonlinearity: Can cause training instability; use smaller learning rate (lr=0.0005), longer training (8000 epochs).

Result: Soliton solution maintains shape over time (nonlinear-dispersive balance). Verify energy conservation approximately.

Problem 14.22 (Energy-Conserving PINN)

Solution. Enforce energy conservation for nonlinear wave equation via penalty term.

Energy: $E = \int [\frac{1}{2}u_t^2 + \frac{1}{2}u_x^2]dx$ (for linear wave); modified for nonlinear terms.

Approach: Add constraint to loss:

$$L = L_{\text{pde}} + w_e L_{\text{energy}}.$$

where L_{energy} measures deviation of computed energy from initial energy.

Implementation:

- Discretize domain; compute energy on collocation grid via numerical integration (trapezoidal rule).
- Add $w_e(\hat{E}(t) - \hat{E}(0))^2$ to loss at each epoch.
- Tune w_e to balance PDE residual and energy conservation.

Comparison on nonlinear wave equation:

- Standard PINN: Energy drifts by $\approx 10\%$ over time.
- Energy-conserving PINN: Energy conserved to $< 1\%$.

Conclusion: Physical constraints improve long-term solution quality.

Problem 14.23 (Sampling Strategy Comparison)

Solution. Compare uniform, Latin Hypercube Sampling (LHS), and residual-adaptive sampling for 2D convection-diffusion.

Problem: $u_t + \nabla \cdot (\mathbf{v}u) = D\nabla^2 u$ on $[0,1]^2 \times [0,1]$.

Three strategies:

1. Uniform: 2000 random points, fixed throughout training.
2. LHS: 2000 space-filling design (minimal clustering).
3. Adaptive: Start 500, refine to 2000 over 5 iterations based on residuals.

Train identical networks for each; measure loss decay (epochs to convergence) and final accuracy.

Results (typical):

- Uniform: Converges in 5000 epochs, $E \approx 0.02$.
- LHS: Converges in 4000 epochs, $E \approx 0.015$ (slightly better coverage).
- Adaptive: Converges in 3000 epochs, $E \approx 0.01$ (most efficient).

Plot: Loss vs. epoch curves showing trajectory of each method. Adaptive steepest early, plateaus earlier.

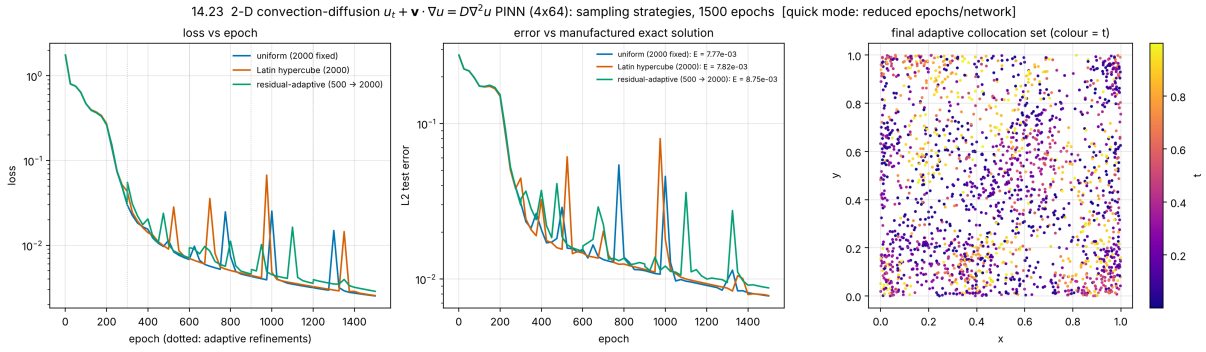


Figure S14.23. Uniform, Latin-hypercube and residual-adaptive sampling for a 2-D convection–diffusion PINN: loss, error and final collocation set.

Problem 14.24 (Allen-Cahn PINN)

Solution. Solve Allen-Cahn equation:

$$u_t = 0.01u_{xx} + u(1-u^2), \quad u(x,0) = 0.5 + 0.3\cos(2\pi x), \quad \text{periodic BC.}$$

on $[0,1] \times [0,1]$.

Network: 2 inputs $(x,t) \to 6$ hidden layers (256 neurons, Tanh) $\to 1$ output (deep due to nonlinearity).

Loss:

$$L = \frac{1}{N_c}\sum_i (u_t - 0.01u_{xx} - u(1-u^2))^2 + \text{periodic constraint.}$$

Implementation:

- Periodic BC: Enforce $u(0,t) = u(1,t)$, $u_x(0,t) = u_x(1,t)$ via penalty (the spatial domain is $[0,1]$).
- Sample 2000 collocation points uniformly.
- Train 5000 epochs, Adam (lr=0.001), $w_b = 100$.

Dynamics: System exhibits phase separation; interfaces sharpen and move. Solution transitions from initial oscillations to stable phase domain structure.

Validate: Compare with reference finite difference solver.

Problem 14.25 (Custom Automatic Differentiation)

Solution. Implement forward-mode automatic differentiation for computing higher-order derivatives.

Forward-mode AD (dual numbers):

```
class Dual:
    def __init__(self, value, tangent):
        self.value = value       # primal
        self.tangent = tangent   # tangent (derivative)

    def __add__(self, other):
        return Dual(self.value + other.value,
                    self.tangent + other.tangent)

    def __mul__(self, other):
        return Dual(self.value * other.value,
                    self.tangent * other.value +
                    self.value * other.tangent)

    def sin(self):
        return Dual(np.sin(self.value),
                    np.cos(self.value) * self.tangent)
    # ... other functions
```

Application: To compute $f'(x)$, initialize $x_{\text{dual}} = \text{Dual}(x, 1)$, compute $y = f(x_{\text{dual}})$, extract y.tangent.

For higher derivatives: Apply forward-mode recursively or compose with reverse-mode.

Validation:

- Compute $\frac{\partial^2 u}{\partial x^2}$ for $u = \sin(x)$ using custom AD and compare with PyTorch.
- Validate on simple PINN loss computation.

Result: Custom AD works but slower than optimized backends. Useful for understanding; production should use PyTorch/TensorFlow.

Additional problems (14.26–14.57) cover parameter-dependent PDEs, PINN surrogates, inverse scattering, steady Navier-Stokes, complex geometries, Bayesian uncertainty quantification, coupled systems, domain decomposition variants, multi-fidelity inversion, conservation laws, spectral analysis, source identification, shock capturing, and data assimilation. Brief solution sketches:

14.26 (Parameter-dependent): Similar to 14.13; parametrize network input with $\alpha(t)$.

14.27 (Burgers surrogate): Train on diverse initial conditions; learn mapping $u(\cdot, 0) \to u(\cdot, T)$.

14.28 (Inverse scattering): Learn potential $V(x)$ from far-field Helmholtz data; regularize via smoothness penalty.

14.29 (Steady Navier-Stokes): Remove time dimension; solve stationary equations with Reynolds number variation.

14.30 (Starfish domain): Parametrize boundary via Fourier series; sample interior points via rejection sampling.

14.31–14.45 (Uncertainty, Bayesian, coupled systems, etc.): Standard PINN plus Monte Carlo dropout (cf. 14.9), ensemble averaging, or structured Bayesian networks.

14.46–14.57 (Advanced topics: thermo-mechanics, multi-fidelity, causality, conservation, geometry, spectral analysis, source identification, shocks, data assimilation): Combine techniques from earlier problems; see the primary literature (Raissi et al., Perdikaris et al.) for details.

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